

COMPASS_Synoptic_SmartChamberFluxes_2025_CO2

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Contents

0.1	Goals of this code:	3
0.2	Set Up & Load Packages	3
0.3	Pull in Smart Chamber Files from the Raw Data folder	3
0.4	Create a unique Plot_ID, check if there is a meta-data file, if not, create a meta-data file . .	16
0.5	Mark Drop column in Metadata for Retaken Fluxes or other reasons to Drop	18
0.6	Match up the metadata to the rest of the data	19
0.7	Change the units from ppm to umol	21
0.8	Calculate fluxes	21
0.9	Visualize fluxes and sort by r2	28
0.10	check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)	29
0.11	Spot check a few fluxes with R2 > 0.9	31
0.12	Take a look at the fluxes <0.9 and a flux estimate below 1	35
0.13	Take a look at the bad fluxes <0.9 and have a high flux estimate	66
1	See if we want to remove these fluxes or if we want to try and recalculate these ones?	66
2	Look at any CO2 fluxes that are negative and remove those plots	84
2.1	Look at the really high or really low ones to see if there is a timing issue or something	87
2.2	Check all fluxes	137
2.3	Plot TIMESTAMP HM81_AIC HM81_flux.estimate HM81_p.value	139
2.4	1 1-Apr-GCW-TR-1 2025-04-16 10:58:18 NA NA NA	139
2.5	2 1-Apr-GCW-TR-2 2025-04-16 11:07:21 415.1026 2.767439 1.620905e-31	139
2.6	3 1-Apr-GCW-TR-3 2025-04-16 11:11:25 417.9395 2.715836 2.236664e-31	139
2.7	4 1-Apr-GCW-TR-4 2025-04-16 11:15:46 350.5254 1.579223 7.544911e-33	139
2.8	5 1-Apr-GCW-TR-5 2025-04-16 11:20:03 407.1443 1.859655 7.860953e-29	139
2.9	6 1-Apr-GCW-UP-1 2025-04-16 11:25:26 NA NA NA	139
2.10	HM81_r.squared HM81_RMSE lin_AIC lin_flux.estimate lin_flux.std.error	139

2.47	Warning in rbind(c("1", "Apr", "GCW", "TR", "1"), c("1", "Apr", "GCW", "TR", :	140
2.48	number of columns of result is not a multiple of vector length (arg 61)	140
2.49	RepNumber Month Site Zone Replicate	140
2.50	1 1 Apr GCW TR 1	140
2.51	2 1 Apr GCW TR 2	140
2.52	3 1 Apr GCW TR 3	140
2.53	4 1 Apr GCW TR 4	140
2.54	5 1 Apr GCW TR 5	140
2.55	6 1 Apr GCW UP 1	140

0.1 Goals of this code:

1. Reads in raw smart chamber
2. If no metadatafile, write one out & fields will be: Label the IDs pulled from the label, Plot *fixed so that if there are any issues in the label they are good here Drop column – false as long as label looks right adjust time – would be initially populated by the start time in the OR blank adjust observation length – initially populated by the obs length OR blank notes / comment
3. Code reads in the meta data and applies it to the data
 - a. i.e. drops fluxes that have the drop column TRUE or FLAG this
 - b. adjusting times as needed
4. Now you compute fluxes
5. Visualize and look for issues
6. If found, report to meta data file then go back to 3 and reanalyze
7. Apply algorithmic qaqc and generate flags

0.2 Set Up & Load Packages

0.3 Pull in Smart Chamber Files from the Raw Data folder

```
# Create a temporary directory
tmp_dir <- tempdir()

# Get list of files inside the zip
zip_contents <- unzip("Raw Data.zip", list = TRUE)

# Keep only JSON files
json_files <- zip_contents$Name[grep("json$", zip_contents$Name, ignore.case = TRUE)]

# Extract only the JSON files
unzip(
  "Raw Data.zip",
  files = json_files,
  exdir = tmp_dir
)
```

```

# List extracted JSON files (full paths)
file.list <- list.files(
  path = tmp_dir,
  pattern = "json$",
  full.names = TRUE,
  recursive = TRUE
)

head(file.list)

```

```

## [1] "C:\\Users\\readz\\AppData\\Local\\Temp\\Rtmp6JoVRq\\Raw Data\\COMPASS-CBSYNOPTIC-GCW-04162025.json"
## [2] "C:\\Users\\readz\\AppData\\Local\\Temp\\Rtmp6JoVRq\\Raw Data\\COMPASS-CBSYNOPTIC-GCW-20250514.json"
## [3] "C:\\Users\\readz\\AppData\\Local\\Temp\\Rtmp6JoVRq\\Raw Data\\COMPASS-CBSYNOPTIC-GCW-20250617.json"
## [4] "C:\\Users\\readz\\AppData\\Local\\Temp\\Rtmp6JoVRq\\Raw Data\\COMPASS-CBSYNOPTIC-GCW-20250716.json"
## [5] "C:\\Users\\readz\\AppData\\Local\\Temp\\Rtmp6JoVRq\\Raw Data\\COMPASS-CBSYNOPTIC-GCW-20250909.json"
## [6] "C:\\Users\\readz\\AppData\\Local\\Temp\\Rtmp6JoVRq\\Raw Data\\COMPASS-CBSYNOPTIC-GCW&MSM-20251112.json"

```

```

# Read in the data
alldat1 <- lapply(file.list, ffi_read_LIsmartchamber)

```

```

## Reading observation 1 label GCW-TR-1

## Reading observation 2 label GCW-TR-2

## Reading observation 3 label GCW-TR-3

## Reading observation 4 label GCW-TR-4

## Reading observation 5 label GCW-TR-5

## Reading observation 6 label GCW-UP-1

## Reading observation 7 label GCW-UP-5

## Reading observation 8 label GCW-UP-2

## Reading observation 9 label GCW-UP-3

## Reading observation 10 label GCW-UP-4

## Reading observation 1 label GCW-TR-1

## Reading observation 2 label GCW-TR-2

## Reading observation 3 label GCW-TR-3

## Reading observation 4 label GCW-TR-3a

```

Reading observation 5 label GCW-TR-4

Reading observation 6 label GCW-TR-5

Reading observation 7 label GCW-UP-1

Reading observation 8 label GCW-UP-5

Reading observation 9 label GCW-UP-5a

Reading observation 10 label GCW-UP-2

Reading observation 11 label GCW-UP-3

Reading observation 12 label GCW-UP-4

Reading observation 1 label GCW-TR-1

Reading observation 2 label GCW-TR-2

Reading observation 3 label GCW-TR-3

Reading observation 4 label GCW-TR-4

Reading observation 5 label GCW-TR-5

Reading observation 6 label GCW-UP-1

Reading observation 7 label GCW-UP-5

Reading observation 8 label GCW-UP-2

Reading observation 9 label GCW-UP-4

Reading observation 10 label GCW-UP-3

Reading observation 1 label GCW-TR-5

Reading observation 2 label GCW-TR-4

Reading observation 3 label GCW-TR-3

Reading observation 4 label GCW-TR-2

Reading observation 5 label GCW-TR-1

Reading observation 6 label GCW-UP-1

Reading observation 7 label GCW-UP-5

Reading observation 8 label GCW-UP-2

Reading observation 9 label GCW-UP-3

Reading observation 10 label GCW-UP-4

Reading observation 1 label GCW-TR-1

Reading observation 2 label GCW-TR-2

Reading observation 3 label GCW-TR-3

Reading observation 4 label GCW-TR-4

Reading observation 5 label GCW-TR-5

Reading observation 6 label GCW-UP-1

Reading observation 7 label GCW-UP-5

Reading observation 8 label GCW-UP-2

Reading observation 9 label GCW-UP-3

Reading observation 10 label GCW-UP-4

Reading observation 1 label GCW-TR-1

Reading observation 2 label GCW-TR-2

Reading observation 3 label GCW-TR-3

Reading observation 4 label GCW-TR-4

Reading observation 5 label GCW-TR-5

Reading observation 6 label GCW-UP-1

Reading observation 7 label GCW-UP-5

Reading observation 8 label GCW-UP-2

Reading observation 9 label GCW-UP-3

Reading observation 10 label GCW-UP-4

Reading observation 11 label MSM-UP-4

Reading observation 12 label MSM-UP-1

Reading observation 13 label MSM-UP-2

Reading observation 14 label MSM-UP-3

Reading observation 15 label MSM-UP-3a

Reading observation 16 label MSM-UP-5

Reading observation 17 label MSM-TR-1

Reading observation 18 label MSM-TR-1a

Reading observation 19 label MSM-TR-2

Reading observation 20 label MSM-TR-3

Reading observation 21 label MSM-TR-4

Reading observation 22 label MSM-TR-5

Reading observation 1 label GWI-TR-1

Reading observation 2 label GWI-TR-2

Reading observation 3 label GWI-TR-3

Reading observation 4 label GWI-TR-4

Reading observation 5 label GWI-TR-5

Reading observation 6 label GWI-UP-1

Reading observation 7 label GWI-UP-2

Reading observation 8 label GWI-UP-3

Reading observation 9 label GWI-UP-3a

Reading observation 10 label GWI-UP-4

Reading observation 11 label GWI-UP-5

Reading observation 1 label GWI-TR-1

Reading observation 2 label GWI-TR-2

Reading observation 3 label GWI-TR-3

Reading observation 4 label GWI-TR-4

Reading observation 5 label GWI-TR-5

Reading observation 6 label GWI-UP-1

Reading observation 7 label GWI-UP-2

Reading observation 8 label GWI-UP-3

Reading observation 9 label GWI-UP-4

Reading observation 10 label GWI-UP-5

Reading observation 1 label GWI-TR-1

Reading observation 2 label GWI-TR-2

Reading observation 3 label GWI-TR-3

Reading observation 4 label GWI-TR-4

Reading observation 5 label GWI-TR-5

Reading observation 6 label GWI-UP-1

Reading observation 7 label GWI-UP-2

Reading observation 8 label GWI-UP-3

Reading observation 9 label GWI-UP-4

Reading observation 10 label GWI-UP-5

Reading observation 1 label GWI-TR-1

Reading observation 2 label GWI-TR-2

Reading observation 3 label GWI-TR-3

Reading observation 4 label GWI-TR-4

Reading observation 5 label GWI-TR-5

Reading observation 6 label GWI-UP-1

Reading observation 7 label GWI-UP-2

Reading observation 8 label GWI-UP-3

Reading observation 9 label GWI-UP-4

Reading observation 10 label GWI-UP-5

Reading observation 1 label GWI-TR-5

Reading observation 2 label GWI-TR-4

Reading observation 3 label GWI-TR-3

Reading observation 4 label GWI-TR-2

Reading observation 5 label GWI-TR-1

Reading observation 6 label GWI-TR-1a

Reading observation 7 label GWI-UP-1

Reading observation 8 label GWI-UP-1a

Reading observation 9 label GWI-UP-2

Reading observation 10 label GWI-UP-3

Reading observation 11 label GWI-UP-3a

Reading observation 12 label GWI-UP-4

Reading observation 13 label GWI-UP-5

Reading observation 1 label 4

Reading observation 2 label MSM-TR-2

Reading observation 3 label MSM-TR-2a

Reading observation 4 label MSM-TR-2b

Reading observation 5 label MSM-TR-3

Reading observation 6 label MSM-TR-1

Reading observation 7 label MSM-TR-4

Reading observation 8 label MSM-TR-5

Reading observation 9 label MSM-UP-1

Reading observation 10 label MSM-UP-2

Reading observation 11 label MSM-UP-3

Reading observation 12 label MSM-UP-5

Reading observation 13 label MSM-UP-4

Reading observation 1 label MSM-TR-1

Reading observation 2 label MSM-TR-2

Reading observation 3 label MSM-TR-3

Reading observation 4 label MSM-TR-4

Reading observation 5 label MSM-TR-5

Reading observation 6 label MSM-UP-5

Reading observation 7 label MSM-UP-4

Reading observation 8 label MSM-UP-3

Reading observation 9 label MSM-UP-2

Reading observation 10 label MSM-UP-1

Reading observation 1 label MSM-TR-1

Reading observation 2 label MSM-TR-2

Reading observation 3 label MSM-TR-3

Reading observation 4 label MSM-TR-4

Reading observation 5 label MSM-TR-5

Reading observation 6 label MSM-UP-5

Reading observation 7 label MSM-UP-4

Reading observation 8 label MSM-UP-3

Reading observation 9 label MSM-UP-2

Reading observation 10 label MSM-UP-1

Reading observation 1 label SWH-TR-6

Reading observation 2 label SWH-TR-6a

Reading observation 3 label SWH-TR-6b

Reading observation 4 label SWH-TR-6c

Reading observation 5 label SWH-TR-1

Reading observation 6 label SWH-TR-2

Reading observation 7 label SWH-TR-7

Reading observation 8 label SWH-TR-7a

Reading observation 9 label SWH-TR-3

Reading observation 10 label SWH-TR-4

Reading observation 11 label SWH-TR-5

Reading observation 12 label SWH-UP-1

Reading observation 13 label SWH-UP-2

Reading observation 14 label SWH-UP-3

Reading observation 15 label SWH-UP-4

Reading observation 16 label SWH-UP-4a

Reading observation 17 label SWH-UP-5

Reading observation 1 label SWH-SWAMP-1

Reading observation 2 label SWH-SWAMP-1a

Reading observation 3 label SWH-SWAMP-2

Reading observation 4 label SWH-SWAMP-3

Reading observation 5 label SWH-SWAMP-3a

Reading observation 6 label SWH-SWAMP-5

Reading observation 7 label SWH-SWAMP-4

Reading observation 8 label SWH-SWAMP-4a

Reading observation 9 label SWH-UP-1

Reading observation 10 label SWH-UP-2

Reading observation 11 label SWH-UP-3

Reading observation 12 label SWH-UP-4

Reading observation 13 label SWH-UP-5

Reading observation 14 label SWH-TR-2

Reading observation 15 label SWH-TR-1

Reading observation 16 label SWH-TR-3

Reading observation 17 label SWH-TR-5

Reading observation 18 label SWH-TR-4

Reading observation 19 label SWH-TR-10

Reading observation 20 label SWH-TR-9

Reading observation 21 label SWH-TR-8

Reading observation 22 label SWH-TR-8a

Reading observation 23 label SWH-TR-7

Reading observation 24 label SWH-TR-7a

Reading observation 25 label SWH-TR-6

Reading observation 26 label SWH-TR-6a

Reading observation 1 label SWH-TR-5

Reading observation 2 label SWH-TR-9

Reading observation 3 label SWH-TR-10

Reading observation 4 label SWH-TR-4

Reading observation 5 label SWH-TR-3

Reading observation 6 label SWH-TR-8

Reading observation 7 label SWH-TR-6

Reading observation 8 label SWH-TR-1

Reading observation 9 label SWH-TR-7

Reading observation 10 label SWH-TR-7a

Reading observation 11 label SWH-TR-2

Reading observation 12 label SWH-UP-1

Reading observation 13 label SWH-UP-2

Reading observation 14 label SWH-UP-3

Reading observation 15 label SWH-UP-4

Reading observation 16 label SWH-UP-5

Reading observation 17 label SWH-SWAMP-1

Reading observation 18 label SWH-SWAMP-2

Reading observation 19 label SWH-SWAMP-3

Reading observation 20 label SWH-SWAMP-4

Reading observation 21 label SWH-SWAMP-5

Reading observation 1 label SWH-UP-1

Reading observation 2 label SWH-UP-2

Reading observation 3 label SWH-UP-3

Reading observation 4 label SWH-UP-3a

Reading observation 5 label SWH-UP-4

Reading observation 6 label SWH-UP-5

Reading observation 7 label SWH-SWAMP-4

Reading observation 8 label SWH-SWAMP-3

Reading observation 9 label SWH-SWAMP-5

Reading observation 10 label SWH-SWAMP-1

Reading observation 11 label SWH-SWAMP-2

Reading observation 12 label SWH-TR-6

Reading observation 13 label SWH-TR-1

Reading observation 14 label SWH-TR-1a

Reading observation 15 label SWH-TR-1b

Reading observation 16 label SWH-TR-7

Reading observation 17 label SWH-TR-2

Reading observation 18 label SWH-TR-3

Reading observation 19 label SWH-TR-3a

Reading observation 20 label SWH-TR-3b

Reading observation 21 label SWH-TR-8

Reading observation 22 label SWH-TR-5

Reading observation 23 label SWH-TR-9

Reading observation 24 label SWH-TR-10

Reading observation 25 label SWH-TR-10a

Reading observation 26 label SWH-TR-4

Reading observation 1 label SWH-TR-7

Reading observation 2 label SWH-TR-6

Reading observation 3 label SWH-TR-8

```

## Reading observation 4 label SWH-TR-4

## Reading observation 5 label SWH-TR-5

## Reading observation 6 label SWH-TR-2

## Reading observation 7 label SWH-TR-1

## Reading observation 8 label SWH-TR-3

## Reading observation 9 label SWH-TR-3a

## Reading observation 10 label SWH-UP-1

## Warning in FUN(X[[i]], ...): There are 0-row data in
## COMPASS-CBSYNOPTIC-SWH-20251104.json at observation 10 label SWH-UP-1
## Warning in FUN(X[[i]], ...): There are 0-row data in
## COMPASS-CBSYNOPTIC-SWH-20251104.json at observation 10 label SWH-UP-1

## Reading observation 11 label SWH-UP-1a

## Warning in FUN(X[[i]], ...): There are 0-row data in
## COMPASS-CBSYNOPTIC-SWH-20251104.json at observation 11 label SWH-UP-1a

## Warning in FUN(X[[i]], ...): There are 0-row data in
## COMPASS-CBSYNOPTIC-SWH-20251104.json at observation 11 label SWH-UP-1a

## Reading observation 12 label SWH-UP-1b

## Warning in FUN(X[[i]], ...): There are 0-row data in
## COMPASS-CBSYNOPTIC-SWH-20251104.json at observation 12 label SWH-UP-1b

## Warning in FUN(X[[i]], ...): There are 0-row data in
## COMPASS-CBSYNOPTIC-SWH-20251104.json at observation 12 label SWH-UP-1b

## Reading observation 13 label SWH-UP-1c

## Reading observation 14 label SWH-UP-2

## Reading observation 15 label SWH-UP-2a

## Reading observation 16 label SWH-UP-3

## Reading observation 17 label SWH-UP-3a

## Reading observation 18 label SWH-UP-3b

## Reading observation 19 label SWH-UP-4

```

```
## Reading observation 20 label SWH-UP-5

## Reading observation 21 label SWH-SWAMP-4

## Reading observation 22 label SWH-SWAMP-4a

## Reading observation 23 label SWH-SWAMP-3

## Reading observation 24 label SWH-SWAMP-3a

## Reading observation 25 label SWH-SWAMP-5

## Reading observation 26 label SWH-SWAMP-3b

## Reading observation 27 label SWH-SWAMP-2

## Reading observation 28 label SWH-SWAMP-1

## Reading observation 29 label SWH-TR-9

## Reading observation 30 label SWH-TR-9a

## Reading observation 31 label SWH-TR-10
```

```
# Combine into one data.table
alldat <- data.table::rbindlist(alldat1, fill = TRUE)

##Remove CO2 values below 0 ppm
alldat <- subset(alldat)
```

0.4 Create a unique Plot_ID, check if there is a meta-data file, if not, create a meta-data file

```
##### only run this first part if you don't already have a meta-data file
#because if you do then all the data for each one will have a plot and you won't be able to cha

#create unique Plot ID for each measurement:
#detach("package:data.table", unload=TRUE)
alldat$month <- as.character(month(alldat$TIMESTAMP, label=TRUE, abbr=TRUE)) #will not run if you have

alldat$year <- year(alldat$TIMESTAMP)

alldat <- alldat %>%
  mutate(Plot_ID = str_c(pmin(month, RepNum),
                           pmax(month, RepNum), sep = '-'))
alldat <- alldat %>%
  mutate(Plot_ID = str_c(pmin(Plot_ID, label),
```

```

        pmax(Plot_ID, label), sep = '-')
#this needs to be Plot ID so that later we ID plot from metadata and not the metadata row

# Path inside the zip
zip_file <- "Raw Data.zip"
csv_inside_zip <- "Raw Data/COMPASS_SmartChamber_Metadata_2025.csv"

# Make sure the destination folder exists
dir.create("Raw Data", showWarnings = FALSE)

# Extract only the metadata CSV if it doesn't already exist
if (!file.exists(csv_inside_zip)) {
  unzip(
    zip_file,
    files = csv_inside_zip,
    exdir = "."
  )
}

# Define the file path where you want to save the CSV file
file_path <- "Raw Data/COMPASS_SmartChamber_Metadata_2025.csv"

# Check if the CSV file exists
if (file.exists(file_path)) {
  # If it exists, read it back into R
  metadata <- read.csv(file_path)
  print("CSV file exists and has been read into the code.")
} else {
  # If it does not exist, create the CSV file
  #create a metadata file from the Plot IDs
  # pull timestart timestamps to put in the metadata and obs lengths
  metadata <- alldat %>%
  group_by(Plot_ID) %>%
  summarise(
    Date = format(min(TIMESTAMP), "%Y-%m-%d"),
    Time_start = format(min(TIMESTAMP), "%H:%M:%S"),
    First_Timestamp = min(TIMESTAMP), # Minimum (earliest) timestamp for each Plot_ID
    Last_Timestamp = max(TIMESTAMP), # Maximum (latest) timestamp for each Plot_ID
    Total_Time = as.numeric(difftime(max(TIMESTAMP), min(TIMESTAMP), units = "secs")), # Time differen
  ) %>%
  ungroup() %>%
  mutate(drop = FALSE) %>% # Add a 'drop' column with 'FALSE' for every row
  #mutate(Notes = " ") # Add a 'drop' column with 'FALSE' for every row
  #mutate(Deadband = 5)
  mutate(Notes = "") %>%
  #mutate(Plot = Plot_ID) #think need this because we need to make a plot column

  # Write df_unique to CSV
  write.csv(metadata, file = file_path, row.names = FALSE)

  print("CSV file does not exist. It has been created.")
}

```

```
## [1] "CSV file exists and has been read into the code."
```

0.5 Mark Drop column in Metadata for Retaken Fluxes or other reasons to Drop

```
##### Check For Plot ID

##Sometimes there are other fluxes in the .json files that we don't want to calculate
#change drop column to TRUE if the plot doesn't include a site code in the Plot_ID
metadata <- metadata %>%
  rename(Plot = Plot_ID) %>%
  mutate(
    drop = ifelse(
      !grepl("SWH|GCW|MSM|GWI|GCRew", Plot),
      # Check if Plot_ID does not contain the specified strings
      TRUE, # Set drop to TRUE if it doesn't contain any of the strings
      drop # Keep the original value of drop if it contains one of the strings
    )
  )

##### Check for retakes

##Sometimes we have to retake a flux in the field due to issues or ebullition
##We need to change the drop column to reflect using only the latest flux
#In order to we expect an "a" after the label, but sometimes this is not the case so we will fix that
metadata <- metadata %>%
  mutate(Plot = case_when(
    str_detect(Plot, "-retake") ~ str_replace(Plot, "-retake", "a"),
    str_detect(Plot, "-Retake") ~ str_replace(Plot, "-Retake", "a"),
    str_detect(Plot, "-b") ~ str_replace(Plot, "-b", "b"),
    TRUE ~ Plot # Keeps the original value if none of the conditions match
  ))

#If the flux was taken more than once, flag the others so that we use the latest measurement
# Extract the base plot (e.g., "GCW-TR-1" from "GCW-TR-1", "GCW-TR-1a", etc.)
metadata$base_plot <- str_extract(metadata$Plot, "^[^-]+-[A-Za-z]+-[A-Za-z]+-[A-Za-z]+")

# Extract the replicate number (the number after the last hyphen)
metadata$replicate <- as.numeric(str_extract(metadata$Plot, "(?<=)[0-9]+(?=[a-zA-Z]?$)"))

# Extract the suffix (if any, the letter after the replicate number)
metadata$suffix <- str_extract(metadata$Plot, "(?<=\\d)[a-zA-Z]$")

# If there's no suffix, set it to an empty string
metadata$suffix[is.na(metadata$suffix)] <- ""

# Sort by base plot, replicate number, and suffix (to keep highest suffix) - #####INSTEAD WE WANT TO CH
metadata <- metadata %>%
  group_by(base_plot, replicate) %>%
  arrange(base_plot, replicate, desc(suffix), .by_group = TRUE) %>%
  mutate(drop = row_number() != 1) %>% #the entry with the latest suffix will keep false all others wil
  #slice_head(n = 1) %>% #if you wanted to remove a the line rather than flag it
  ungroup()
```

```
##### Check Observation lengths

#if the observation length is over 5 mins then change drop column to true
metadata <- metadata %>%
  mutate(drop = if_else(Total_Time > 300, TRUE, drop))

##### Remove columns with drop = TRUE
metadata <- metadata %>%
  filter(drop == "FALSE")
```

##Dont need to pull L1 temp data or any other temps like we do for macrophyte because the smartchamber records the temp

0.6 Match up the metadata to the rest of the data

```
#match the altered metadata to the concentration data with fluxfinder ffi_metadata_match
alldat$metadat_row <- ffi_metadata_match(
  data_timestamps = alldat$TIMESTAMP,
  start_dates = metadata$Date,
  start_times = metadata$Time_start,
  obs_lengths = metadata$Total_Time)
```

YYYY-MM-DD format failed for start_dates; trying MM/DD/YYYY

```
head(alldat)
```

```
##      label remark  Chamber Version InstrumentModel InstrumentSerialNumber
##      <char> <char>   <char>   <char>          <char>          <char>
## 1: GCW-TR-1      82s-1050   1.24      LI-7810      TG10-01087
## 2: GCW-TR-1      82s-1050   1.24      LI-7810      TG10-01087
## 3: GCW-TR-1      82s-1050   1.24      LI-7810      TG10-01087
## 4: GCW-TR-1      82s-1050   1.24      LI-7810      TG10-01087
## 5: GCW-TR-1      82s-1050   1.24      LI-7810      TG10-01087
## 6: GCW-TR-1      82s-1050   1.24      LI-7810      TG10-01087
##  RepNum DeadBand  Area Offset ChamVolume IrgaVolume TotalVolume  gps_time
##    <int>   <int> <int>  <int>      <num>      <num>      <num>      <int>
## 1:     1       5   318    8      4244.1      46.96     6835.06 1744815467
## 2:     1       5   318    8      4244.1      46.96     6835.06 1744815467
## 3:     1       5   318    8      4244.1      46.96     6835.06 1744815467
## 4:     1       5   318    8      4244.1      46.96     6835.06 1744815467
## 5:     1       5   318    8      4244.1      46.96     6835.06 1744815467
## 6:     1       5   318    8      4244.1      46.96     6835.06 1744815467
##  latitude longitude gps_hdop altitude gps_sats      TimeZone chamber_p
##    <num>    <num>   <num>    <num>   <int>      <char>      <num>
## 1: 38.87443 -76.55125  1.15    -4.3      9 America/New_York 101.286
## 2: 38.87443 -76.55125  1.15    -4.3      9 America/New_York 101.293
## 3: 38.87443 -76.55125  1.15    -4.3      9 America/New_York 101.300
## 4: 38.87443 -76.55125  1.15    -4.3      9 America/New_York 101.299
```

```

## 5: 38.87443 -76.55125 1.15 -4.3 9 America/New_York 101.302
## 6: 38.87443 -76.55125 1.15 -4.3 9 America/New_York 101.309
## chamber_p_t chamber_t soil_t soilp_c soilp_m soilp_t ch4 co2 h2o
## <num> <num> <int> <num> <num> <num> <num> <num> <num>
## 1: 24.4500 18.2998 9999 0.033 0.506 17.7 2034.13 424.550 6.09496
## 2: 24.4375 18.2697 9999 0.033 0.506 17.7 2034.25 424.096 6.06980
## 3: 24.4688 18.2395 9999 0.033 0.506 17.7 2034.28 424.071 6.08350
## 4: 24.4688 18.2133 9999 0.033 0.506 17.7 2034.21 424.680 6.12490
## 5: 24.4792 18.1903 9999 0.033 0.506 17.7 2034.00 424.959 6.18503
## 6: 24.5208 18.1681 9999 0.033 0.506 17.7 2034.20 424.965 6.25992
## err ch4_F_o ch4_F_cv ch4_t_o ch4_C_o ch4_a ch4_C_x ch4_iter
## <int> <num> <num> <num> <num> <num> <num> <int>
## 1: 0 -0.323731 2.66239 5.95681 2034.21 1.7817e-05 0 10
## 2: 0 -0.323731 2.66239 5.95681 2034.21 1.7817e-05 0 10
## 3: 0 -0.323731 2.66239 5.95681 2034.21 1.7817e-05 0 10
## 4: 0 -0.323731 2.66239 5.95681 2034.21 1.7817e-05 0 10
## 5: 0 -0.323731 2.66239 5.95681 2034.21 1.7817e-05 0 10
## 6: 0 -0.323731 2.66239 5.95681 2034.21 1.7817e-05 0 10
## ch4_sei ch4_ses ch4_r2 ch4_slope ch4_domain ch4_n co2_F_o co2_F_cv
## <num> <num> <num> <num> <int> <int> <num> <num>
## 1: 0.0288185 0.000886724 0.969814 -0.0362435 53 54 6.1572 2.74702
## 2: 0.0288185 0.000886724 0.969814 -0.0362435 53 54 6.1572 2.74702
## 3: 0.0288185 0.000886724 0.969814 -0.0362435 53 54 6.1572 2.74702
## 4: 0.0288185 0.000886724 0.969814 -0.0362435 53 54 6.1572 2.74702
## 5: 0.0288185 0.000886724 0.969814 -0.0362435 53 54 6.1572 2.74702
## 6: 0.0288185 0.000886724 0.969814 -0.0362435 53 54 6.1572 2.74702
## co2_t_o co2_C_o co2_a co2_C_x co2_iter co2_sei co2_ses co2_r2
## <num> <num> <num> <num> <int> <num> <num> <num>
## 1: 11.8262 424.168 6.89626e-07 1e+06 10 0.568689 0.0174981 0.96758
## 2: 11.8262 424.168 6.89626e-07 1e+06 10 0.568689 0.0174981 0.96758
## 3: 11.8262 424.168 6.89626e-07 1e+06 10 0.568689 0.0174981 0.96758
## 4: 11.8262 424.168 6.89626e-07 1e+06 10 0.568689 0.0174981 0.96758
## 5: 11.8262 424.168 6.89626e-07 1e+06 10 0.568689 0.0174981 0.96758
## 6: 11.8262 424.168 6.89626e-07 1e+06 10 0.568689 0.0174981 0.96758
## co2_slope co2_domain co2_n TIMESTAMP month year Plot_ID
## <num> <int> <int> <POS> <char> <num> <char>
## 1: 0.689334 53 54 2025-04-16 10:57:47 Apr 2025 1-Apr-GCW-TR-1
## 2: 0.689334 53 54 2025-04-16 10:57:48 Apr 2025 1-Apr-GCW-TR-1
## 3: 0.689334 53 54 2025-04-16 10:57:49 Apr 2025 1-Apr-GCW-TR-1
## 4: 0.689334 53 54 2025-04-16 10:57:50 Apr 2025 1-Apr-GCW-TR-1
## 5: 0.689334 53 54 2025-04-16 10:57:51 Apr 2025 1-Apr-GCW-TR-1
## 6: 0.689334 53 54 2025-04-16 10:57:52 Apr 2025 1-Apr-GCW-TR-1
## metadat_row
## <num>
## 1: 1
## 2: 1
## 3: 1
## 4: 1
## 5: 1
## 6: 1

```

```

#alldat$Deadband <- metadata$Deadband[alldat$metadat_row] #flux calcs function cannot handle a column a
alldat$Plot <- metadata$Plot[alldat$metadat_row]

```

0.7 Change the units from ppm to umol

```
#change the units
alldat$CO2_umol <- ffi_ppm_to_umol(alldat$co2,
                                   volume = (alldat$TotalVolume / 1000000), # have to divide to get to m3
                                   temp = alldat$chamber_t) # degrees C

## Assuming atm = 101325 Pa

## Using R = 8.31446261815324 m3 Pa K-1 mol-1

#> Assuming atm = 101325 Pa
#> Using R = 8.31446261815324 m3 Pa K-1 mol-1

alldat$CO2_umol_m2 <- alldat$CO2_umol / 0.0318 #adjust for the area
```

0.8 Calculate fluxes

[illegible]


```
## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
```

[illegible]


```
## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
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## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
```

```
## Warning in rlm.default(x, y, weights, method = method, wt.method = wt.method, :
## 'rlm' failed to converge in 20 steps
```

```
## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
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## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
## NOTE: HM81_flux.estimate is not NA, implying nonlinear data
```

```
head(fluxes)
```

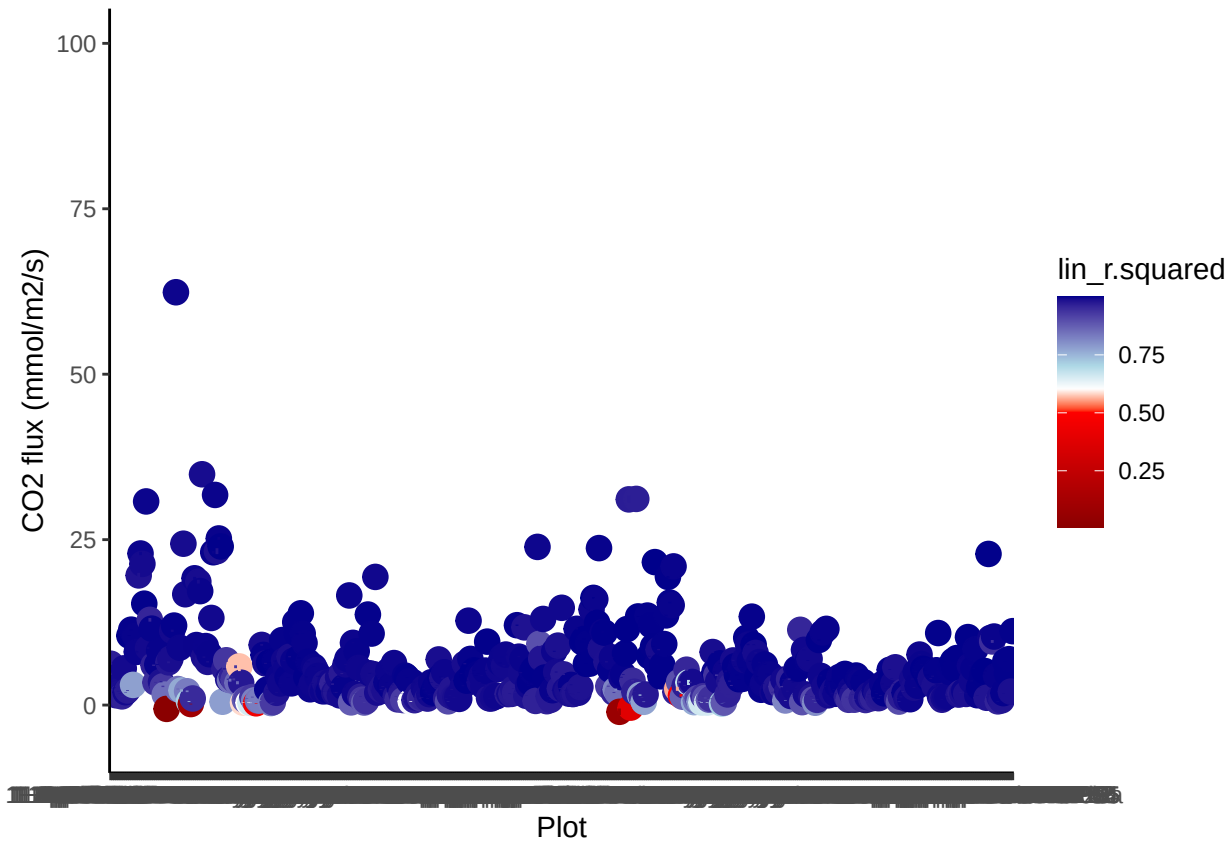
```
##          Plot      TIMESTAMP HM81_AIC HM81_flux.estimate HM81_p.value
## 1 1-Apr-GCW-TR-1 2025-04-16 10:58:18      NA              NA          NA
## 2 1-Apr-GCW-TR-2 2025-04-16 11:07:21 415.1026          2.767439 1.620905e-31
## 3 1-Apr-GCW-TR-3 2025-04-16 11:11:25 417.9395          2.715836 2.236664e-31
## 4 1-Apr-GCW-TR-4 2025-04-16 11:15:46 350.5254          1.579223 7.544911e-33
## 5 1-Apr-GCW-TR-5 2025-04-16 11:20:03 407.1443          1.859655 7.860953e-29
## 6 1-Apr-GCW-UP-1 2025-04-16 11:25:26      NA              NA          NA
##   HM81_r.squared HM81_RMSE  lin_AIC lin_flux.estimate lin_flux.std.error
## 1              NA        NA 471.5188          6.246571          0.16032365
## 2      0.9288755 10.890805 299.8162          2.560309          0.03269938
## 3      0.9279904 11.180673 314.5950          2.609335          0.03749459
## 4      0.9367803  5.989341 308.6082          1.477810          0.03547271
```

```
## 5      0.9098118 10.117133 302.4367      2.107285      0.03350251
## 6      NA      NA 292.1896      1.791610      0.03046990
##   lin_int.estimate lin_int.std.error  lin_p.value lin_r.squared  lin_RMSE
## 1      3745.930      5.634565 3.725386e-40      0.9668802 18.362133
## 2      3820.197      1.149218 1.227954e-55      0.9915894  3.745114
## 3      3865.564      1.317745 5.325900e-53      0.9893771  4.294318
## 4      3821.678      1.246686 1.273818e-41      0.9709106  4.062748
## 5      3867.925      1.177444 9.630206e-51      0.9870270  3.837098
## 6      3823.874      1.070863 3.055808e-49      0.9851825  3.489769
##   poly_AIC poly_r.squared poly_RMSE  rob_AIC rob_converged rob_flux.estimate
## 1 277.0248      0.9991611  2.980210 472.0052      TRUE      6.340314
## 2 285.7144      0.9939849  3.229907 299.8216      TRUE      2.558427
## 3 275.5805      0.9952104  2.940620 314.5963      TRUE      2.609710
## 4 292.6486      0.9798994  3.444087 308.7654      TRUE      1.466874
## 5 296.0562      0.9892958  3.554485 302.4814      TRUE      2.106470
## 6 295.6443      0.9853314  3.540957 292.2965      TRUE      1.784351
##   rob_flux.std.error  rob_RMSE      TIMESTAMP_min      TIMESTAMP_max
## 1      0.15971609 21.553589 2025-04-16 10:57:52 2025-04-16 10:58:45
## 2      0.03185804  4.236081 2025-04-16 11:06:55 2025-04-16 11:07:48
## 3      0.03863647  5.165895 2025-04-16 11:10:59 2025-04-16 11:11:52
## 4      0.03421619  3.837129 2025-04-16 11:15:20 2025-04-16 11:16:13
## 5      0.03435700  4.056659 2025-04-16 11:19:37 2025-04-16 11:20:30
## 6      0.03321909  3.737283 2025-04-16 11:25:00 2025-04-16 11:25:53
```

```
allflux <- fluxes
```

0.9 Visualize fluxes and sort by r2

```
#take a look at the fluxes we calculated with R2
ggplot(allflux, aes(Plot, lin_flux.estimate, color = lin_r.squared)) +
  geom_point(size=4) + theme_classic() + ylim(-5,100) +
  geom_linerange(aes(ymin = lin_flux.estimate- lin_flux.std.error,
                    ymax = lin_flux.estimate+ lin_flux.std.error)) +
  scale_color_gradientn(colours = c("darkred", "red", "white", "lightblue", "darkblue"),
                        values = c(0, 0.5, 0.6, 0.7, 1)) +
  ylab("CO2 flux (mmol/m2/s)")
```



```
#sort fluxes by R2
fluxes_good <- as.data.frame(subset(allflux, lin_r.squared >= .90, select = Plot:TIMESTAMP_max))
fluxes_lowr2 <- as.data.frame(subset(allflux, lin_r.squared < .90 & abs(lin_flux.estimate) < 1, select = Plot:TIMESTAMP_max))
fluxes_weird <- as.data.frame(subset(allflux, lin_r.squared < .90 & lin_flux.estimate > 1, select = Plot:TIMESTAMP_max))
fluxes_negative <- as.data.frame(subset(allflux, lin_flux.estimate < 0, select = Plot:TIMESTAMP_max))

#find fluxes that represent the first and fourth quartile of the data
low <- quantile(allflux$lin_flux.estimate, 0.05)
high <- quantile(allflux$lin_flux.estimate, 0.95)

fluxes_low <- as.data.frame(subset(allflux, lin_flux.estimate <= low, select = Plot:TIMESTAMP_max))
fluxes_high <- as.data.frame(subset(allflux, lin_flux.estimate >= high, select = Plot:TIMESTAMP_max))

tails <- rbind(fluxes_low, fluxes_high)
```

0.10 check the fluxes for which I changed the start time due to lags or ebullition (recorded in notes of metadata)

```
tobechecked <- c(
  "1-Jul-SWH-TR-7a"
)

colnames <- c("Plot")
```

```

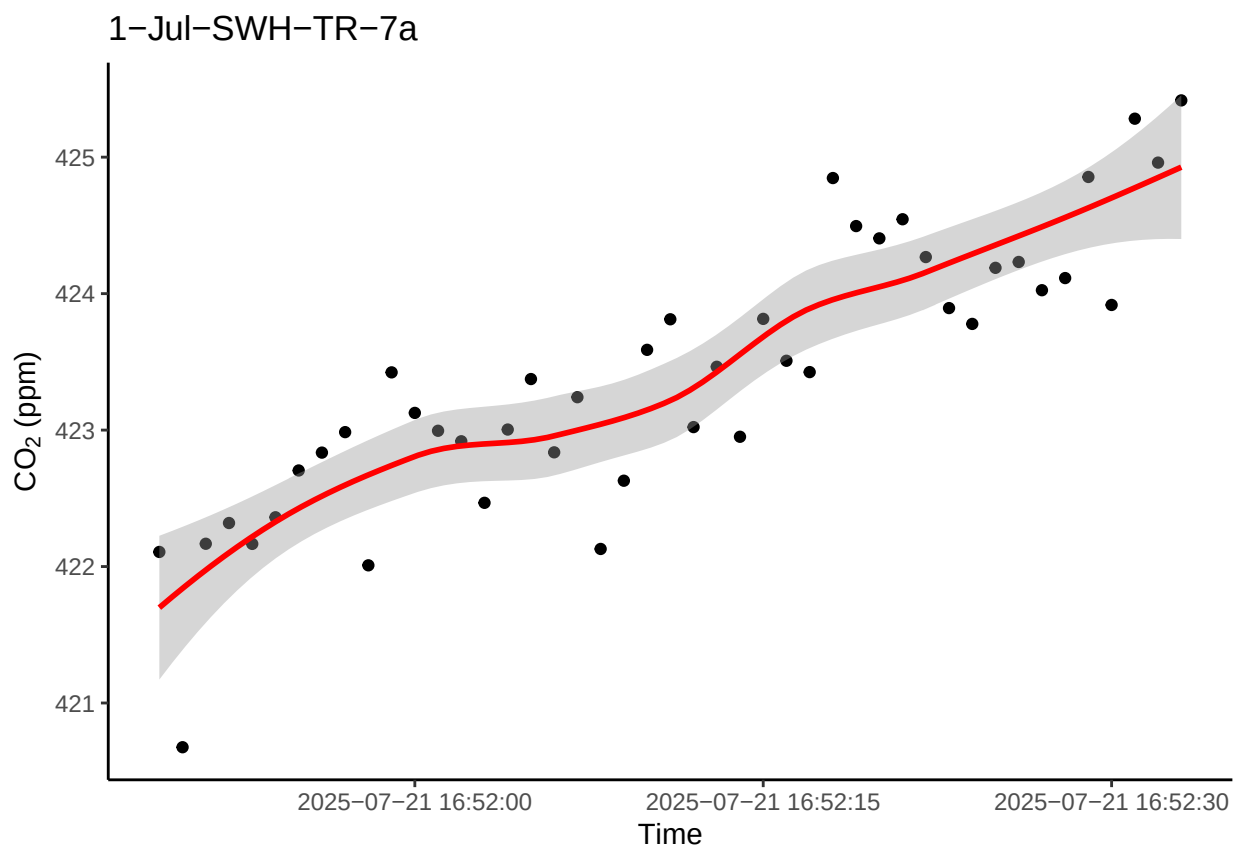
#Loop to run five random plots subset above
for(P in tobechecked){
  r1 <- subset(alldat, Plot == P)
  p1 <- ggplot(r1, aes(as.POSIXct(TIMESTAMP), co2)) + geom_point() +
  scale_x_datetime(breaks = "15 sec") +
  theme_classic() +
  ylab(expression(paste( CO [2], " (ppm)"))) +
  xlab("Time") + labs(title = P) +
  geom_smooth(color="red")
  print(p1)
}

```

```

## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'

```



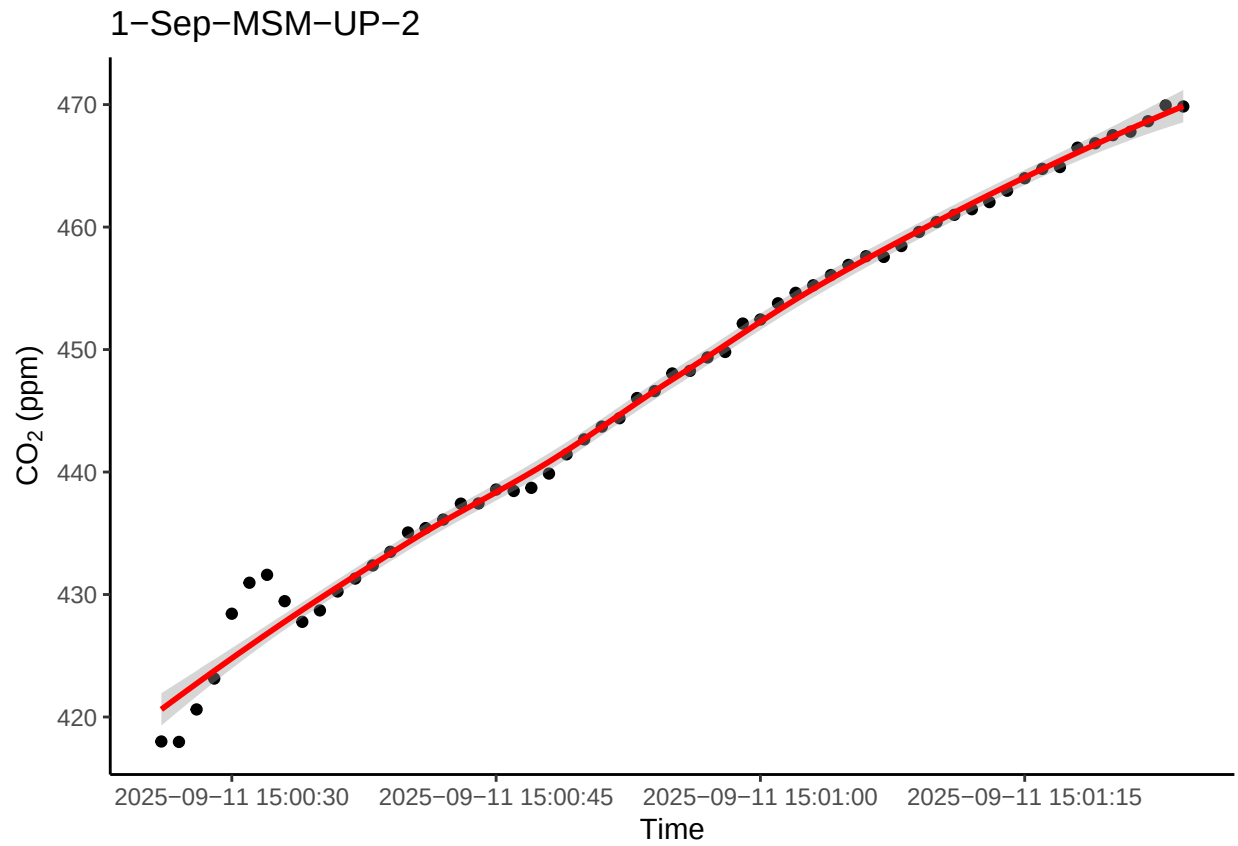
```

#list of the ones that I changed and checked and looked good?
fixedplots <- c("2-Sep-SWH-TR-7", "2-Jul-SWH-TR-3", "1-May-SWH-TR-6a", "1-Jul-SWH-TR-3",
  "1-Jul-SWH-TR-10", "2-Jul-SWH-TR-2", "2-Mar-GWI-TR-5", "2-Mar-GWI-UP-3a",
  "2-May-SWH-TR-6a", "1-Mar-GWI-UP-3a", "1-May-SWH-TR-7a",
  "1-Jul-SWH-TR-9", "1-May-SWH-TR-1", "2-Mar-SWH-UP-1", "2-Mar-SWH-UP-2",
  "2-Mar-SWH-UP-5", "2-Nov-SWH-TR-2", "2-Nov-SWH-TR-4", "2-Nov-SWH-TR-5",
  "1-Jul-SWH-TR-2", "1-Jul-SWH-TR-6", "1-Jul-SWH-TR-7a", "2-May-MSM-UP-2",
  "2-May-MSM-UP-4", "2-Jul-GWI-UP-5", "1-Jul-SWH-TR-3", "2-Jul-SWH-TR-6",
  "2-May-MSM-UP-2")

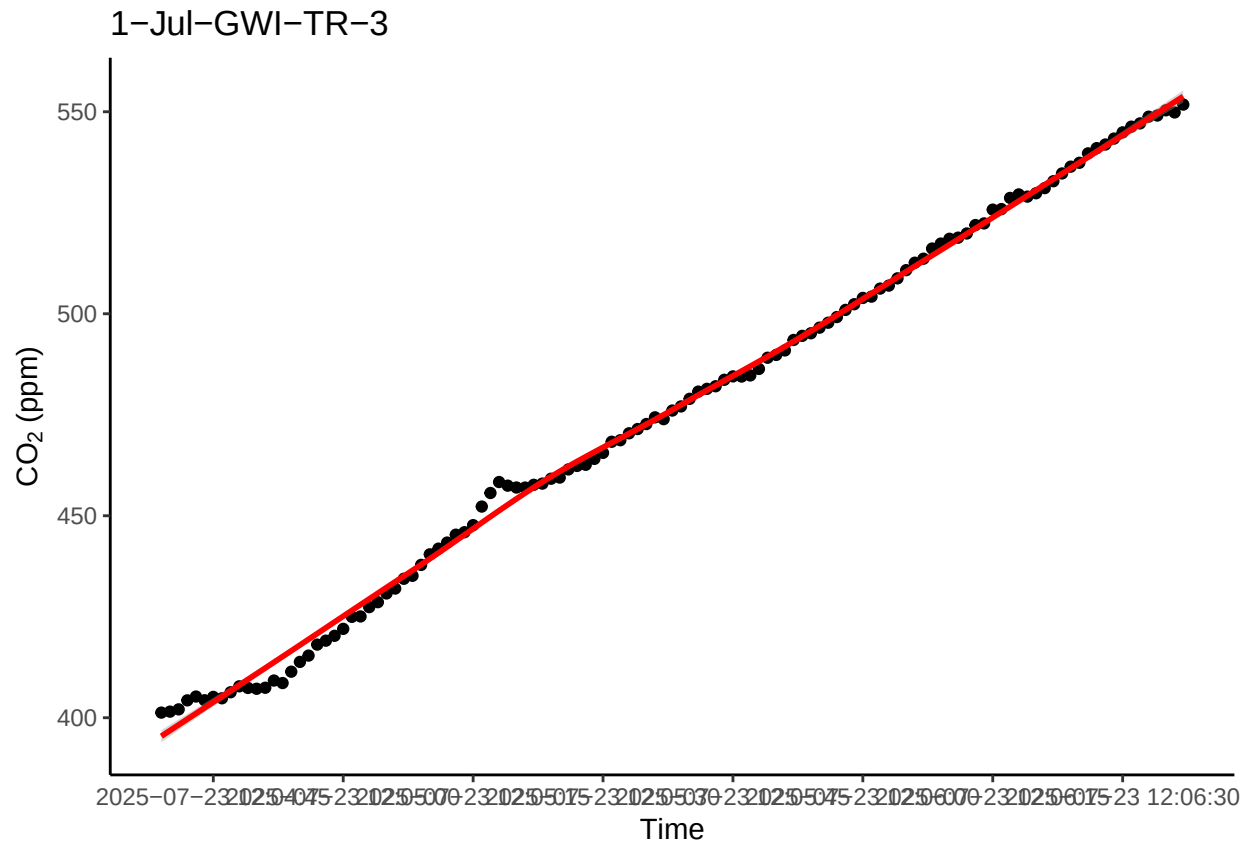
```

0.11 Spot check a few fluxes with $R^2 > 0.9$

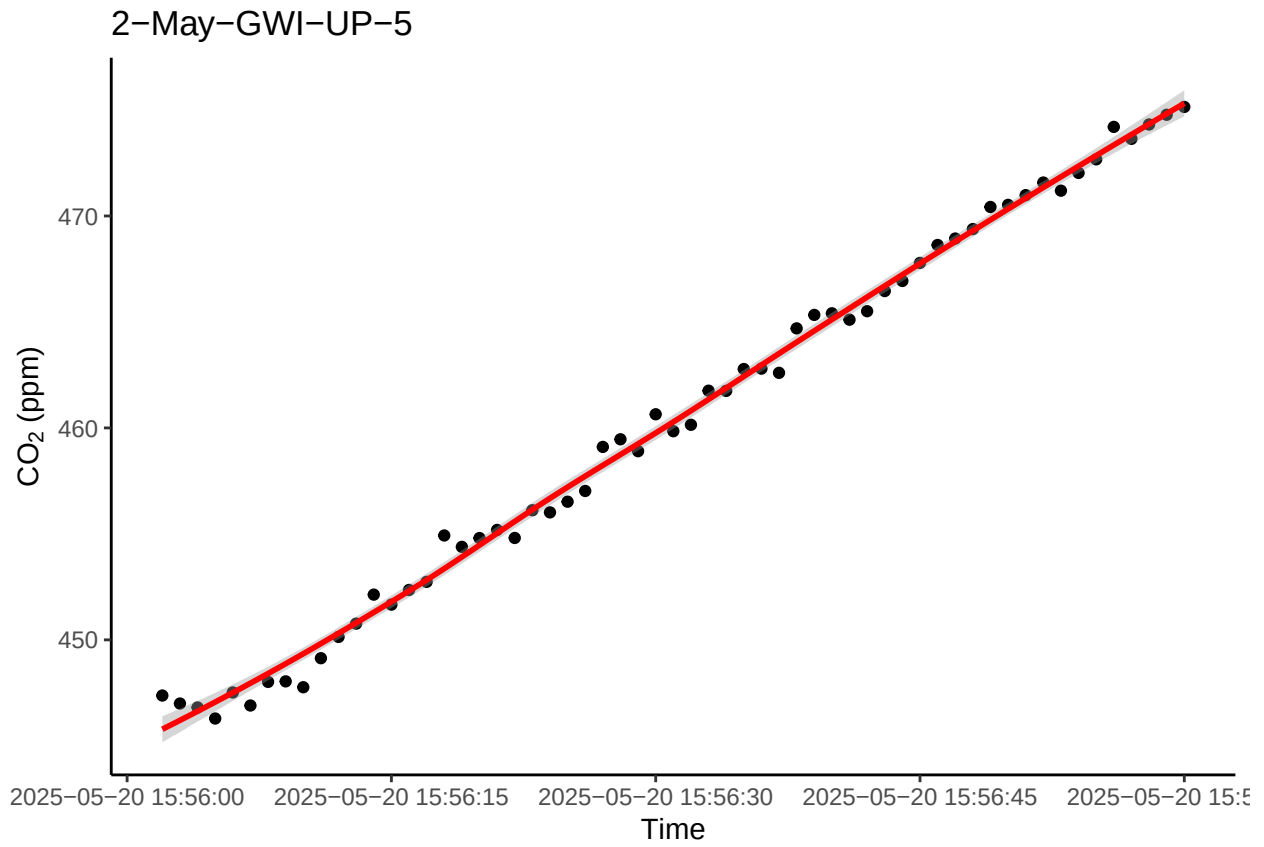
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



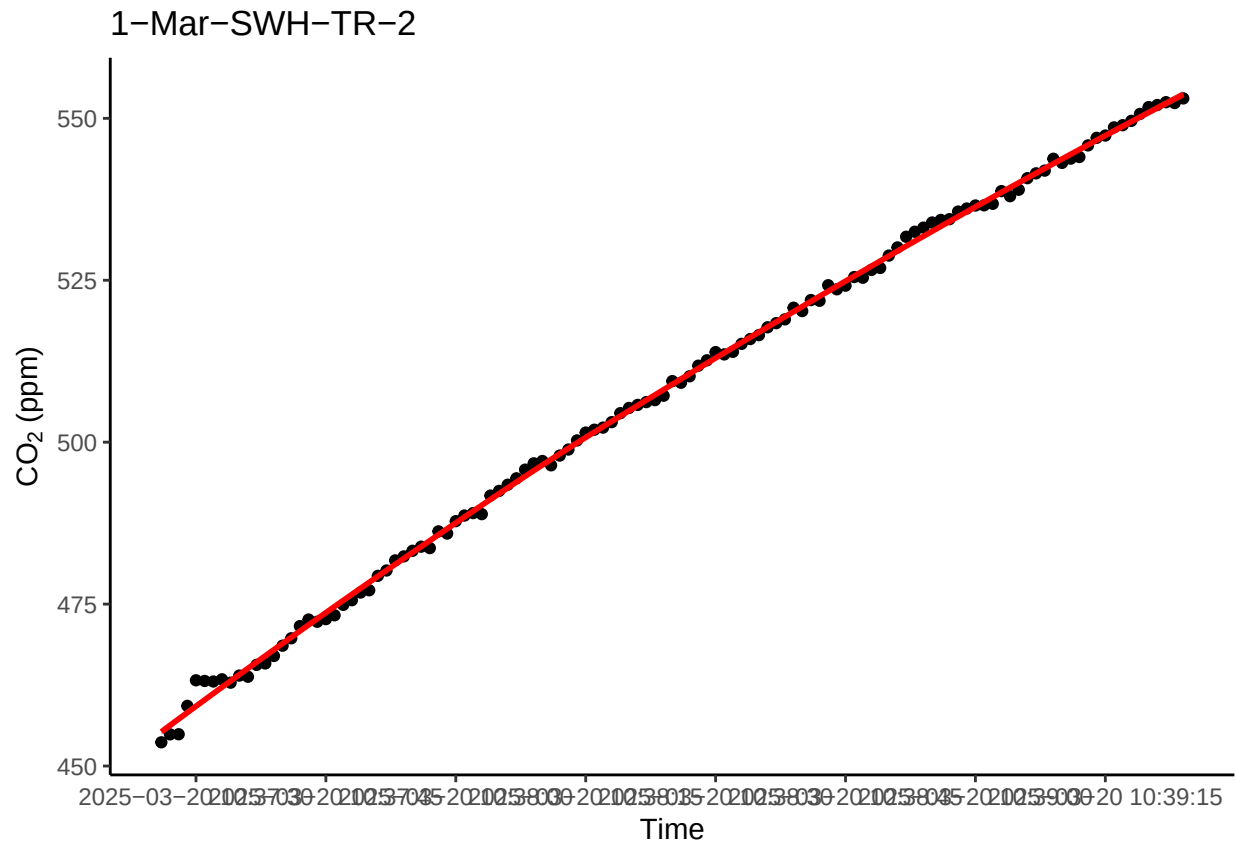
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



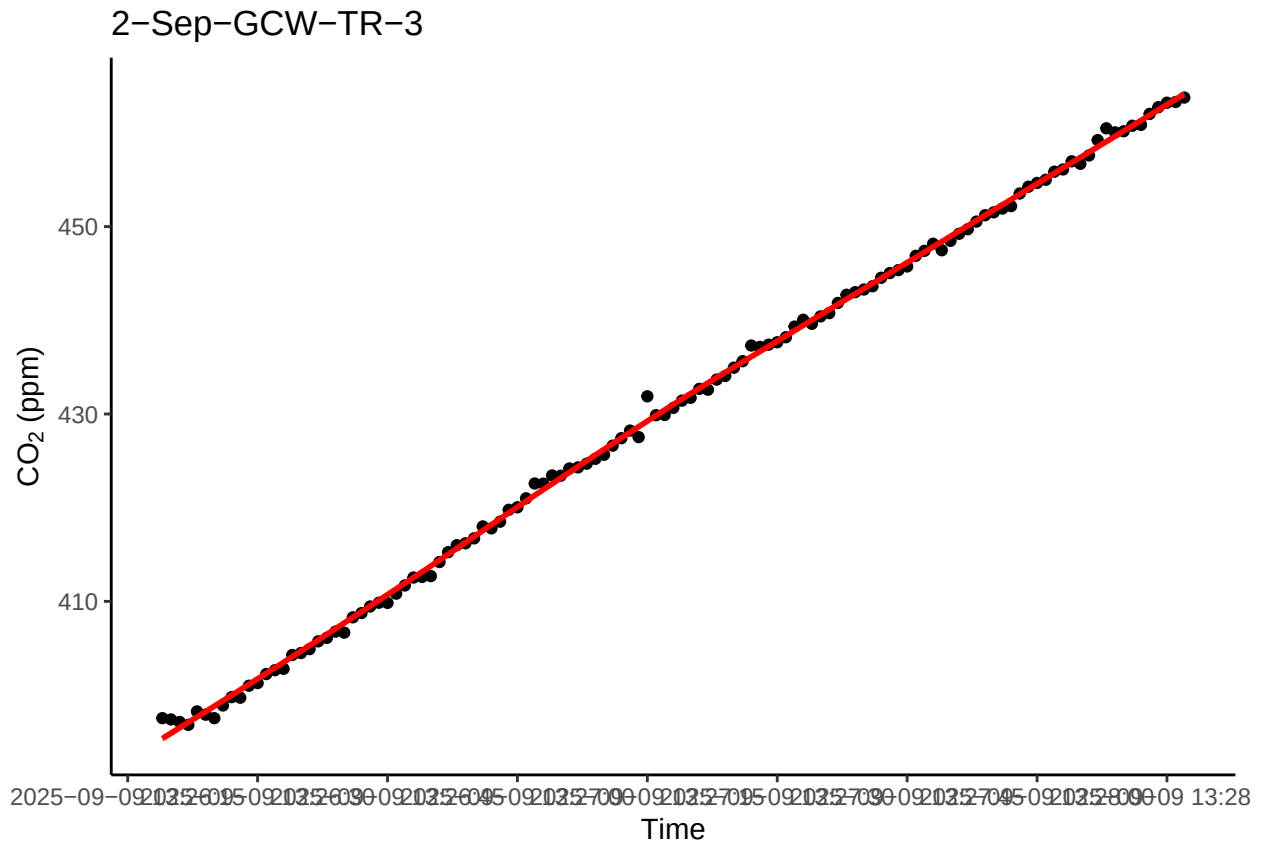
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

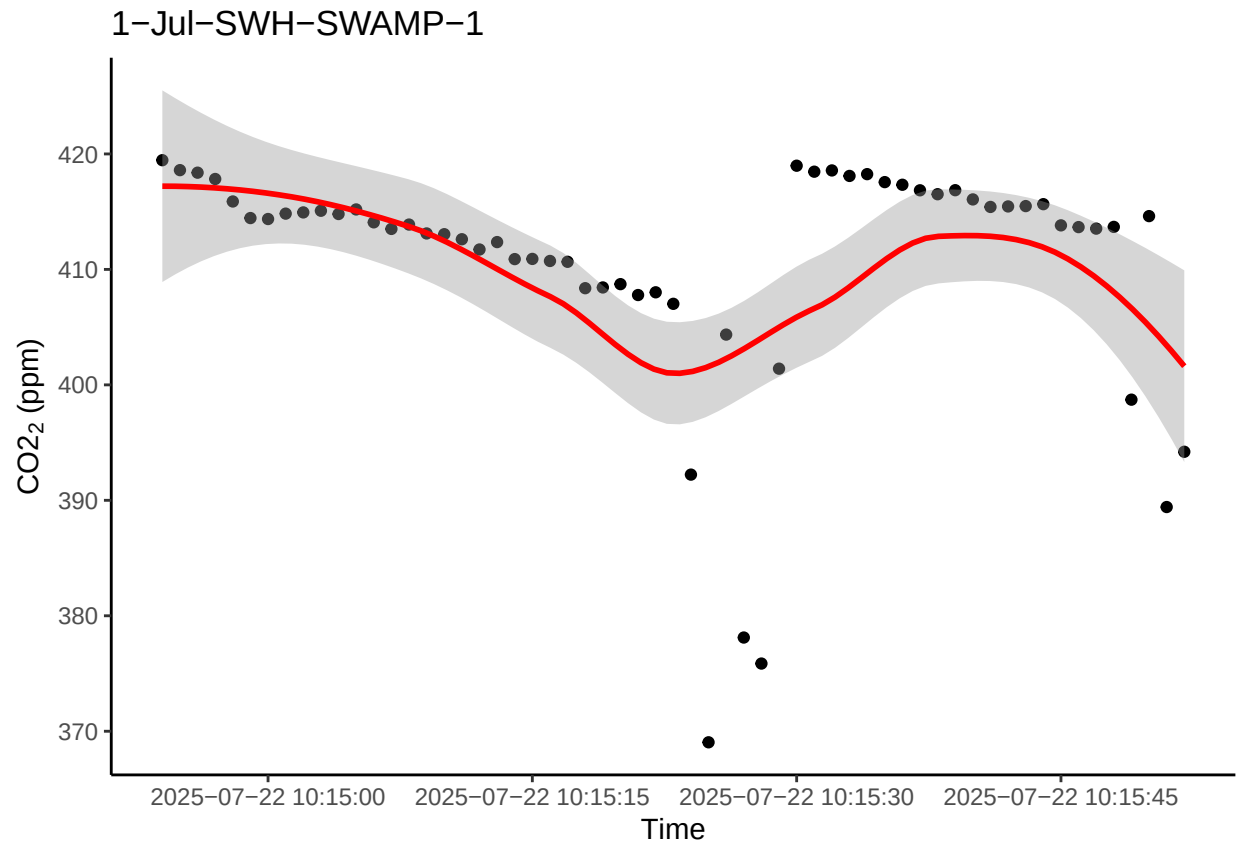


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

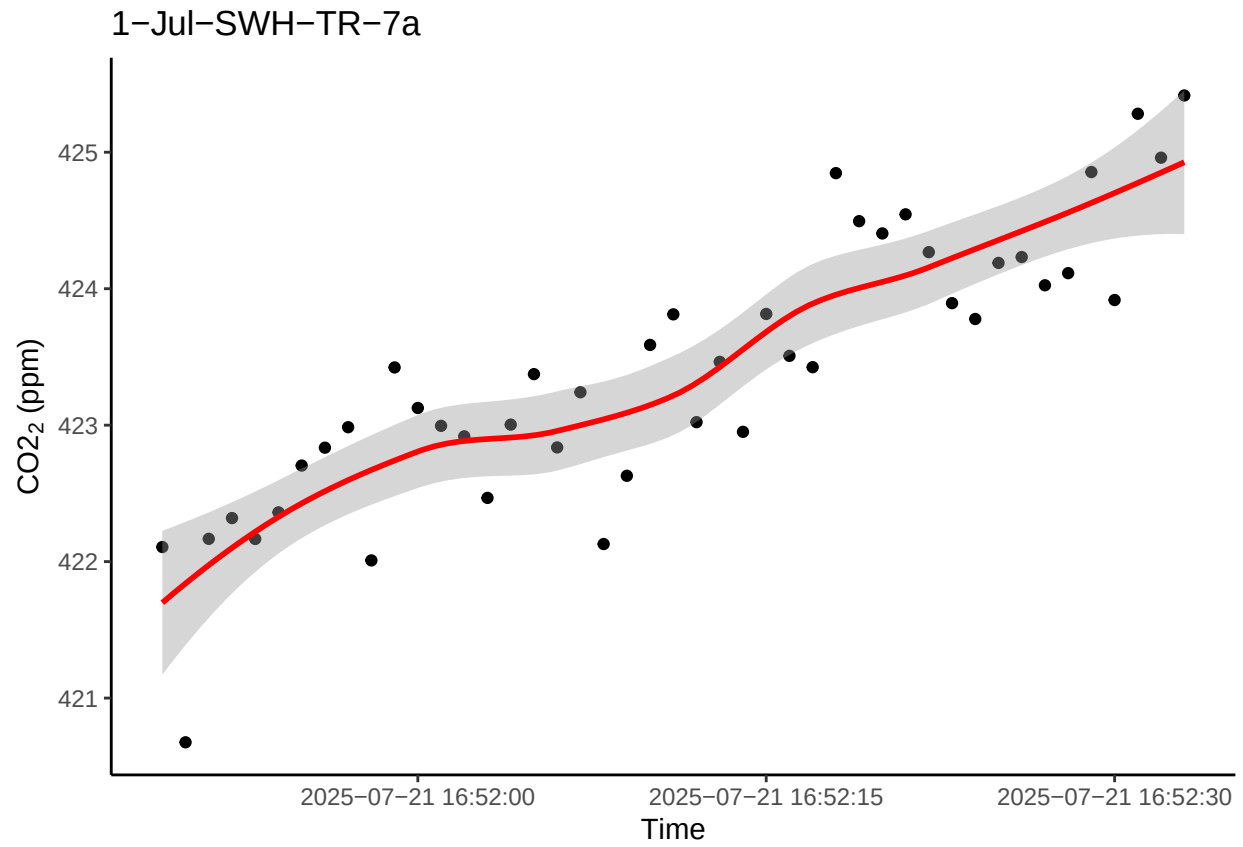


0.12 Take a look at the fluxes <0.9 and a flux estimate below 1

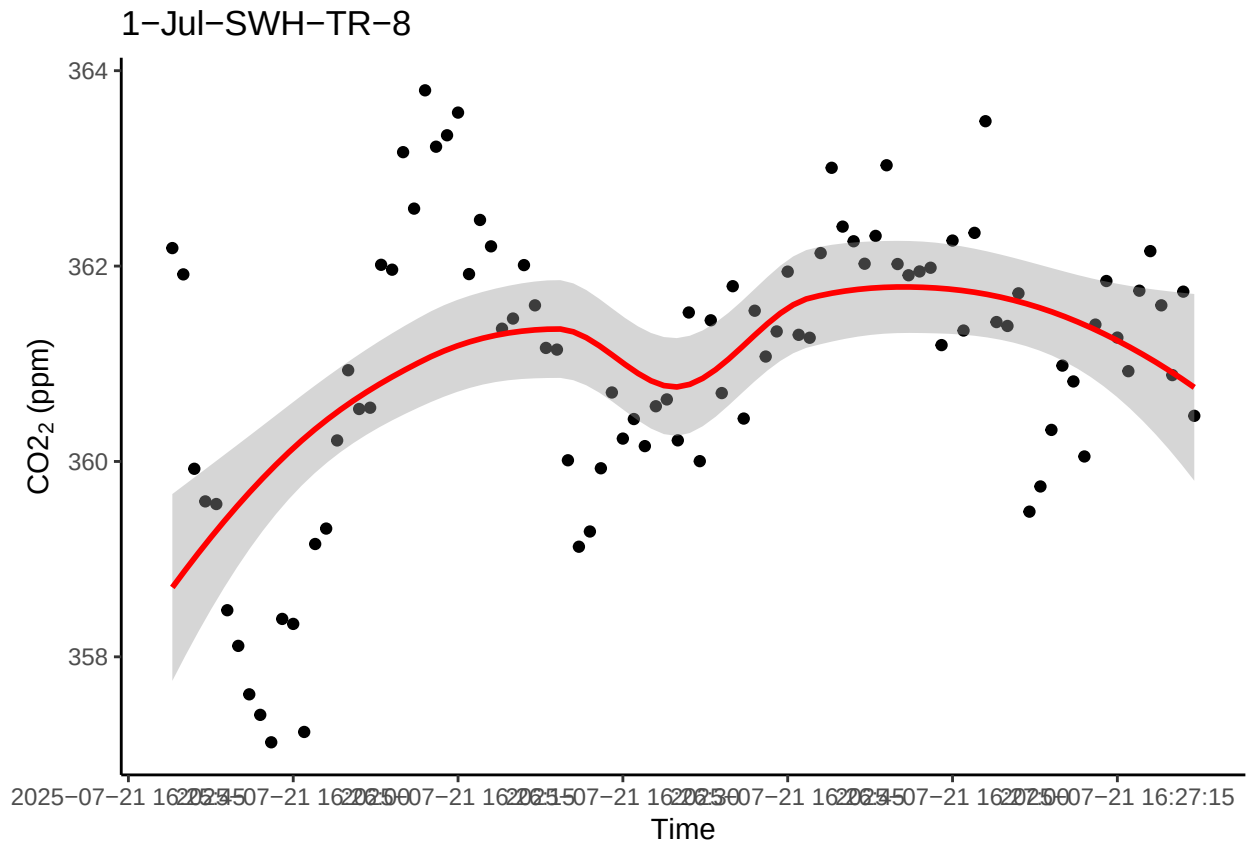
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



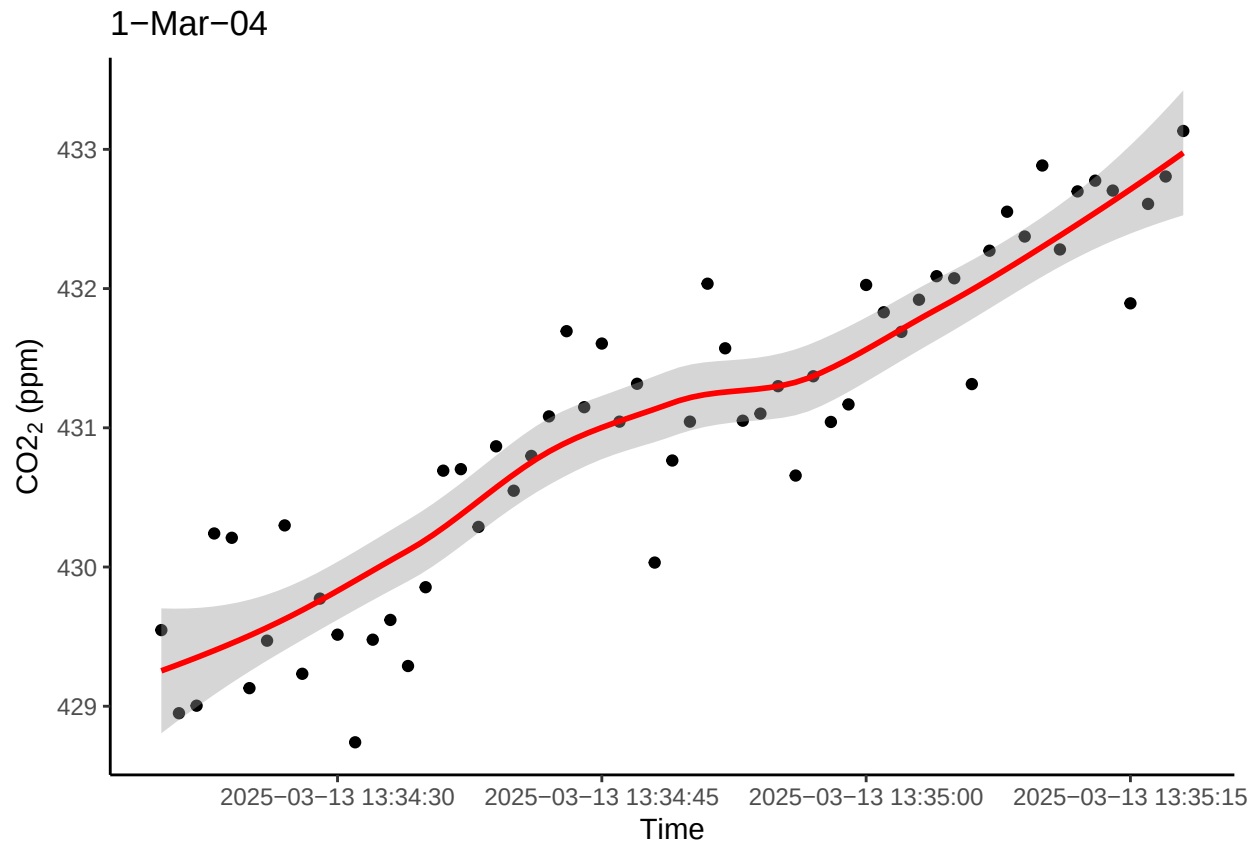
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



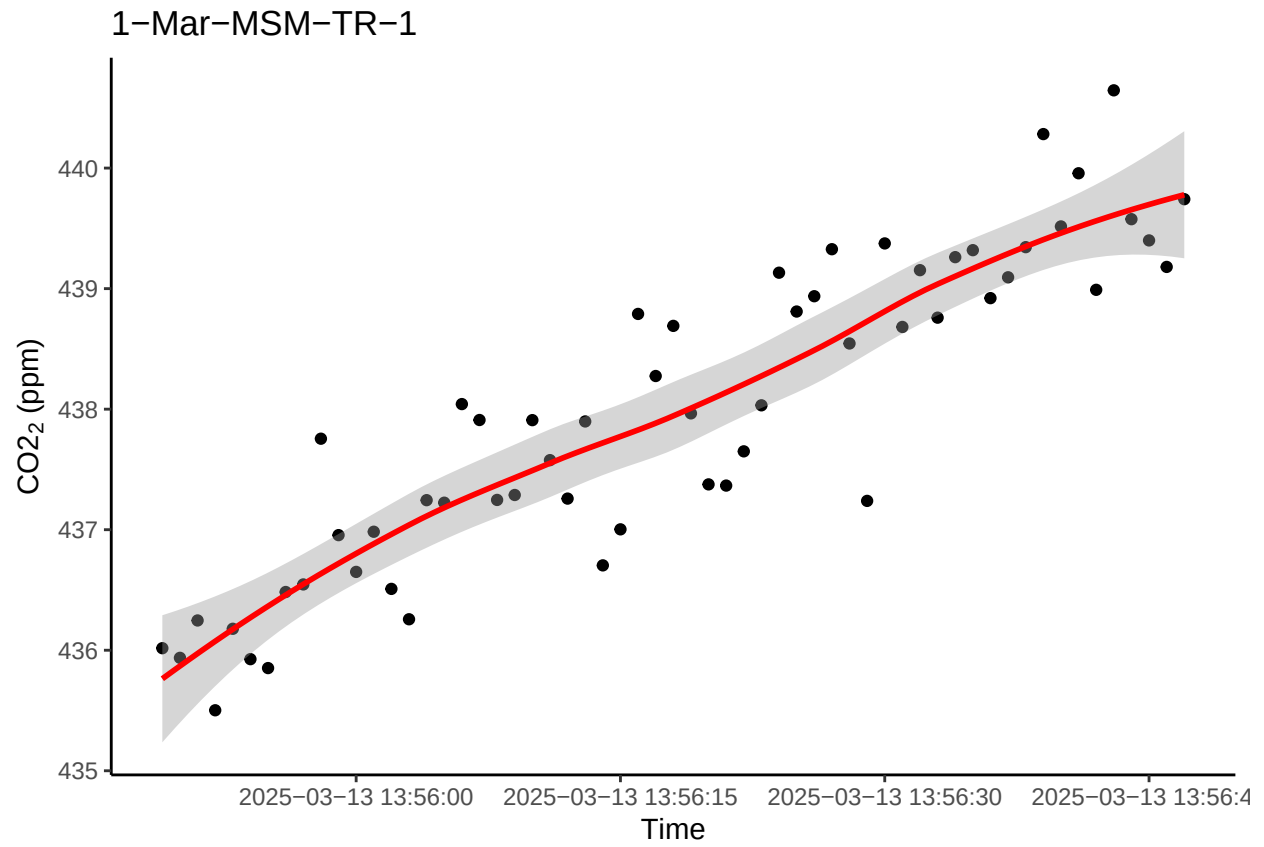
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



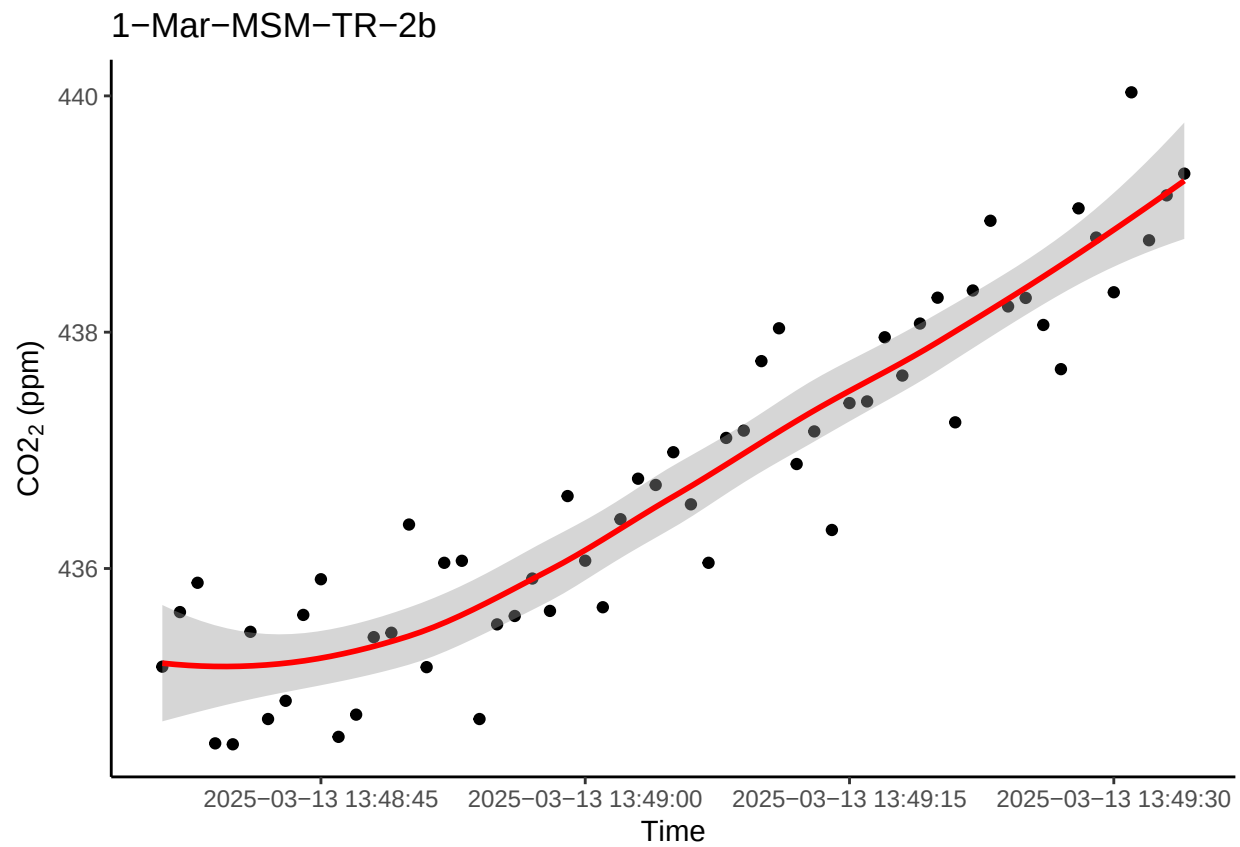
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



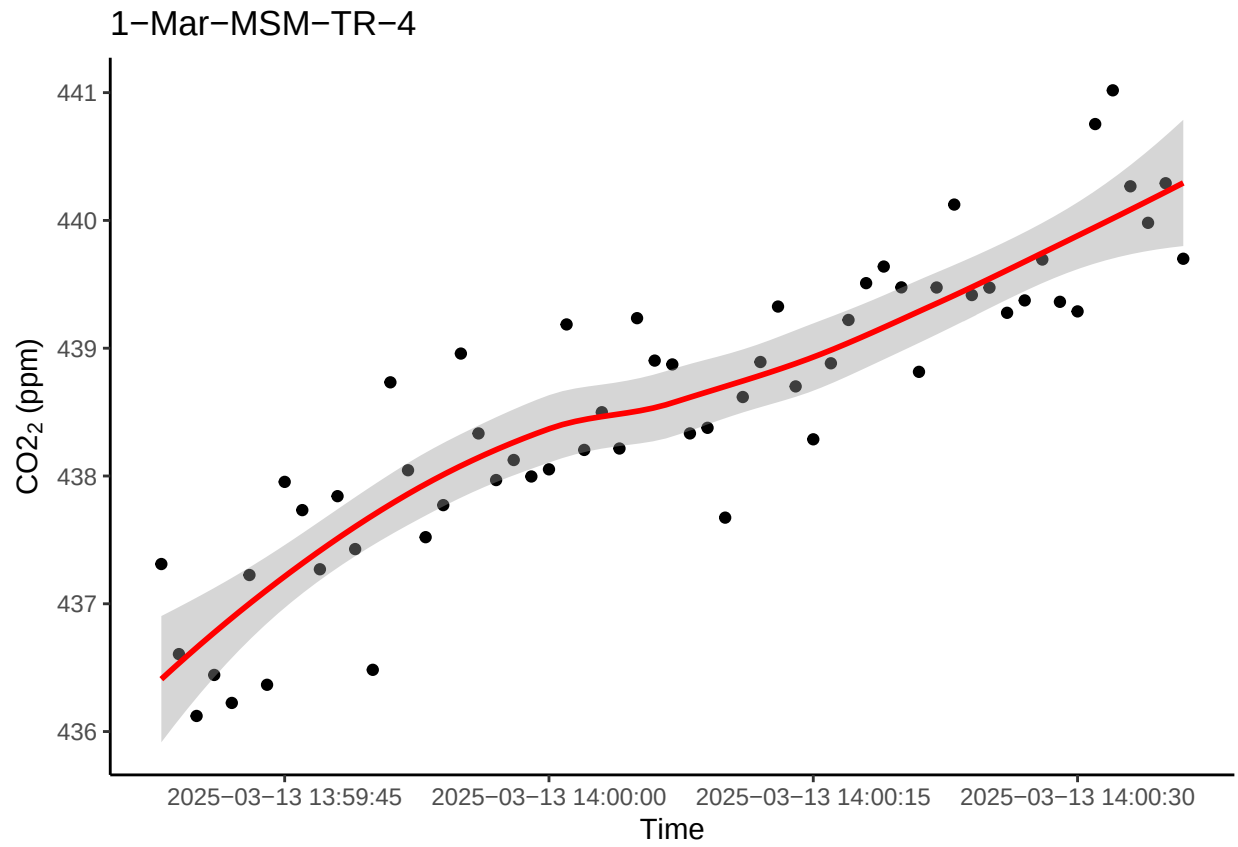
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



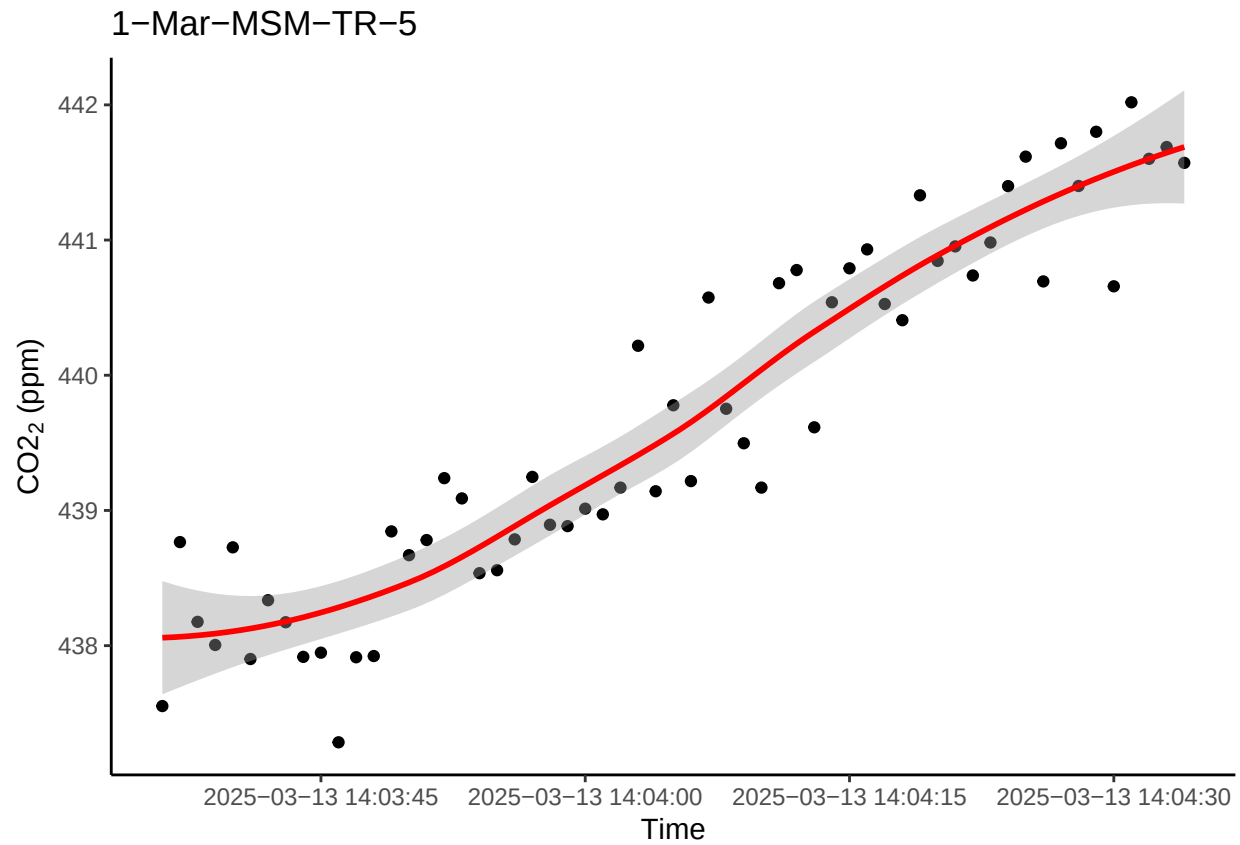
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



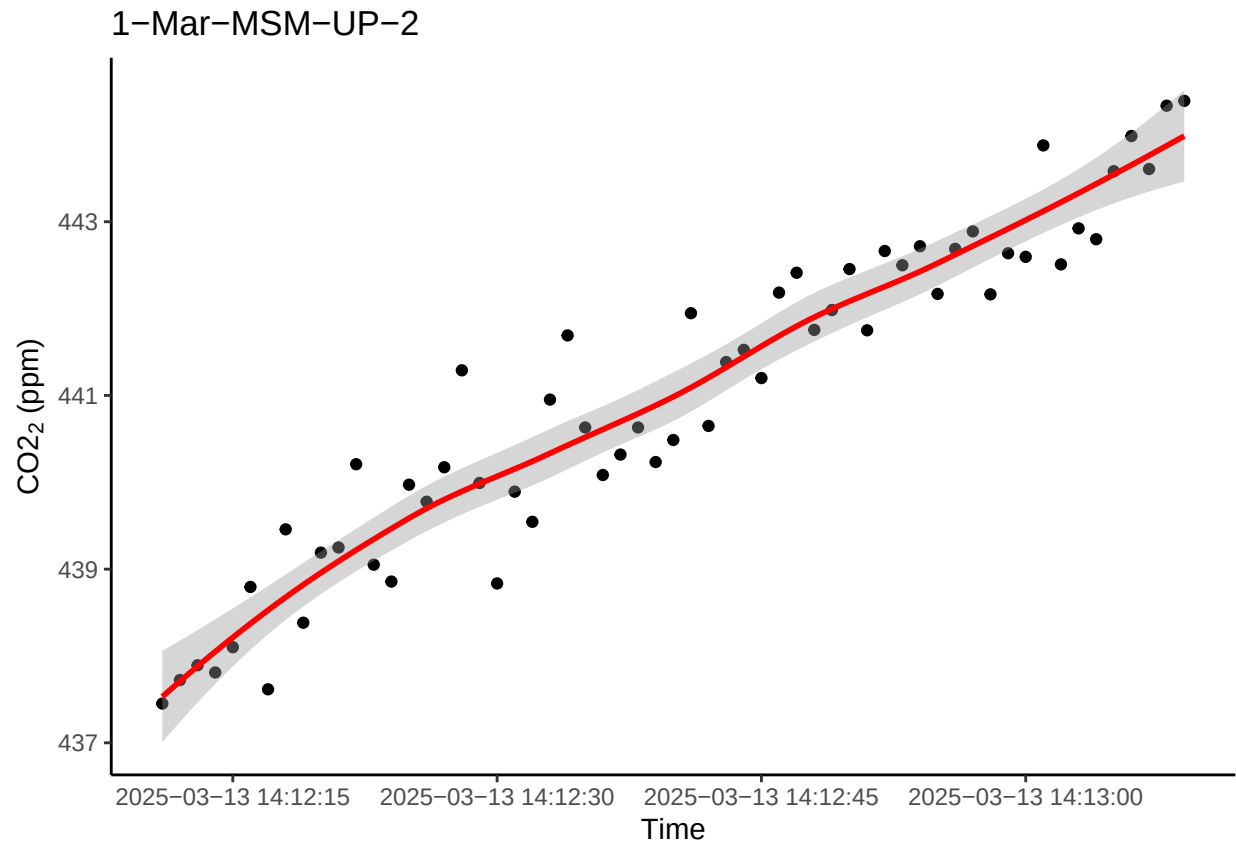
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



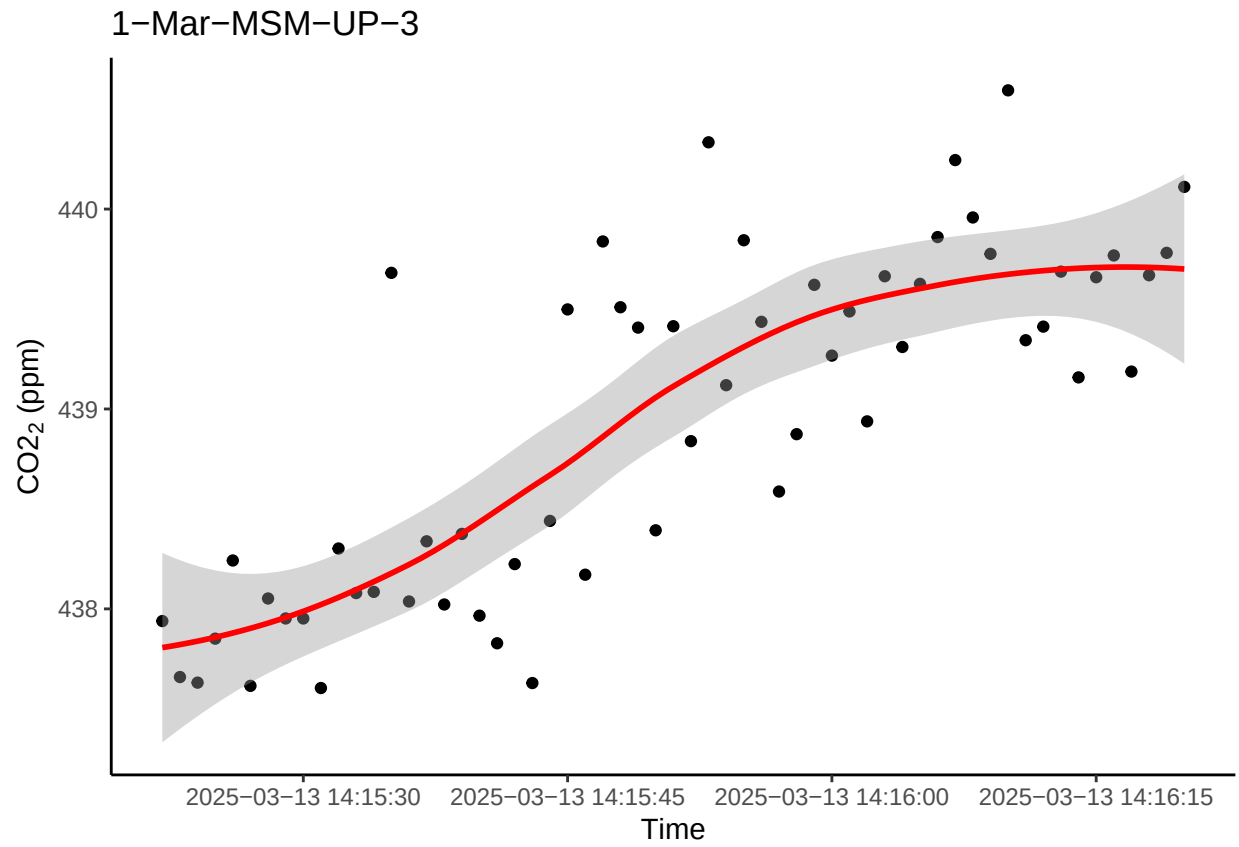
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



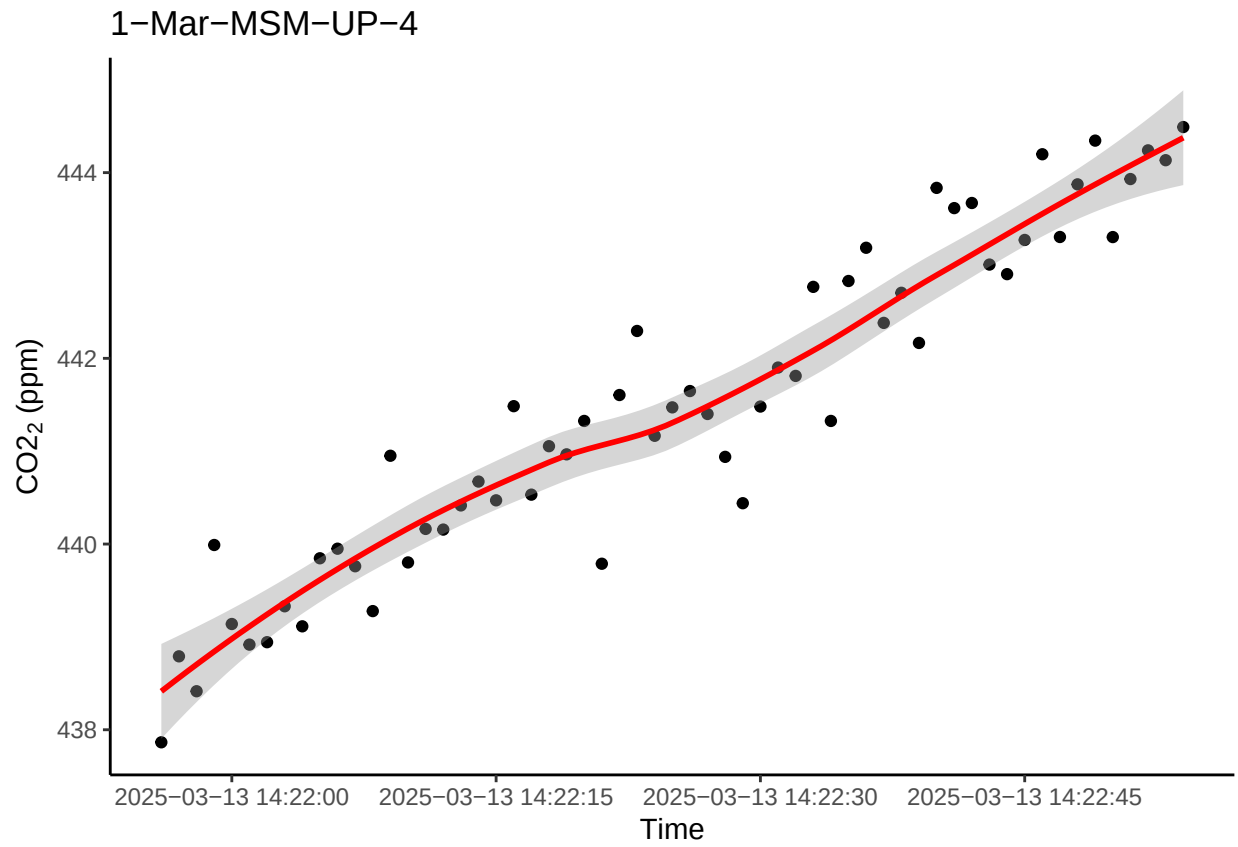
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



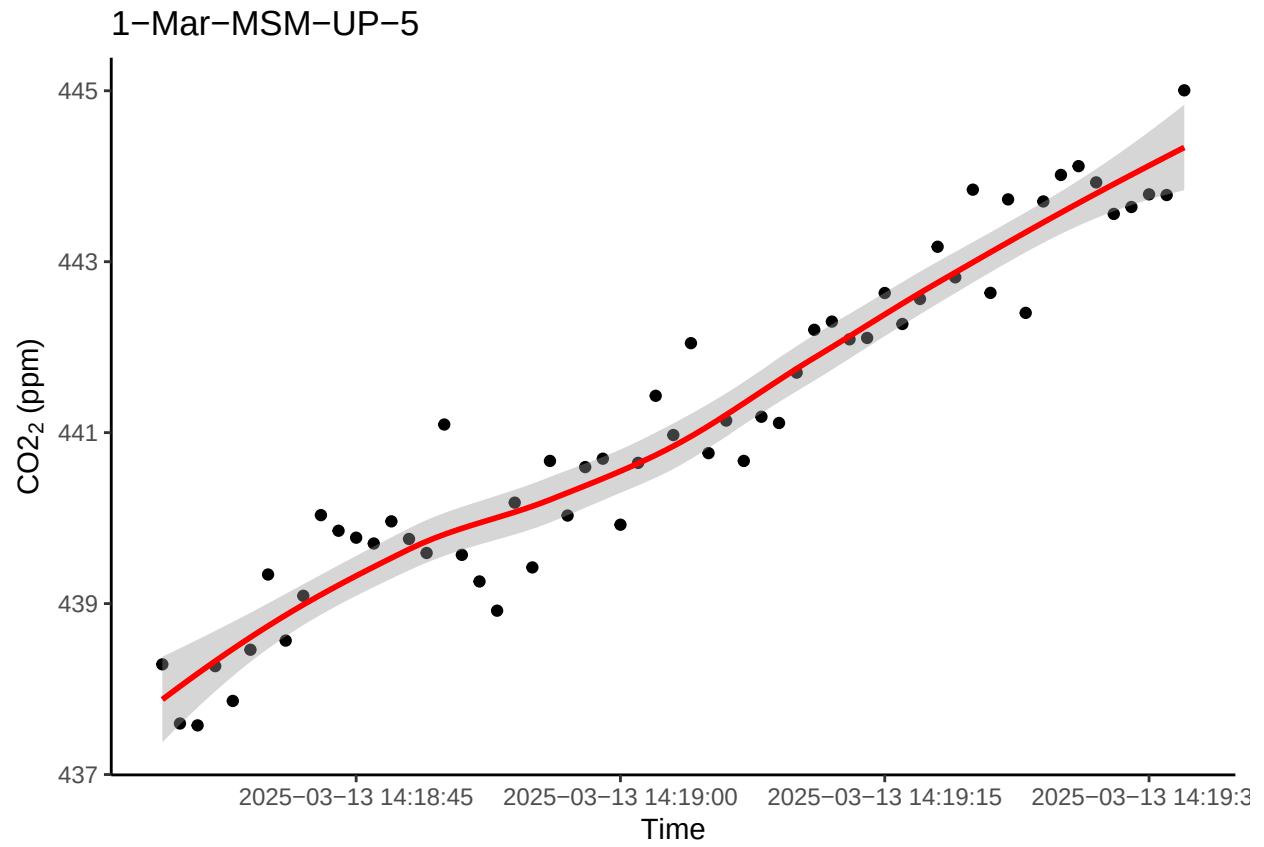
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



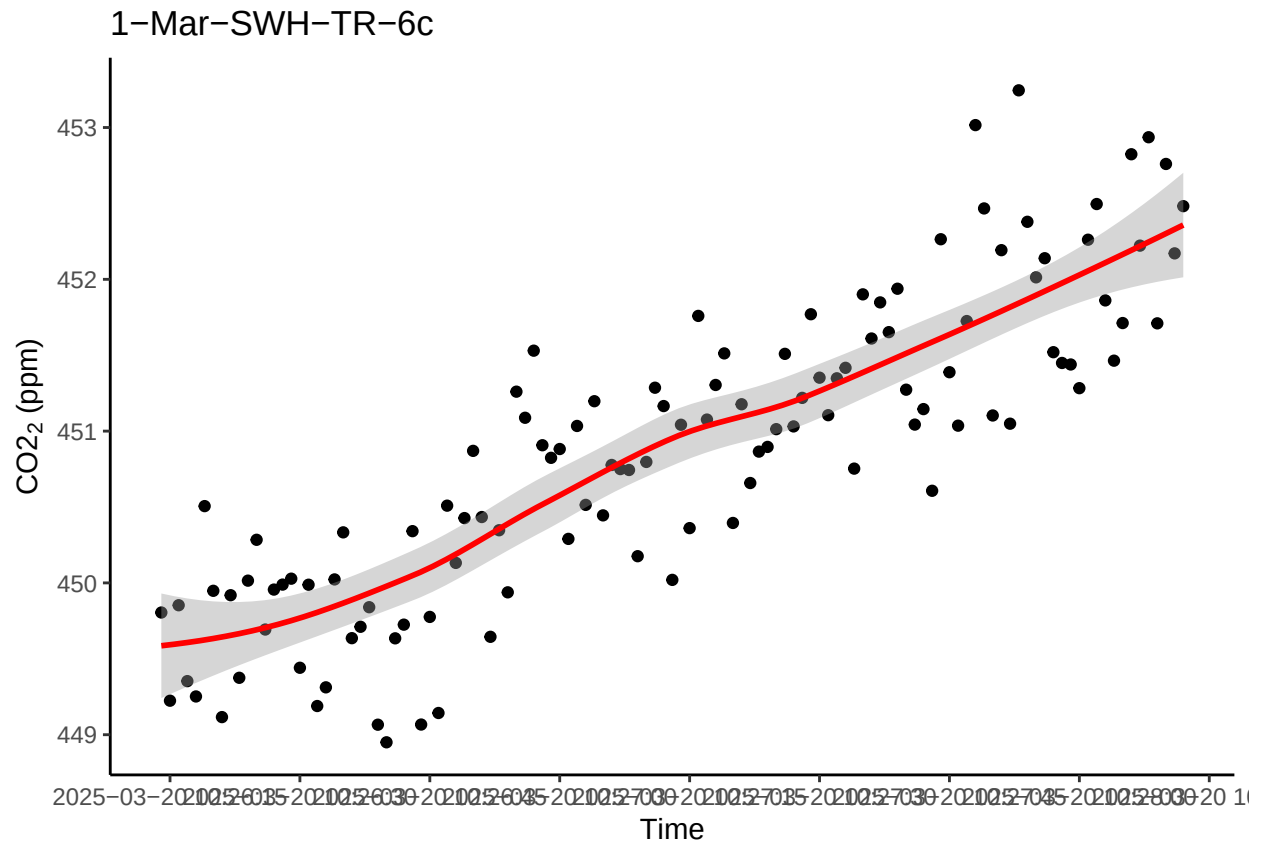
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



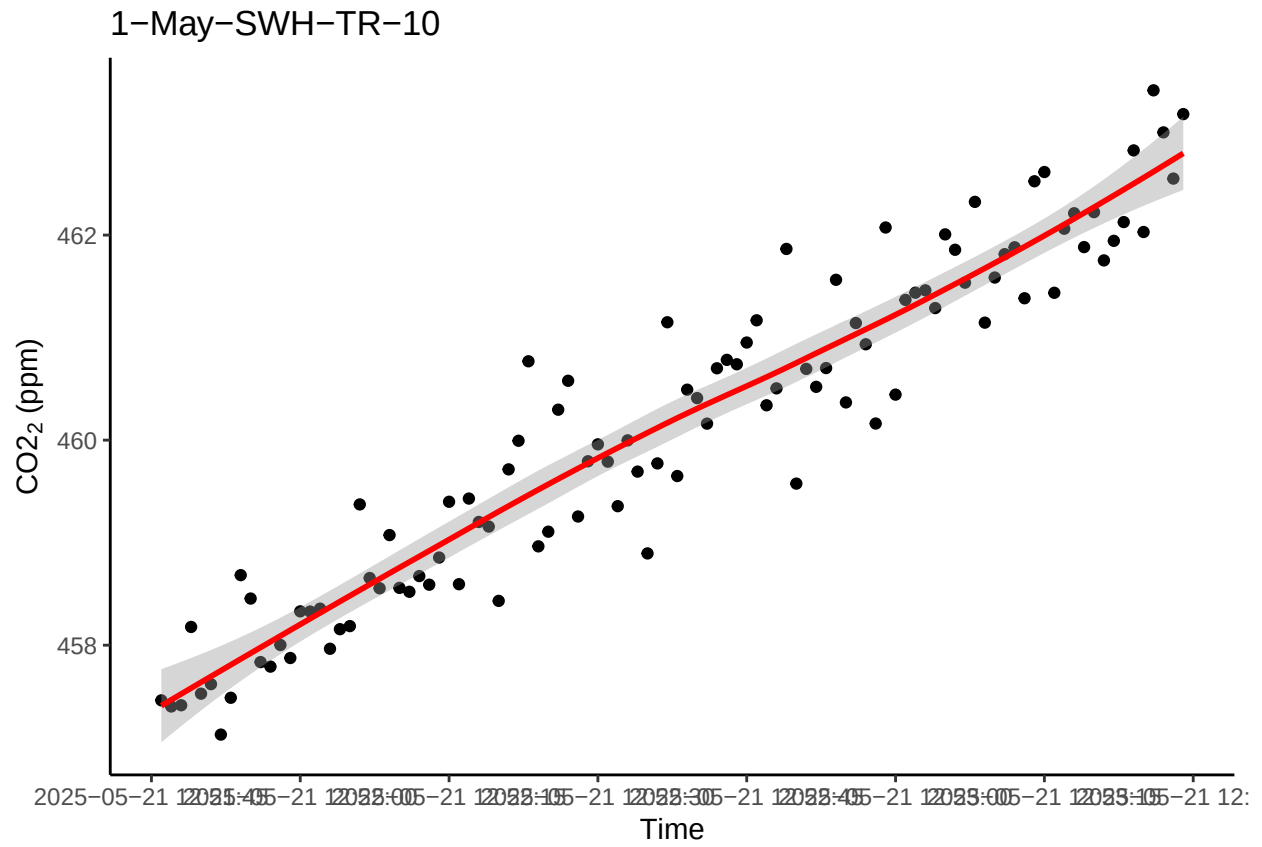
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



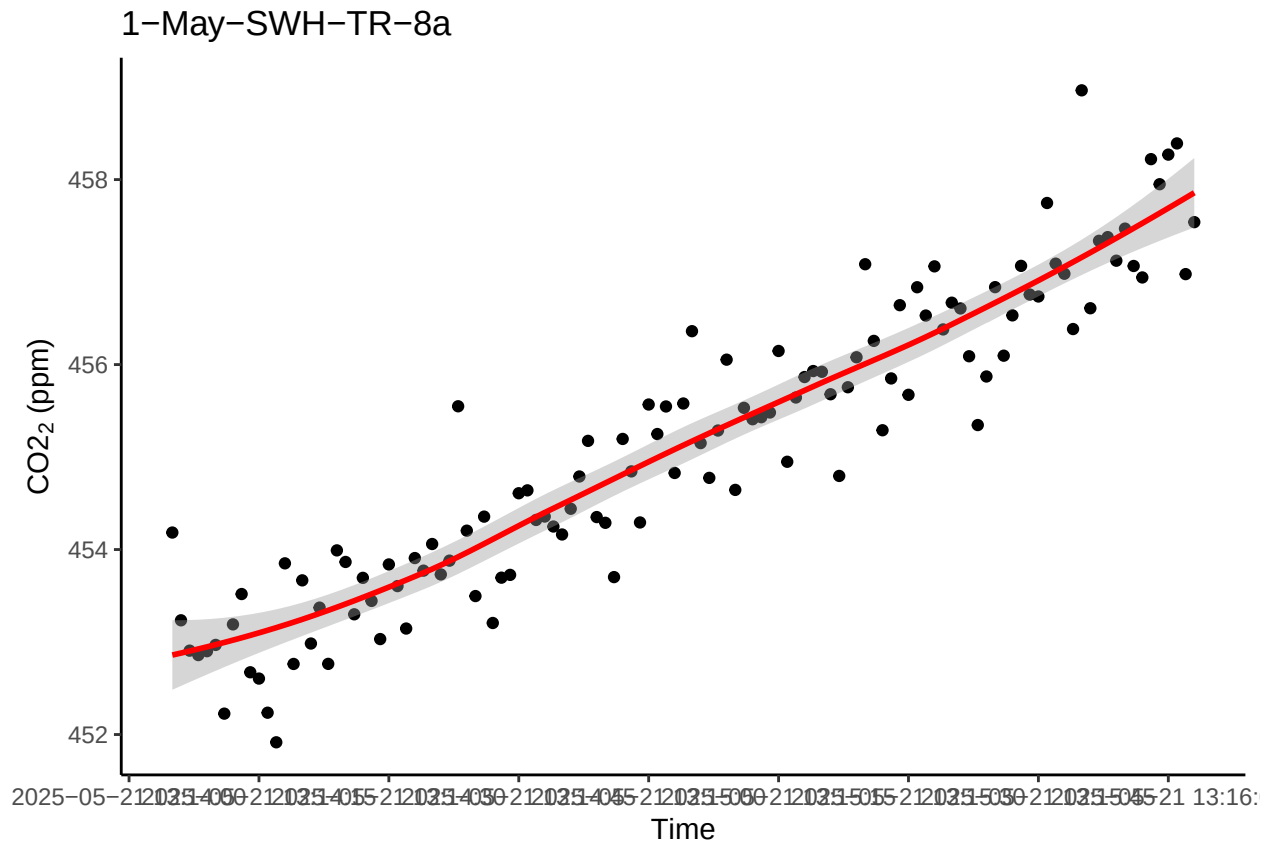
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



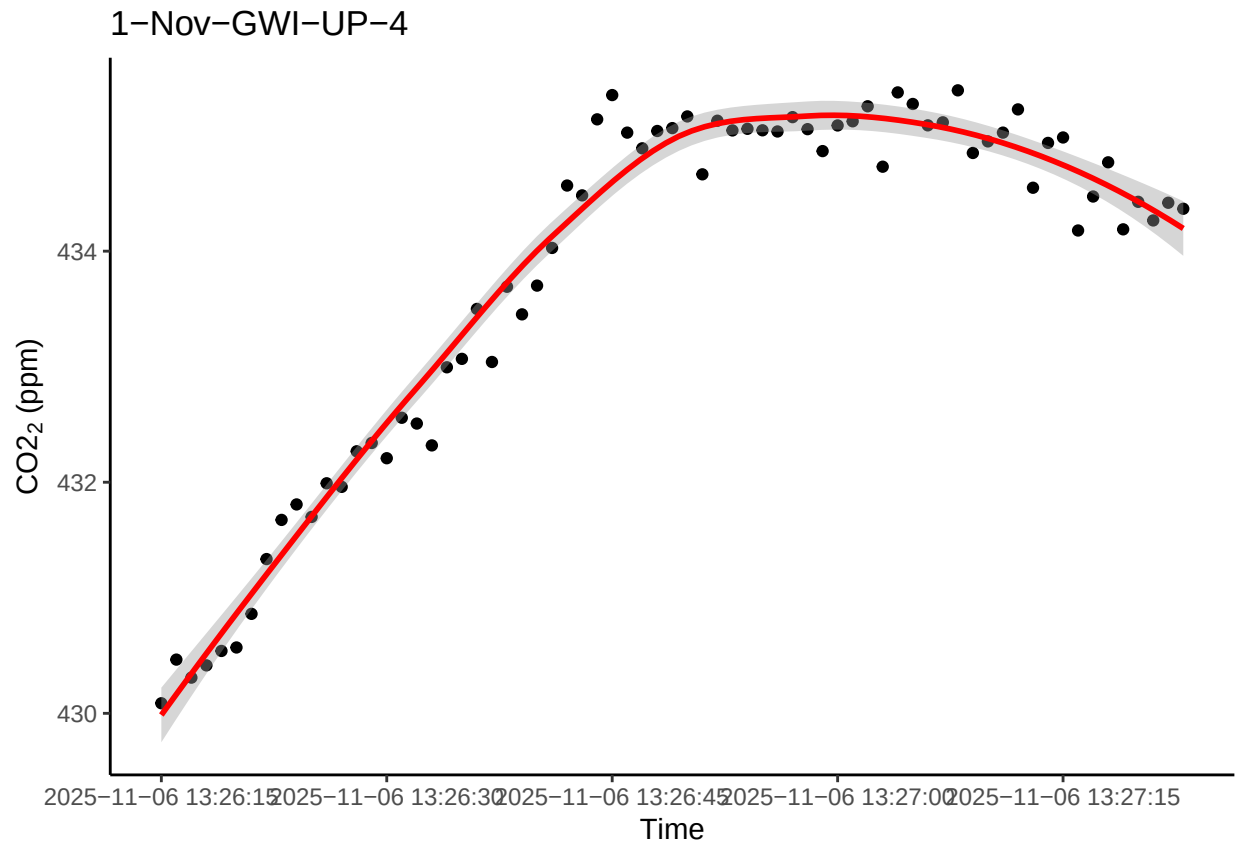
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



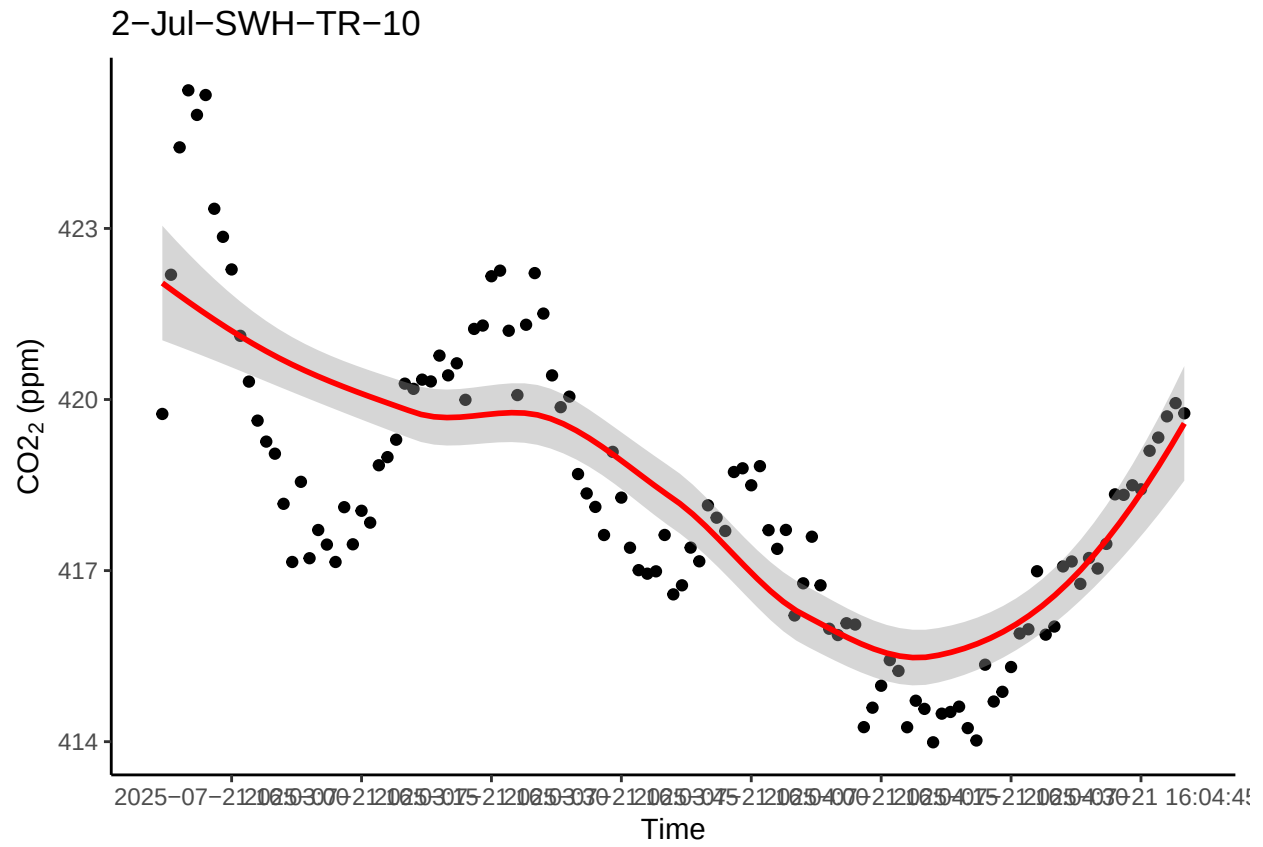
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



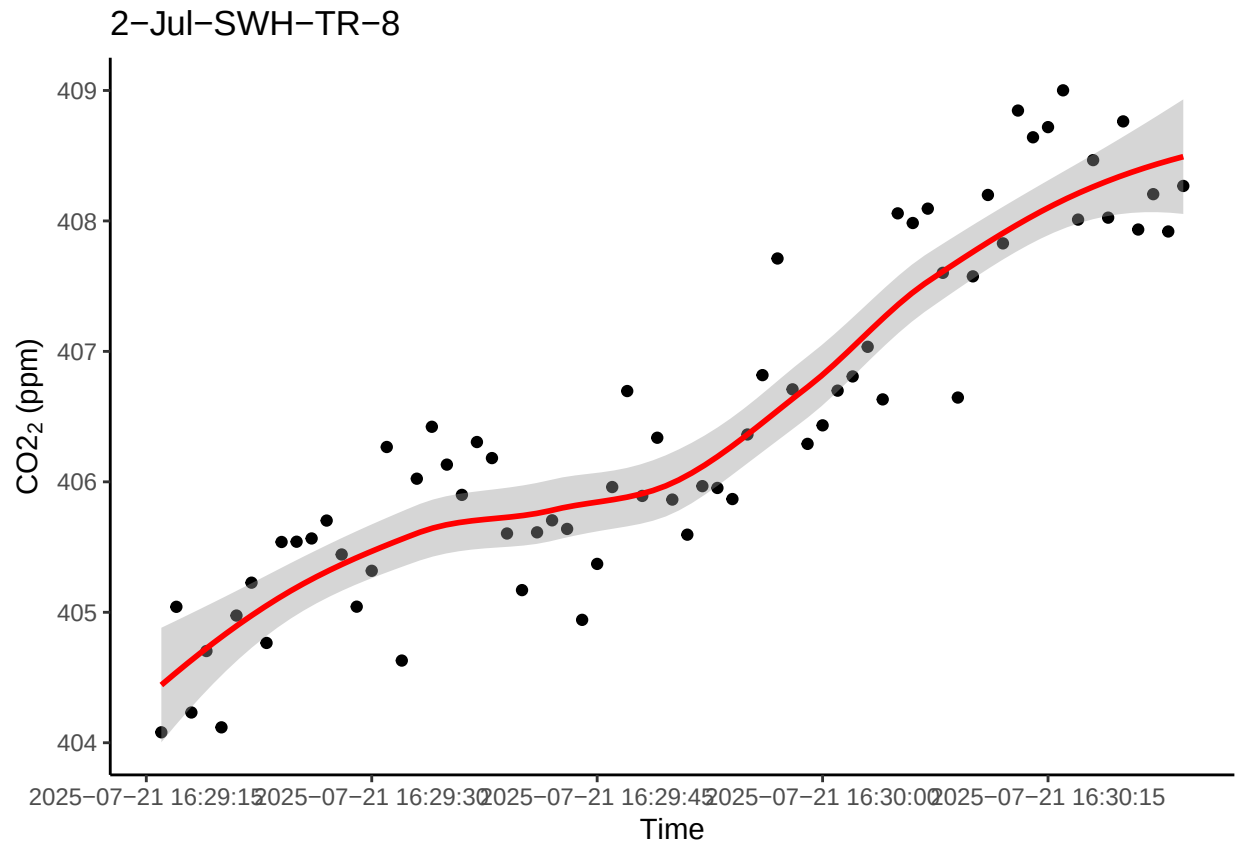
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



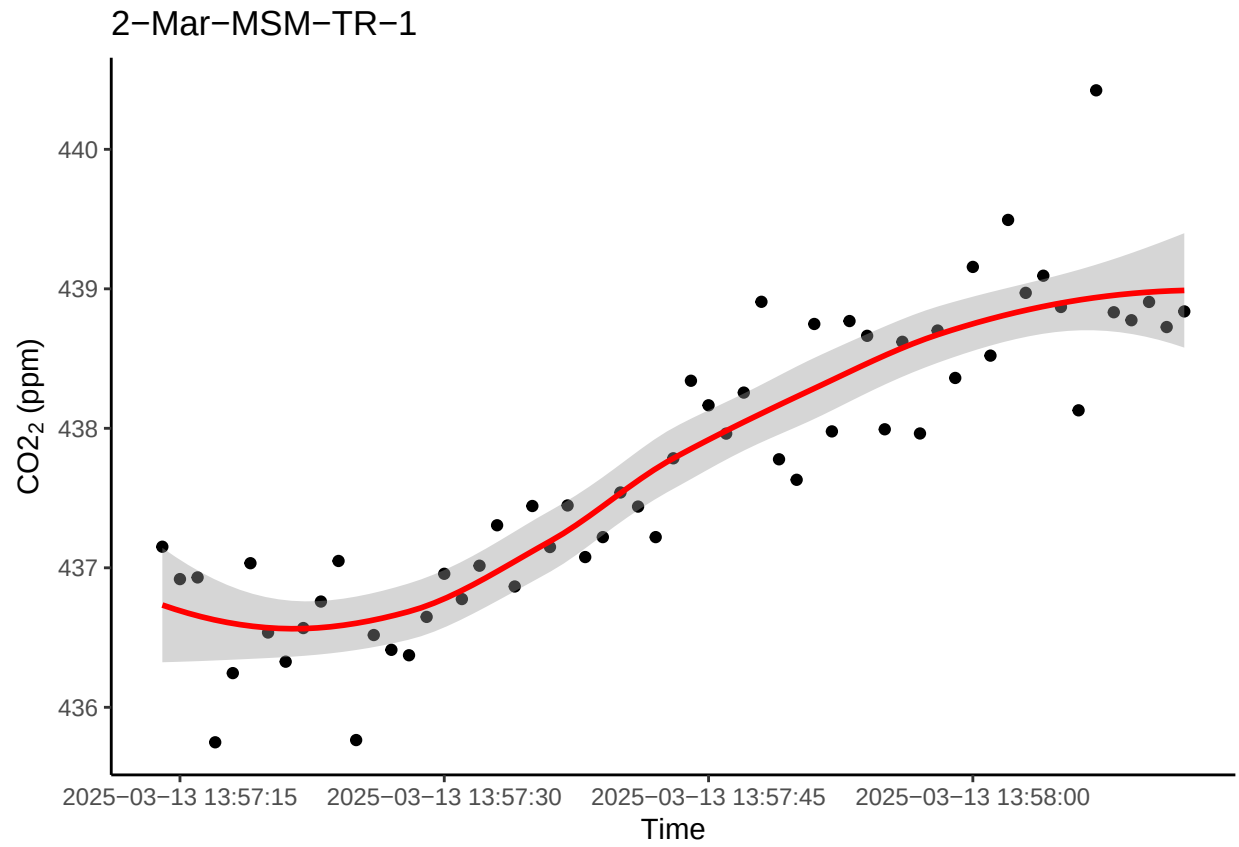
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



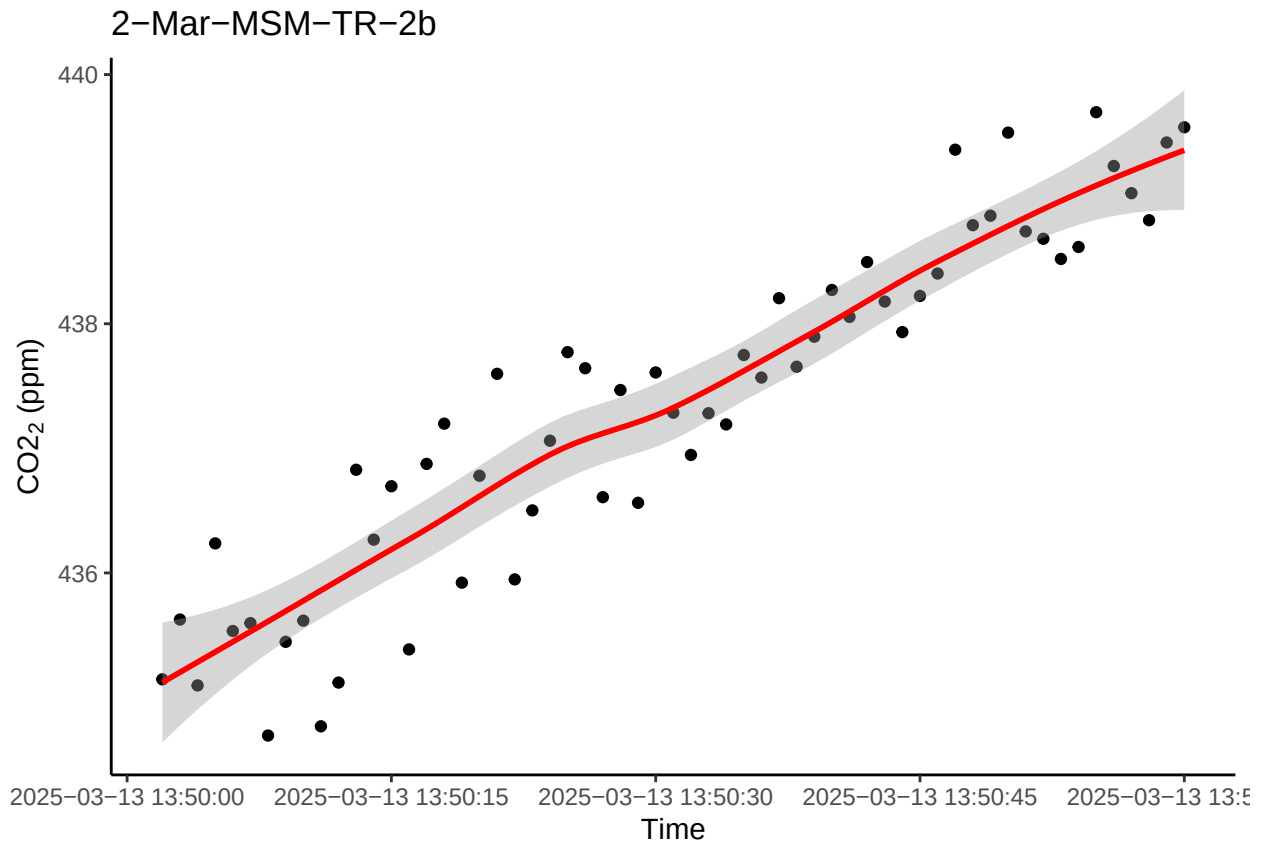
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



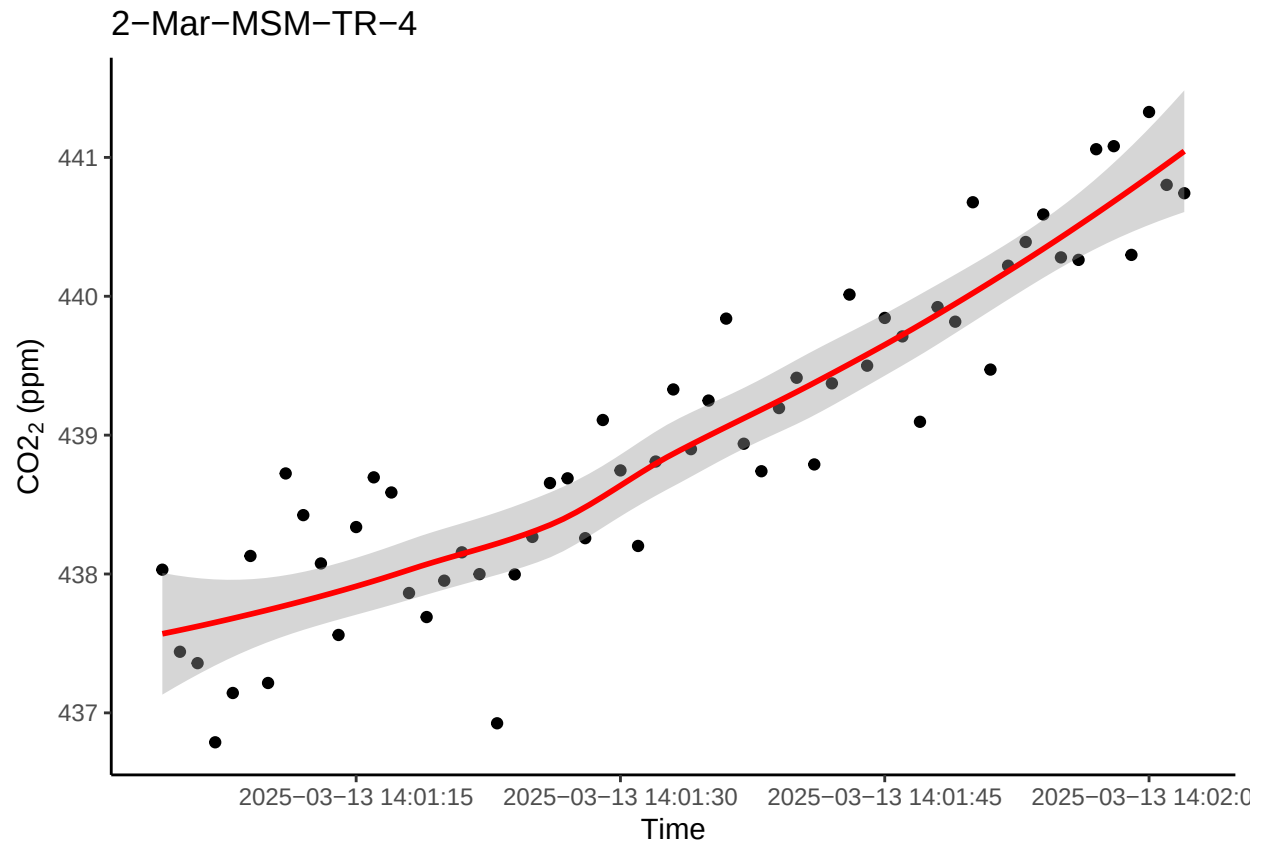
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



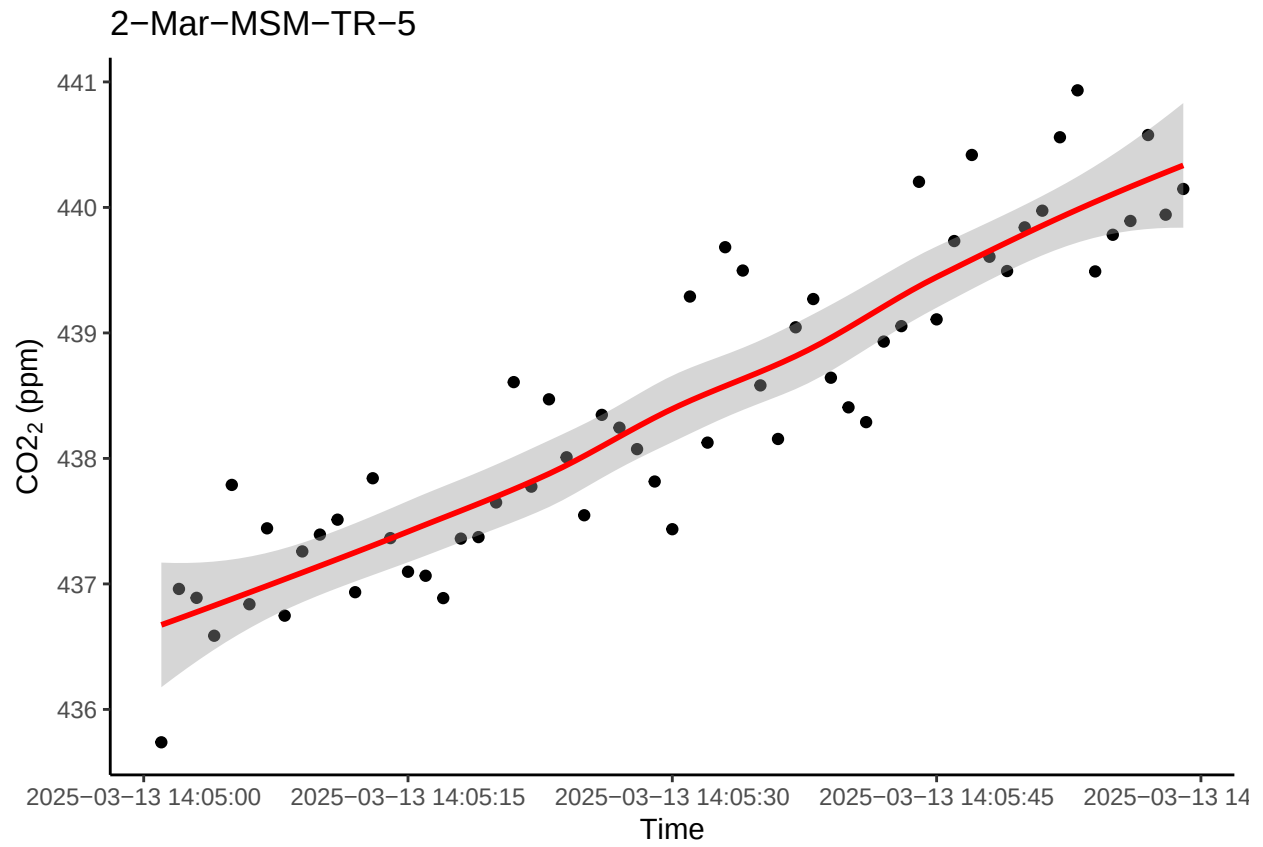
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



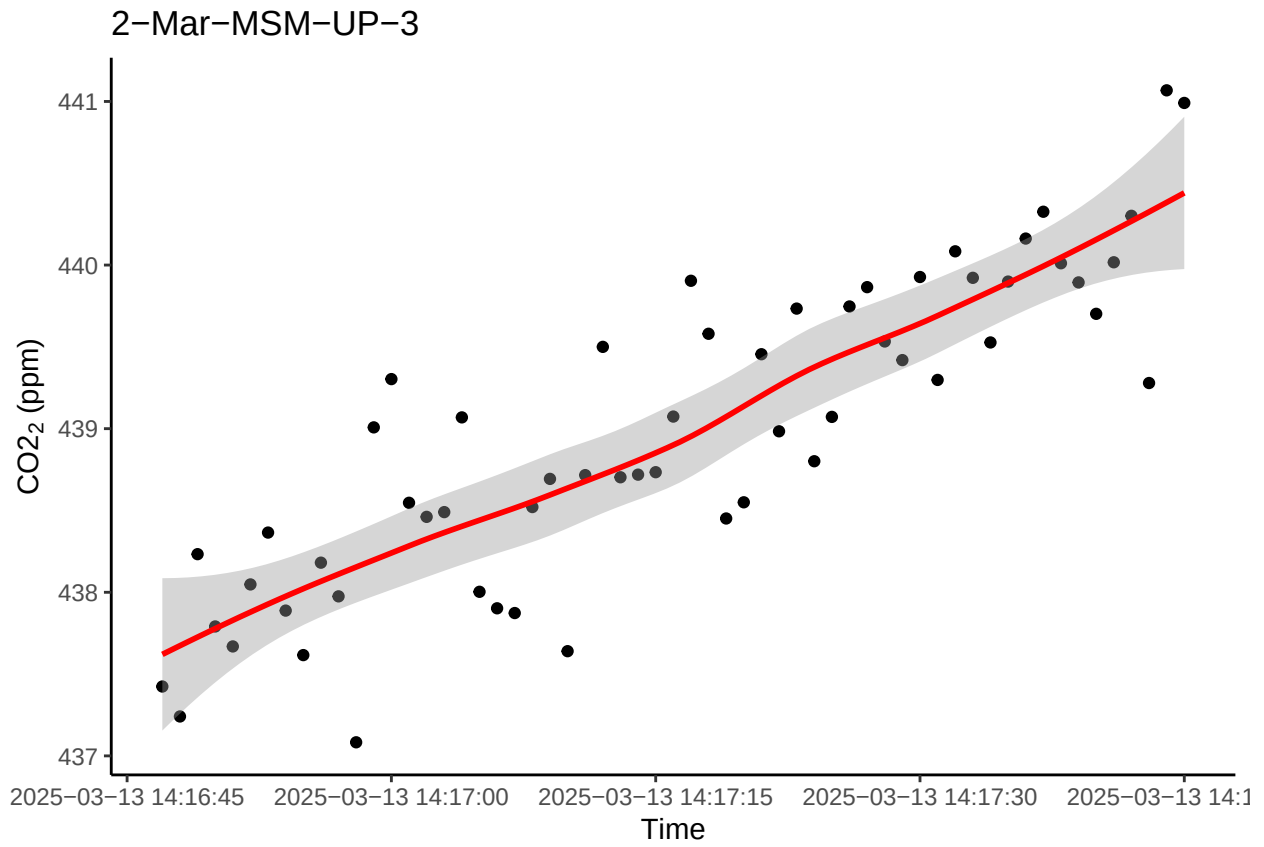
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



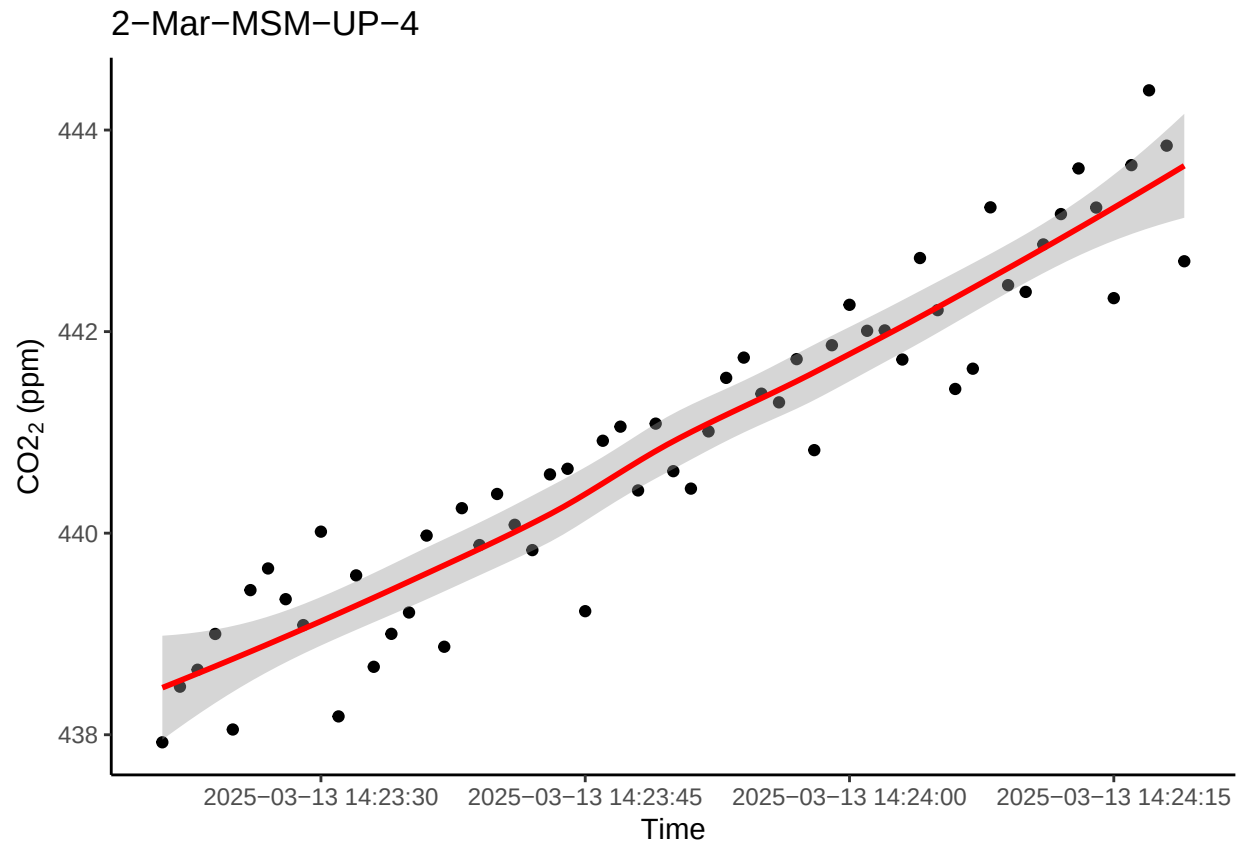
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



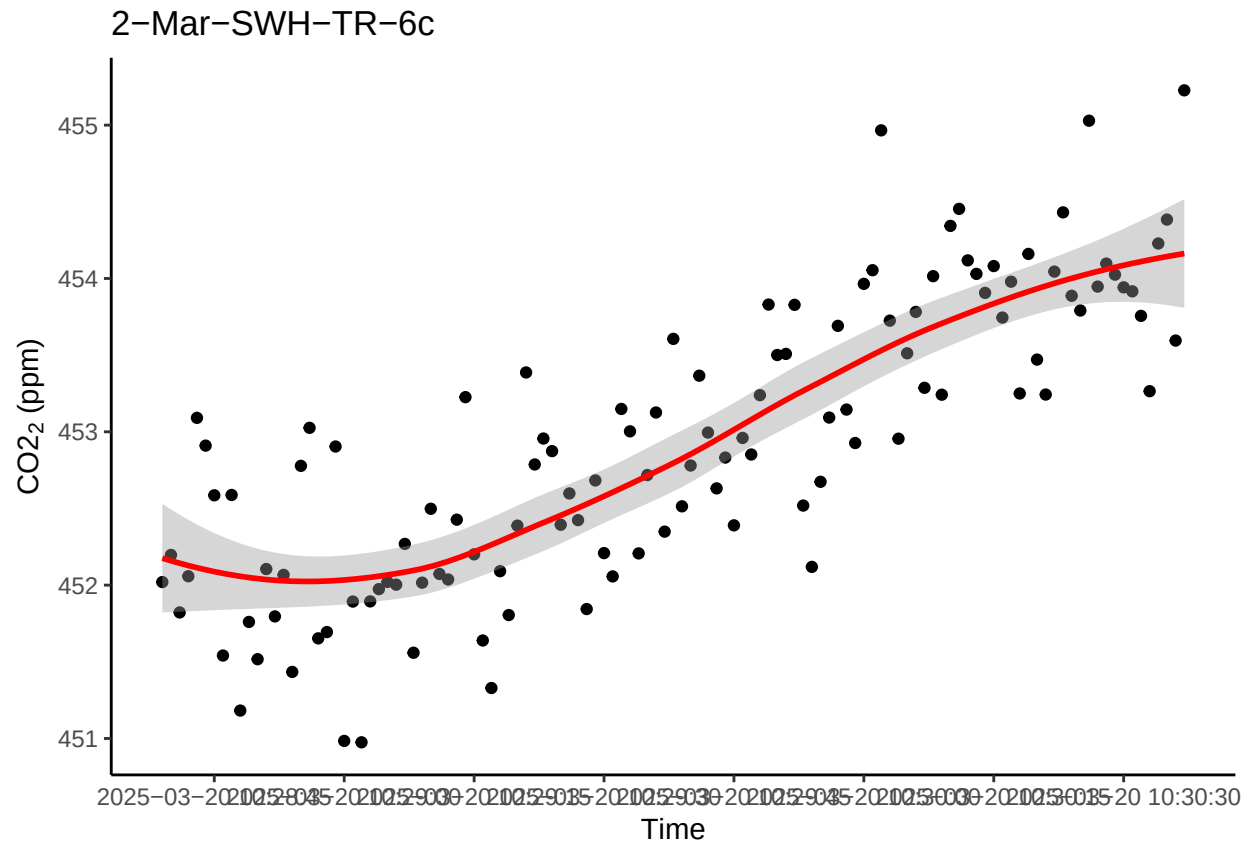
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



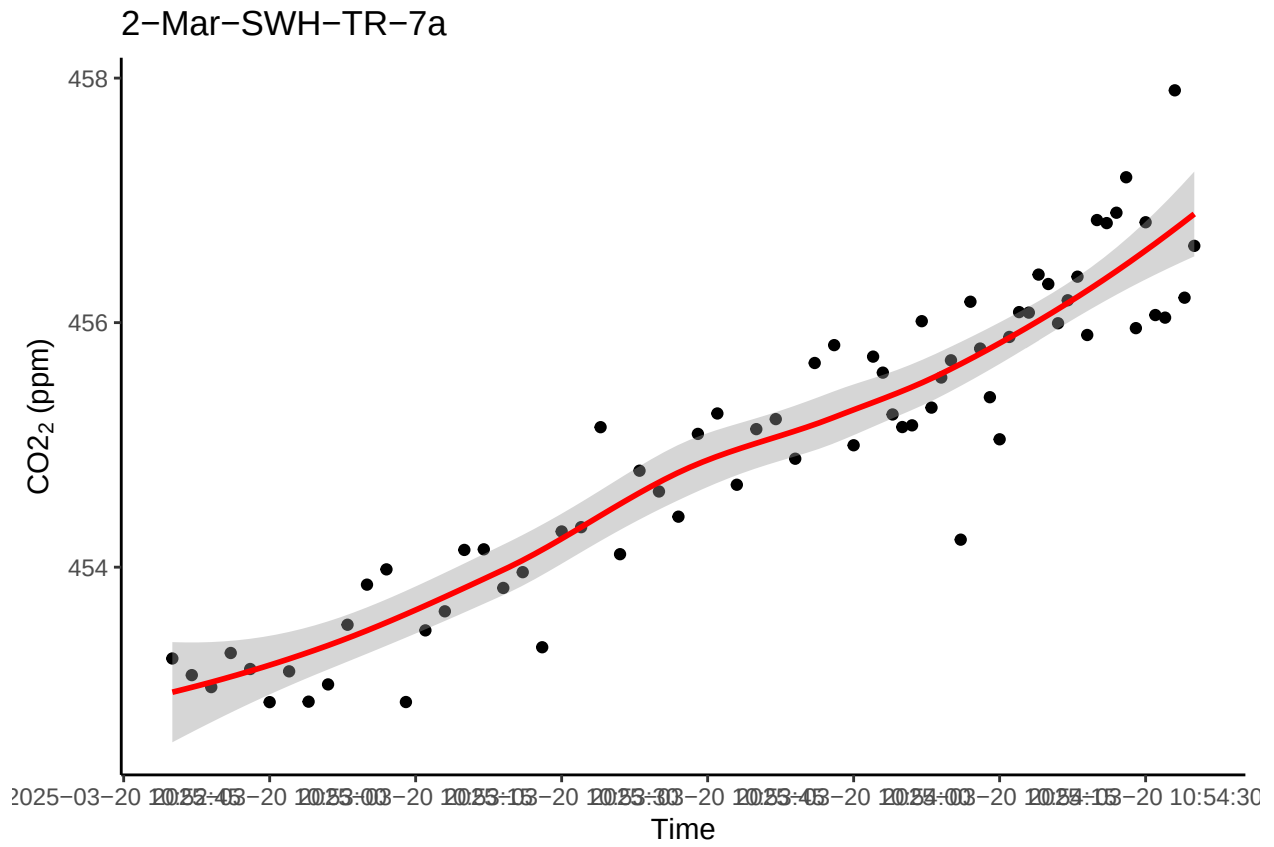
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



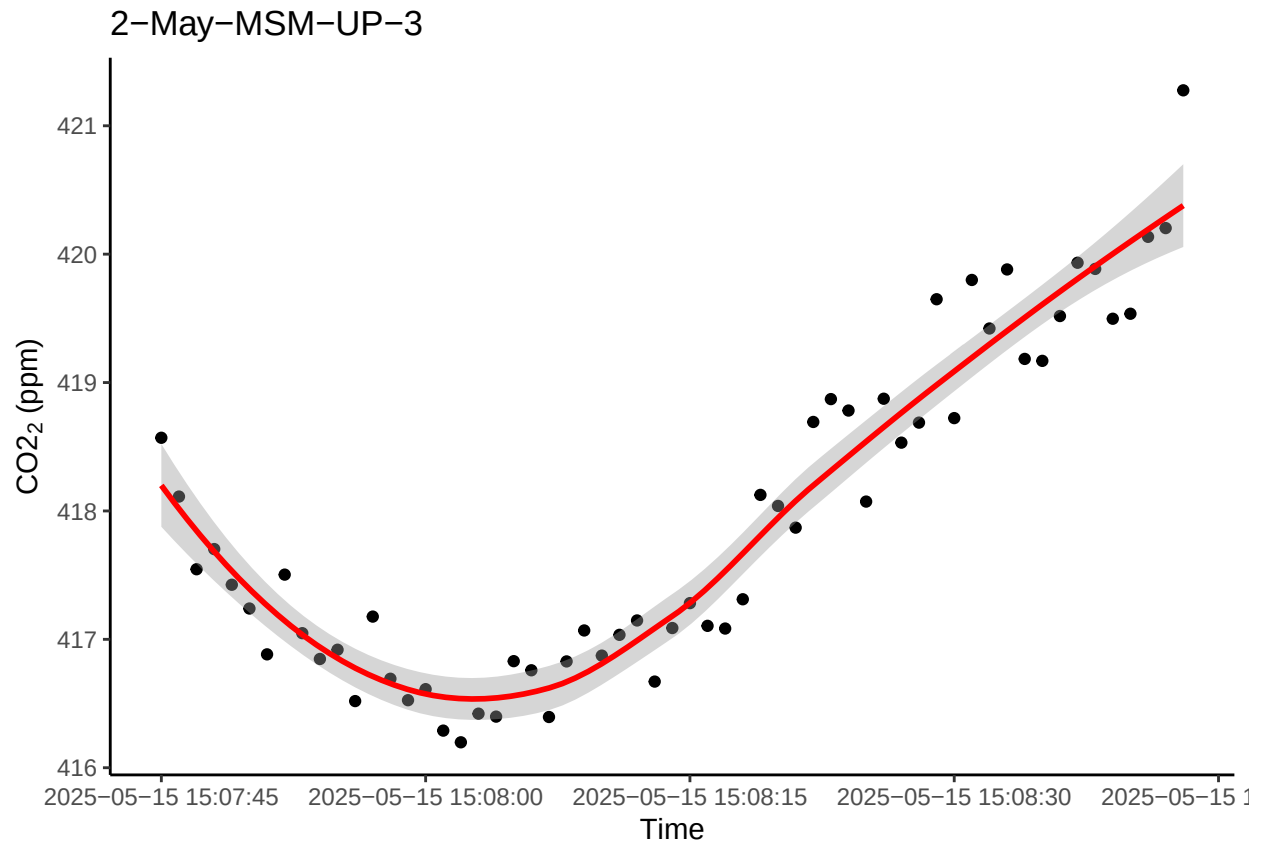
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



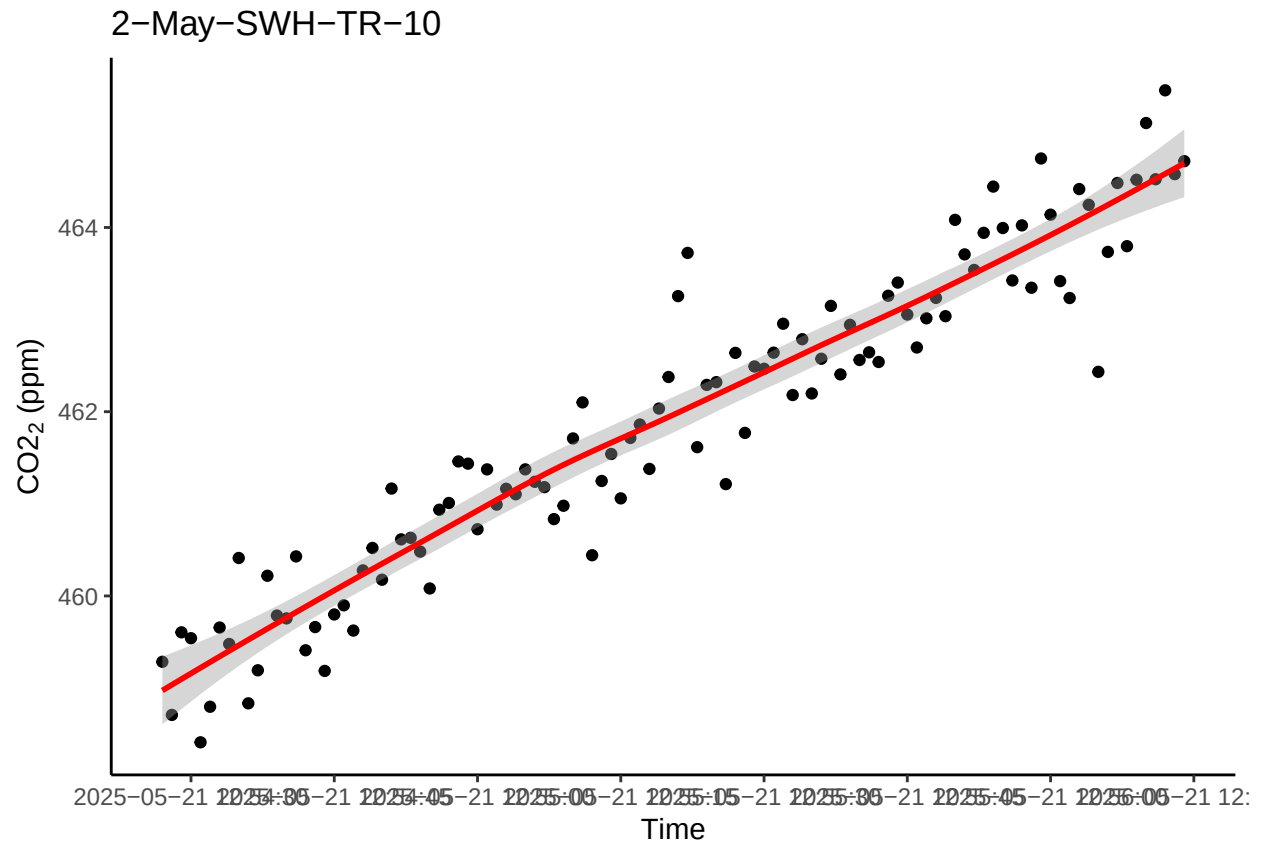
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



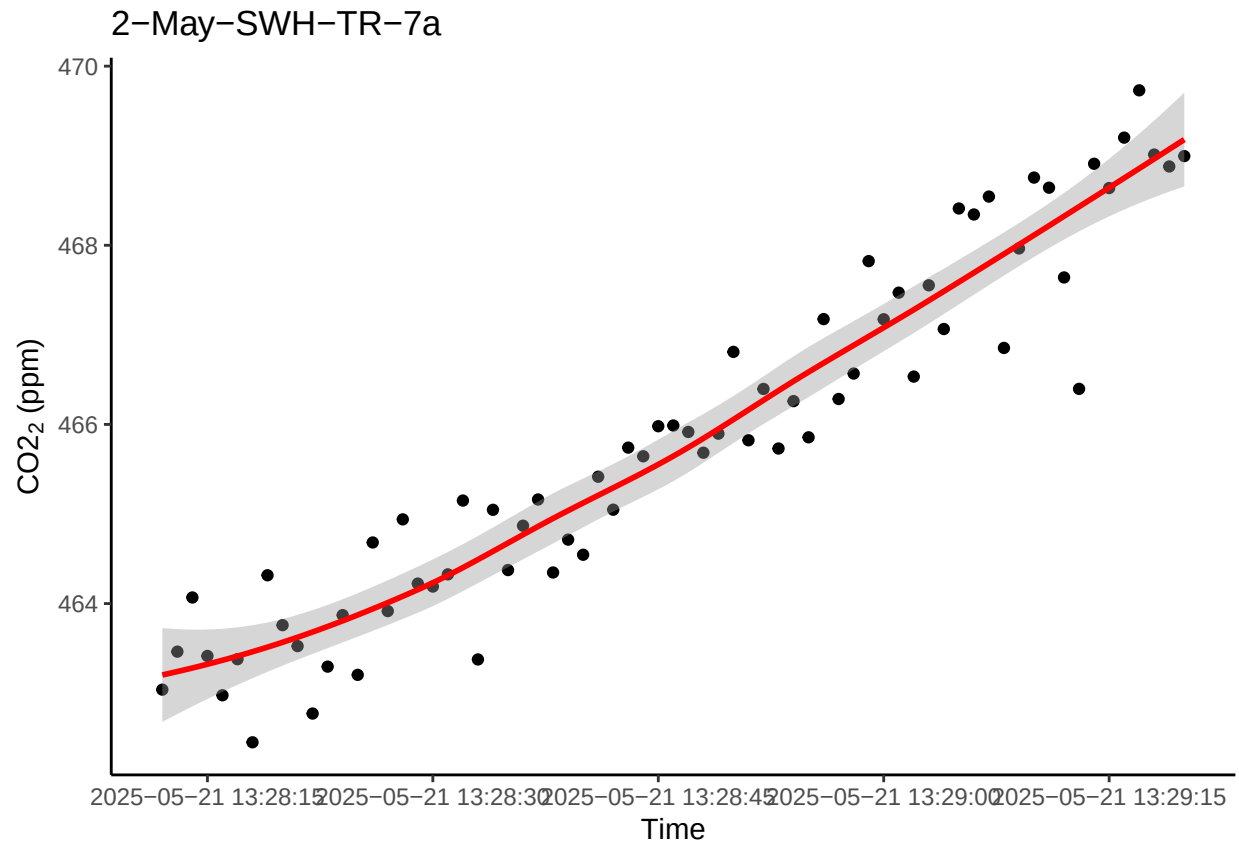
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



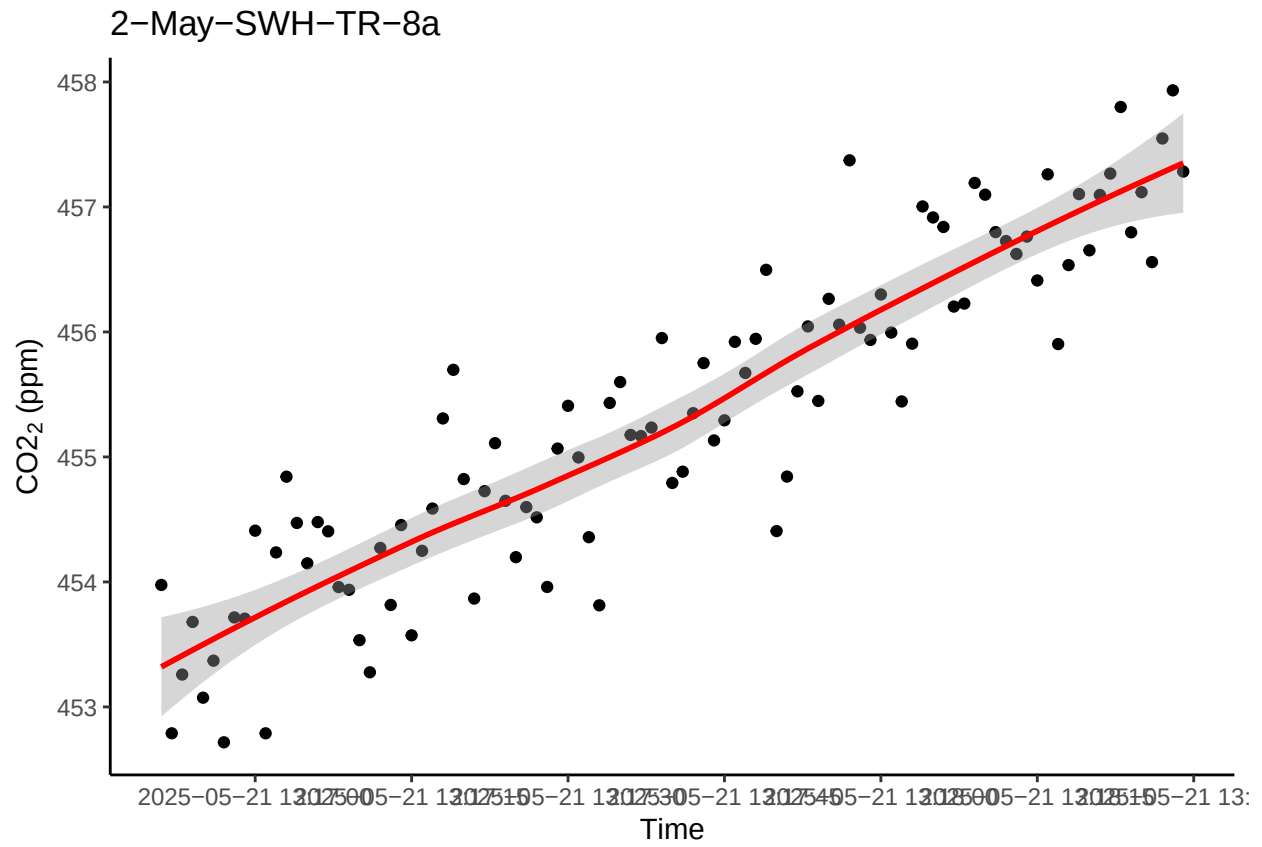
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



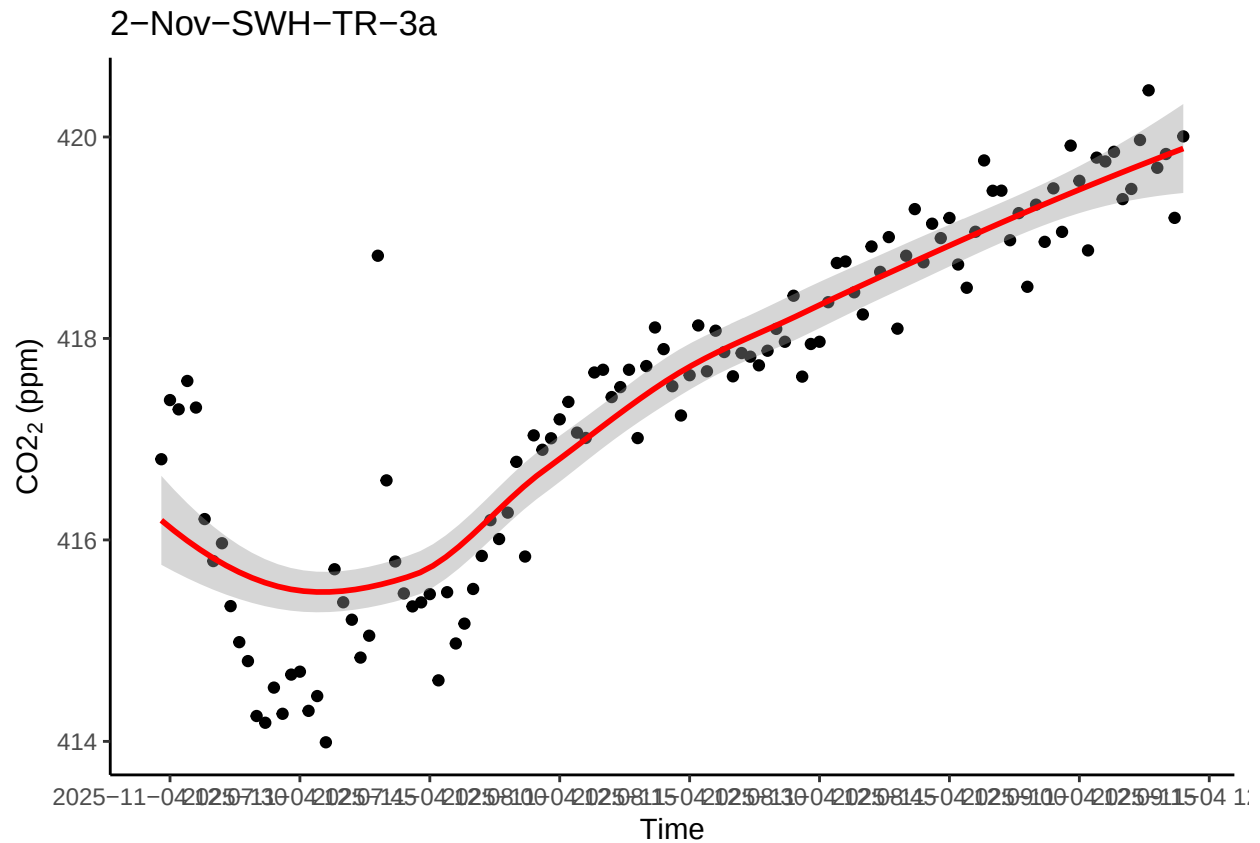
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



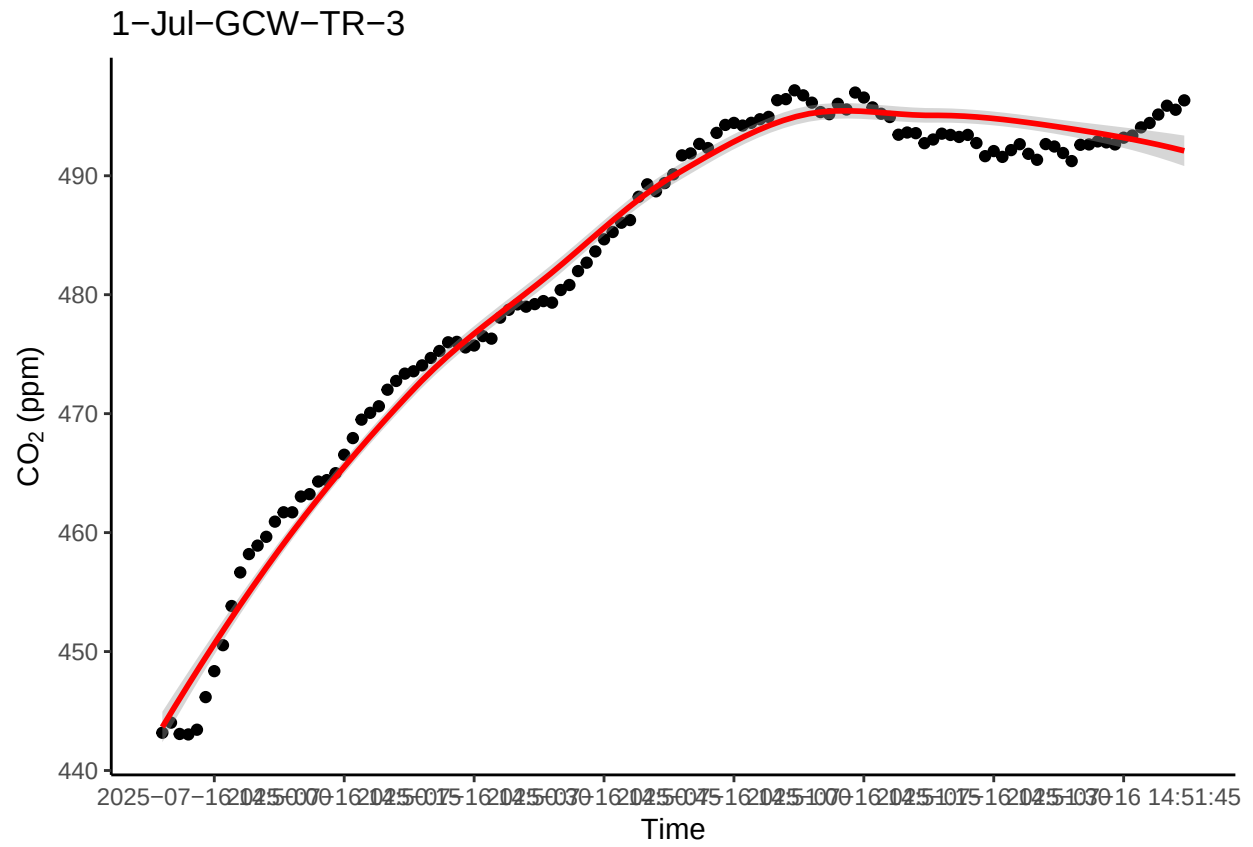
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



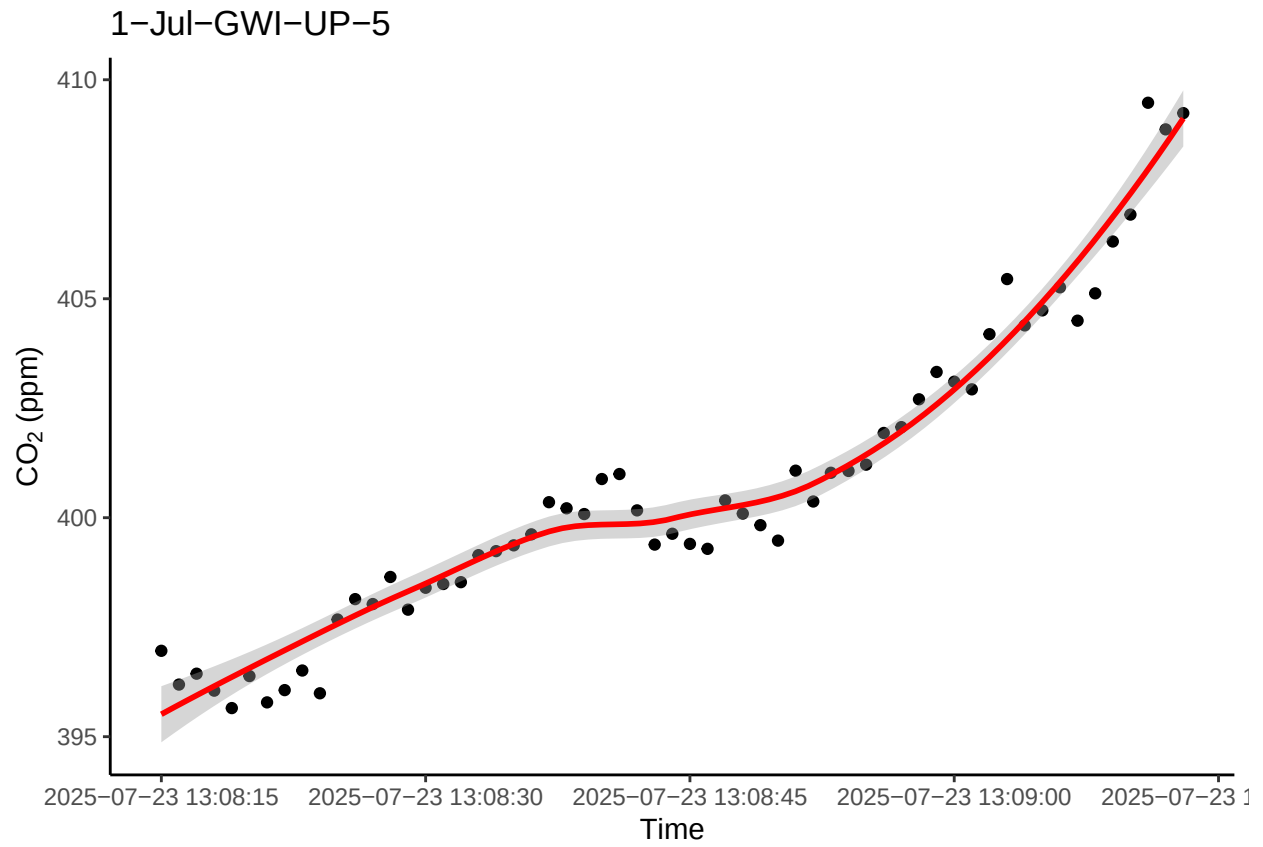
0.13 Take a look at the bad fluxes < 0.9 and have a high flux estimate

- 1 See if we want to remove these fluxes or if we want to try and recalculate these ones?

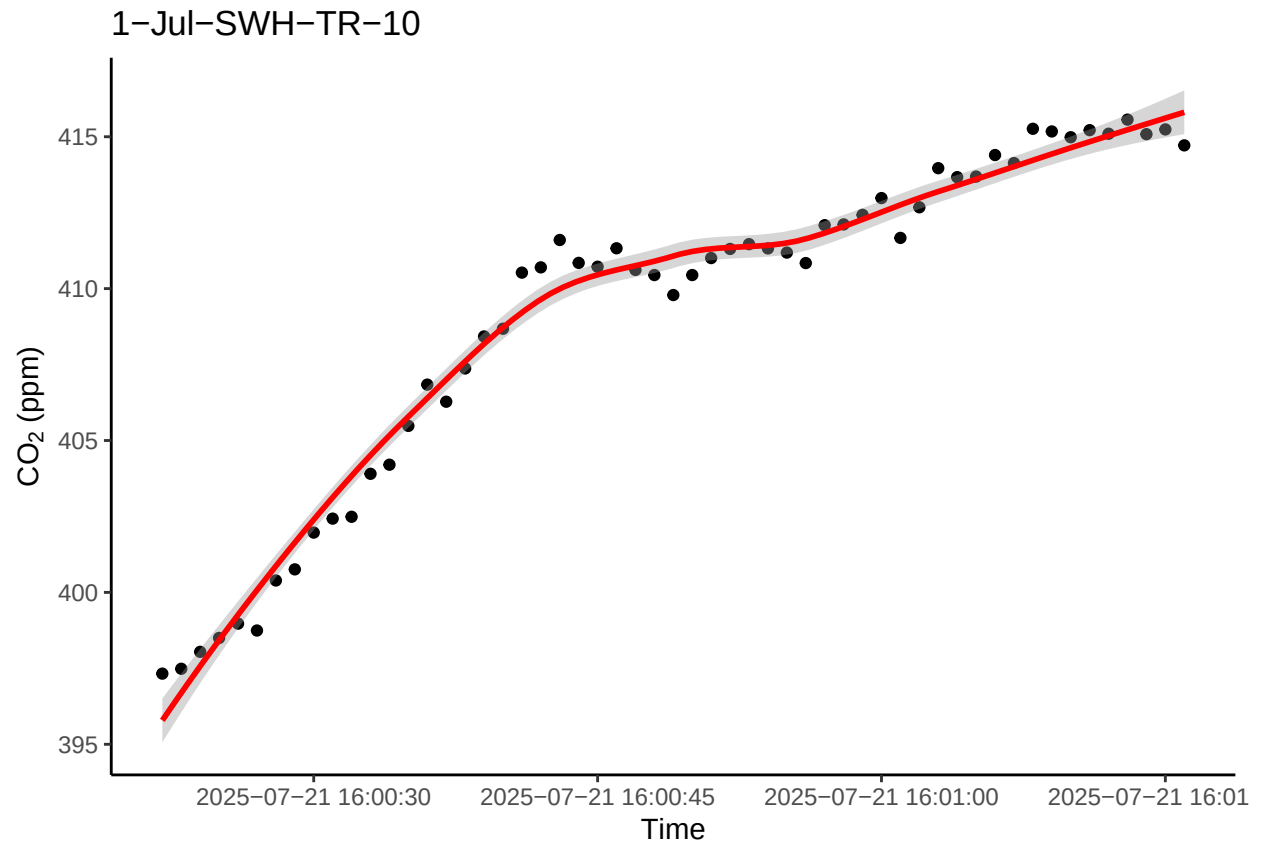
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



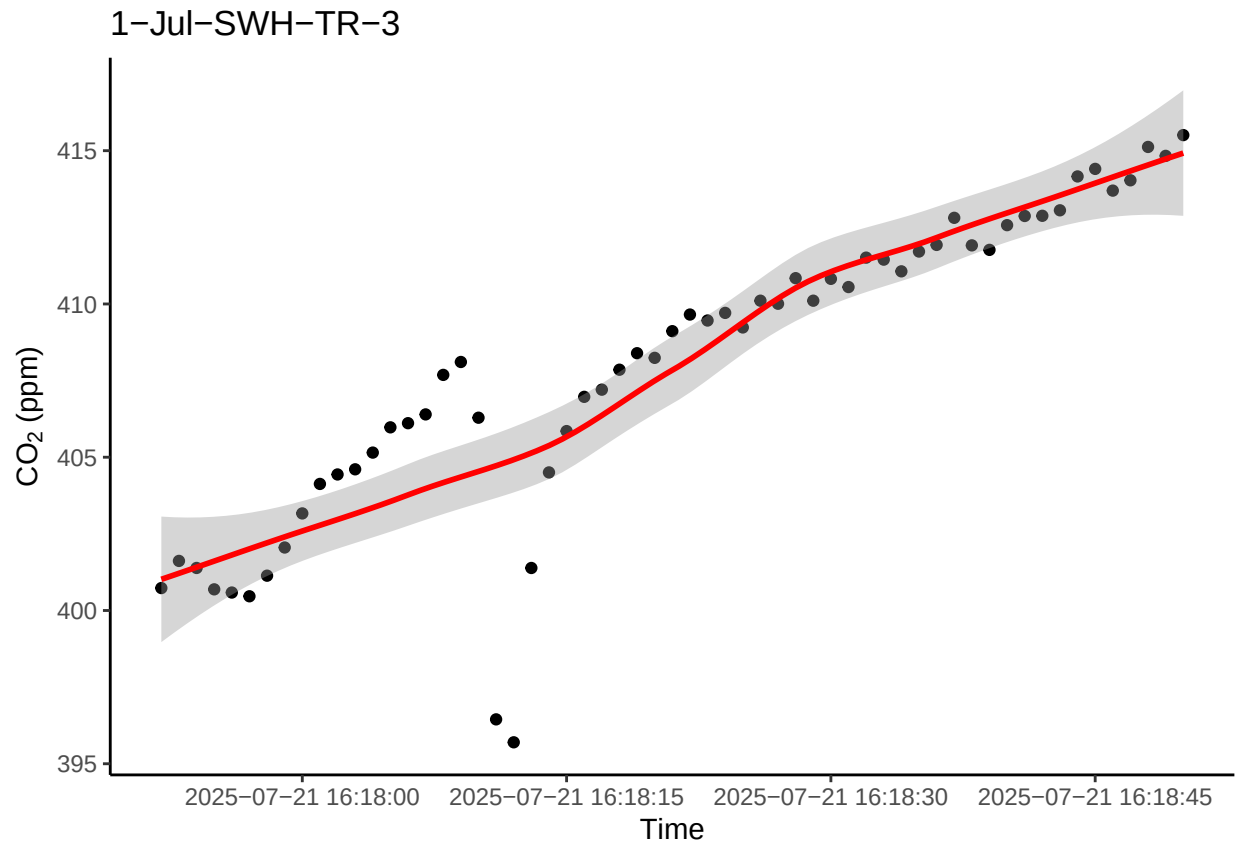
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



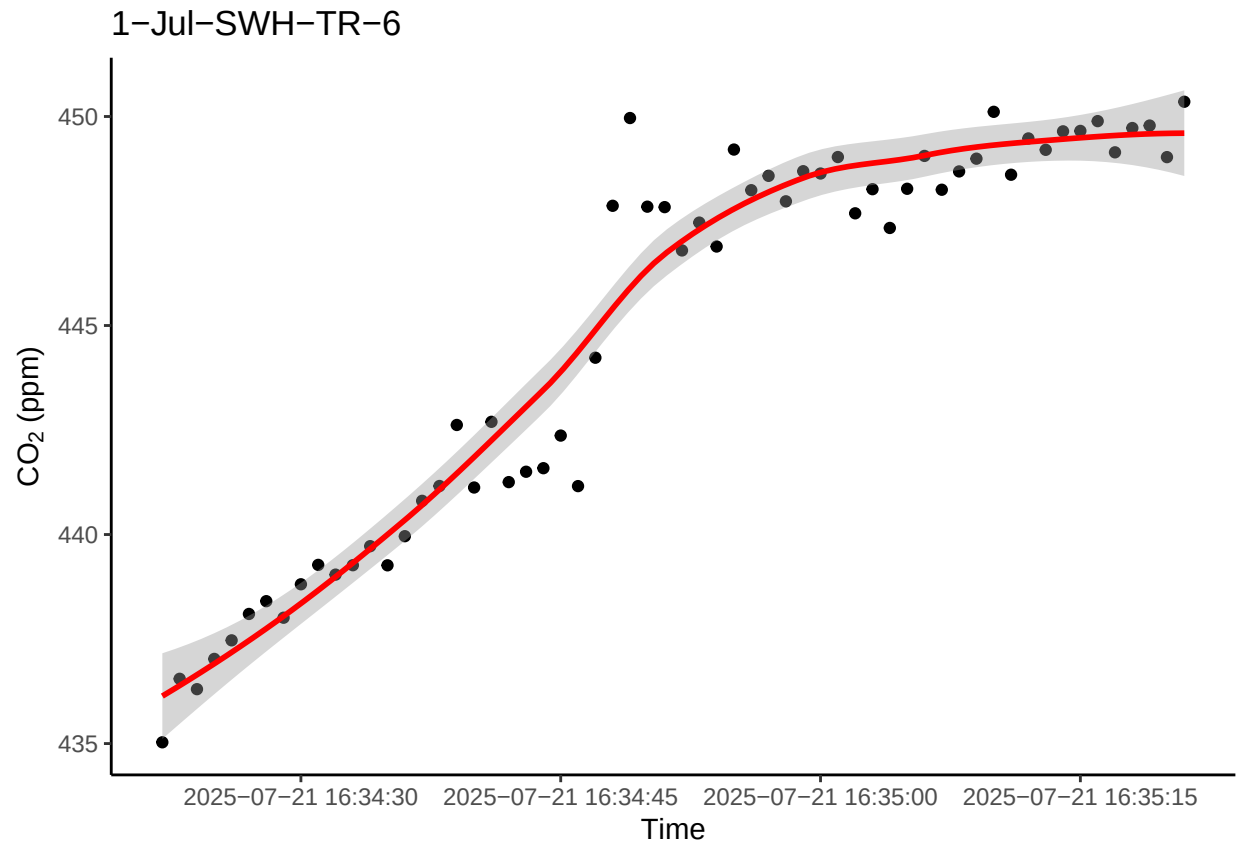
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



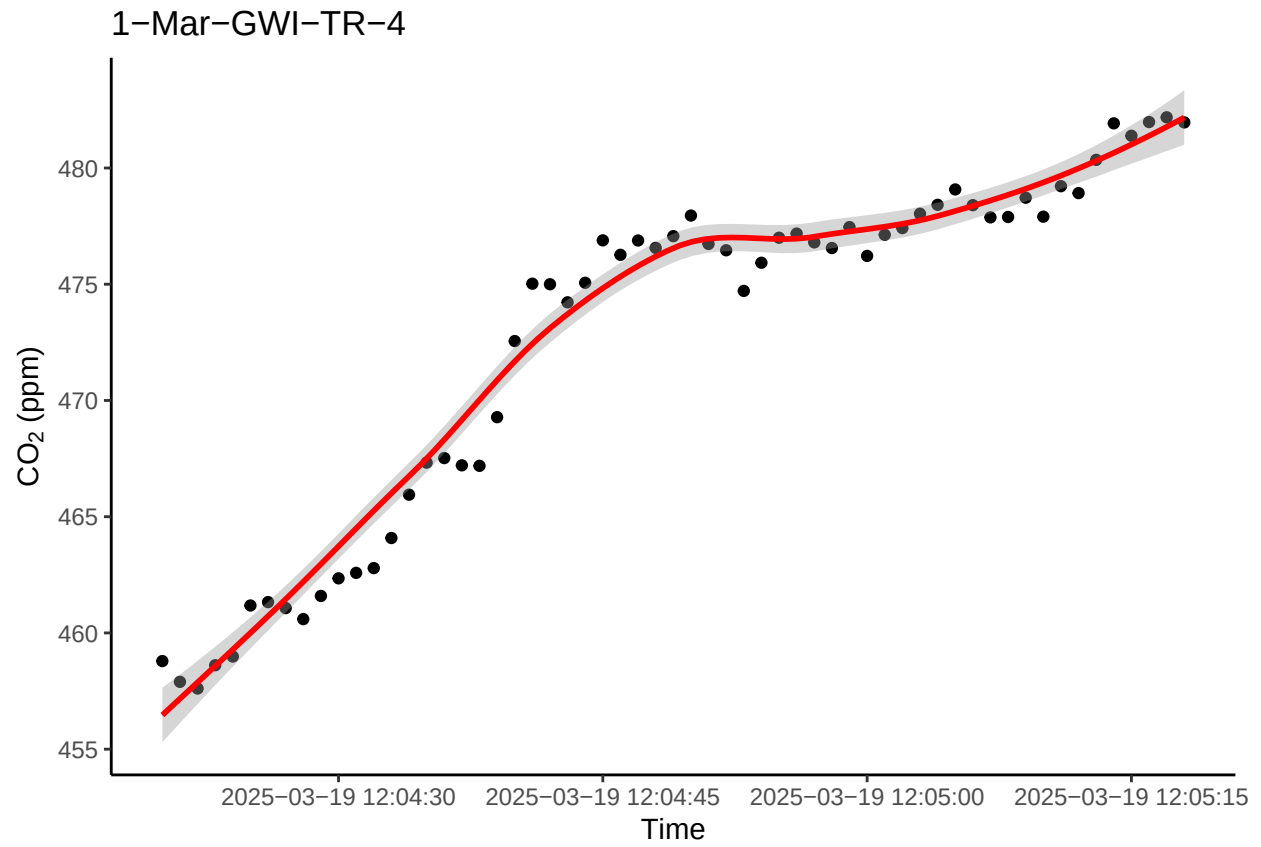
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



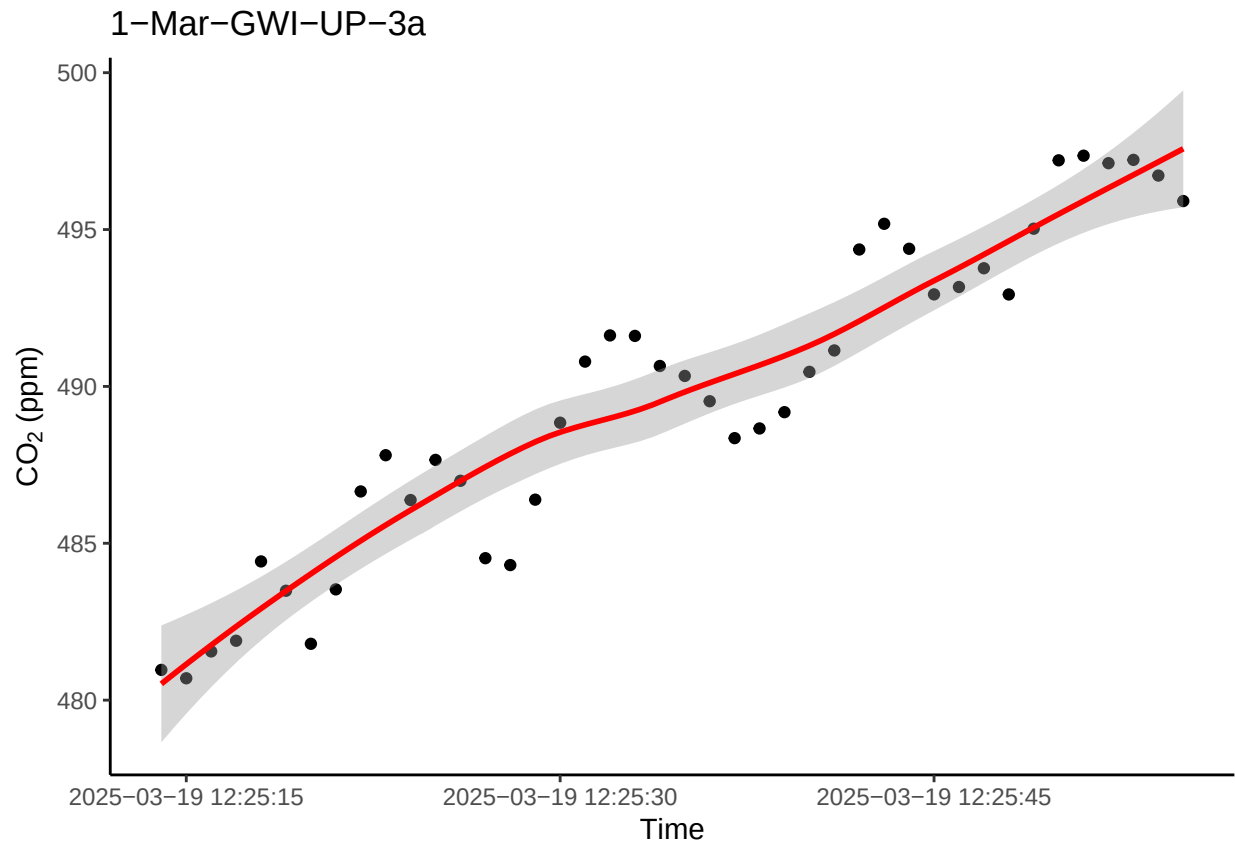
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



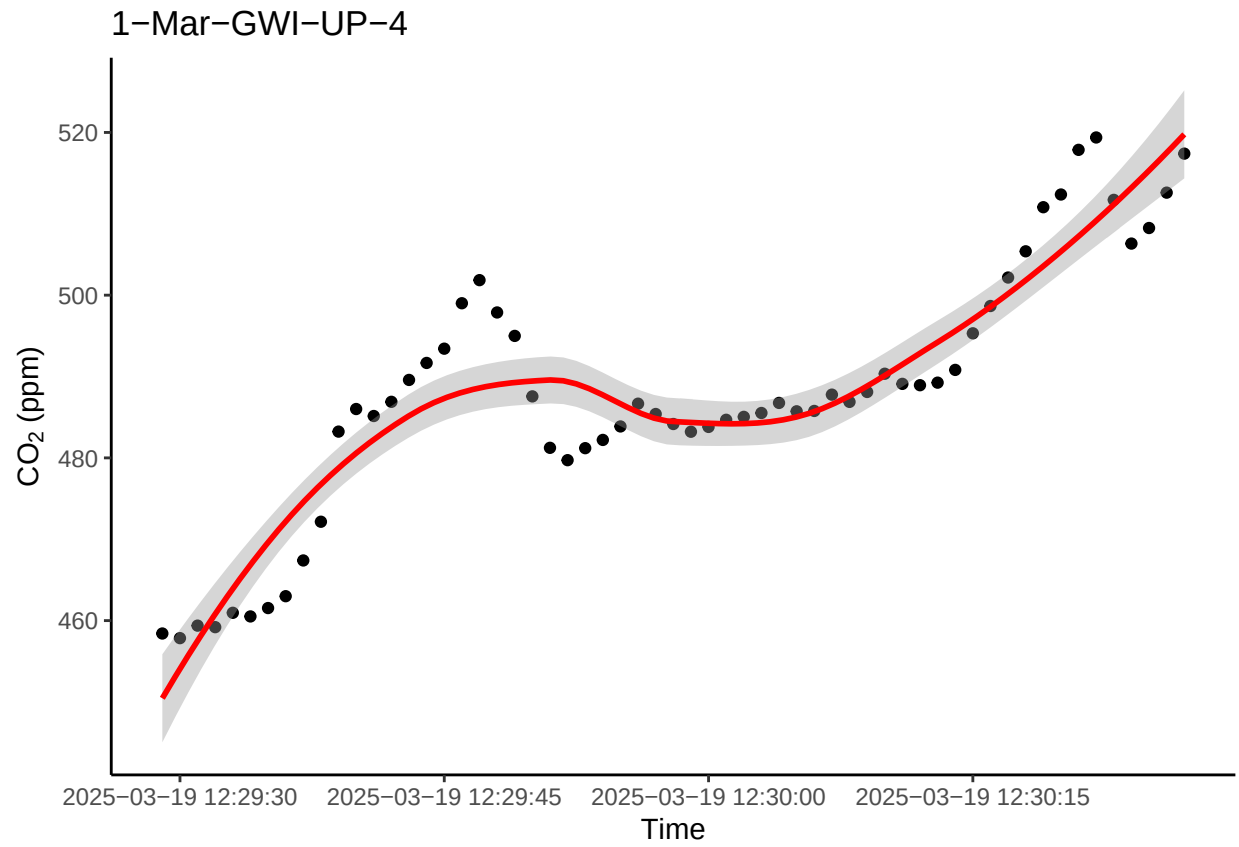
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



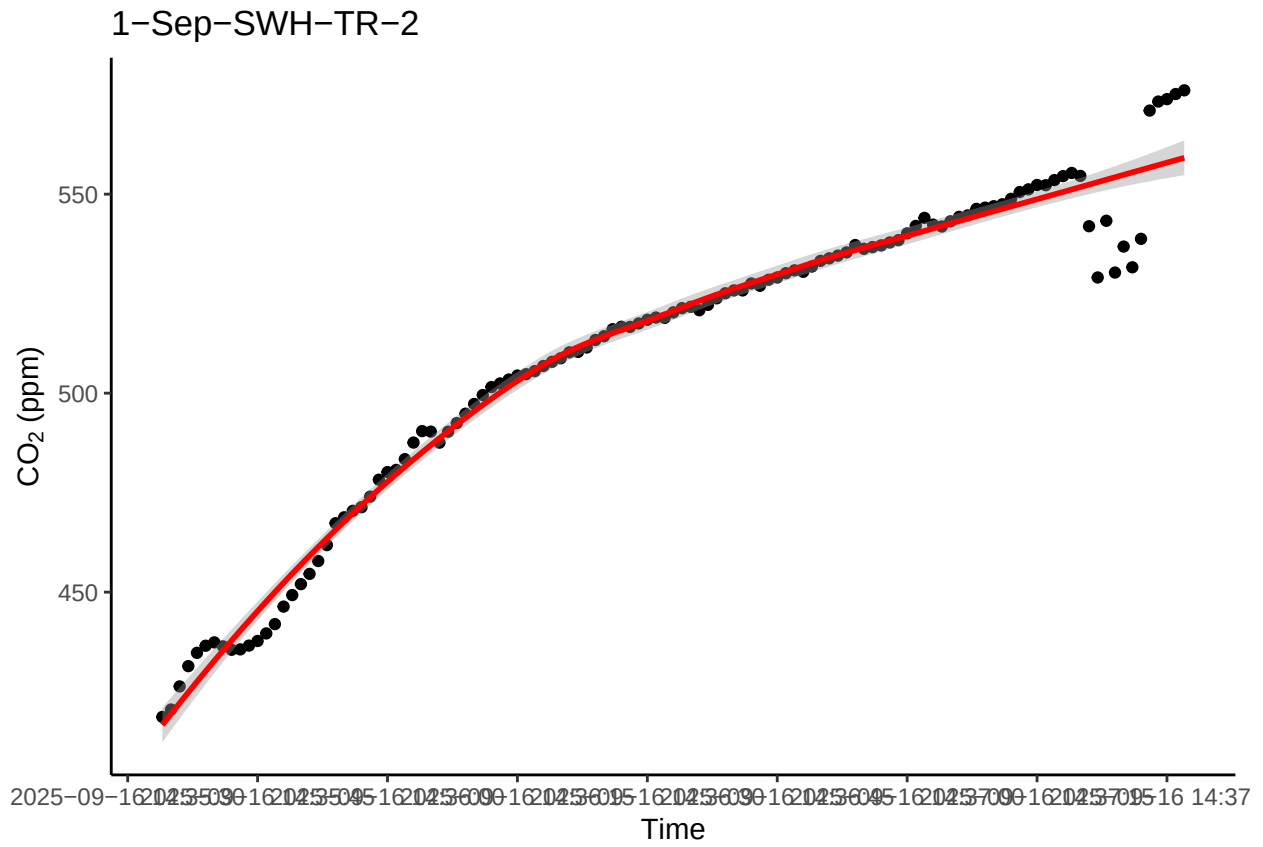
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



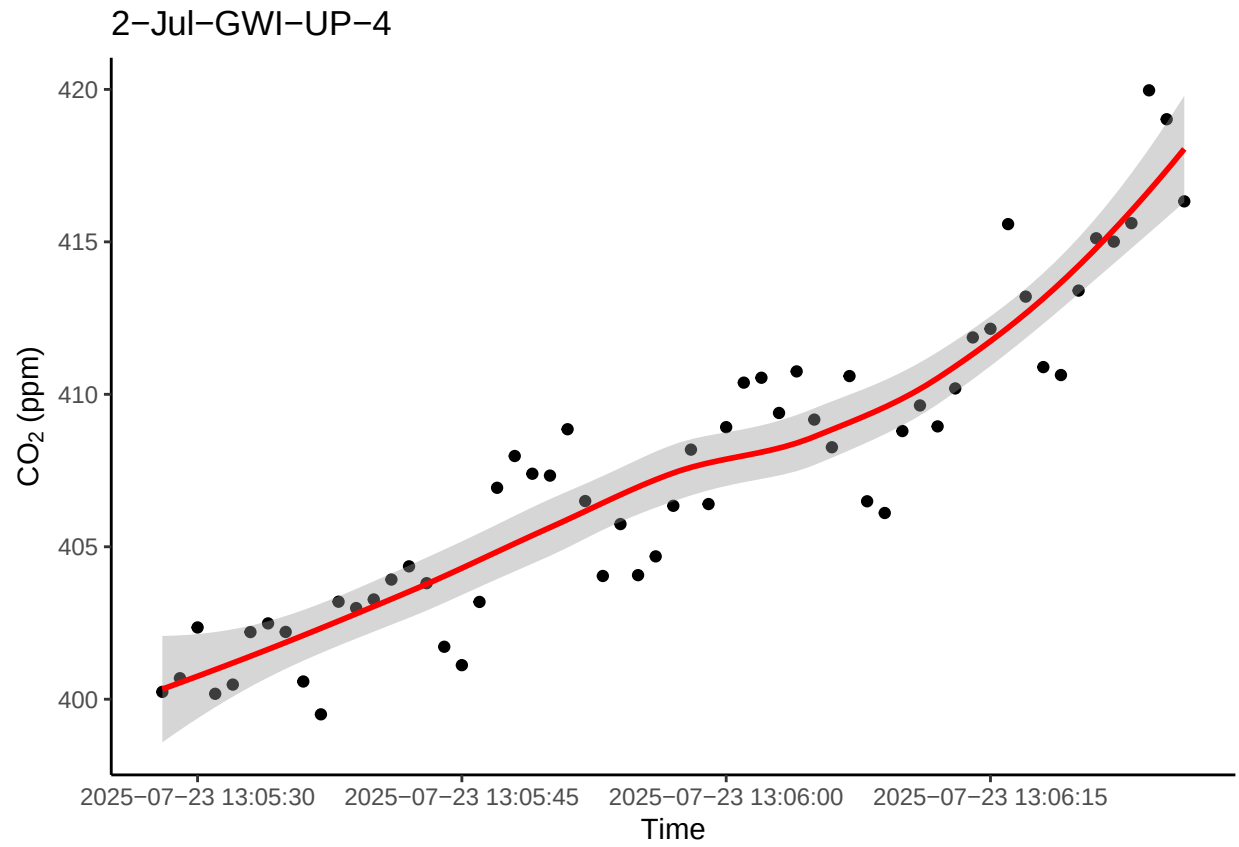
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



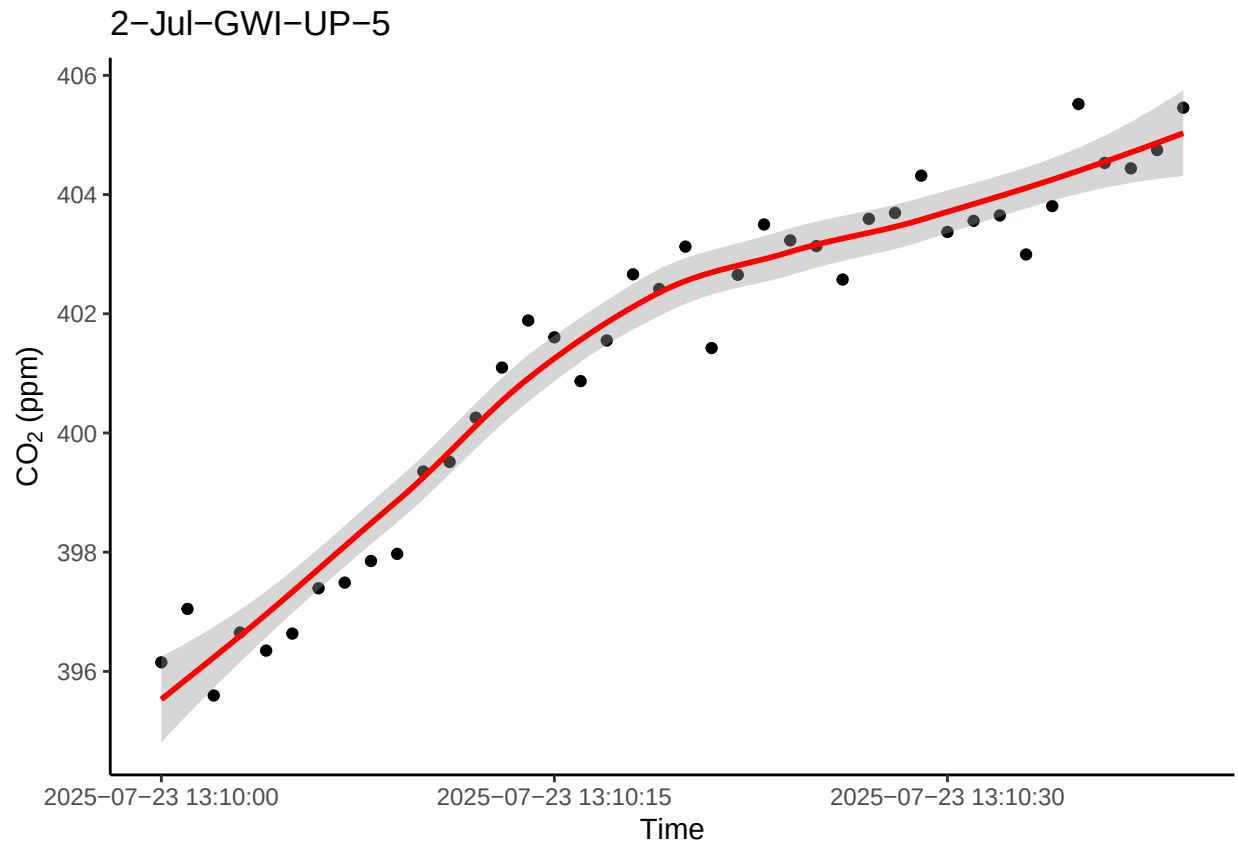
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



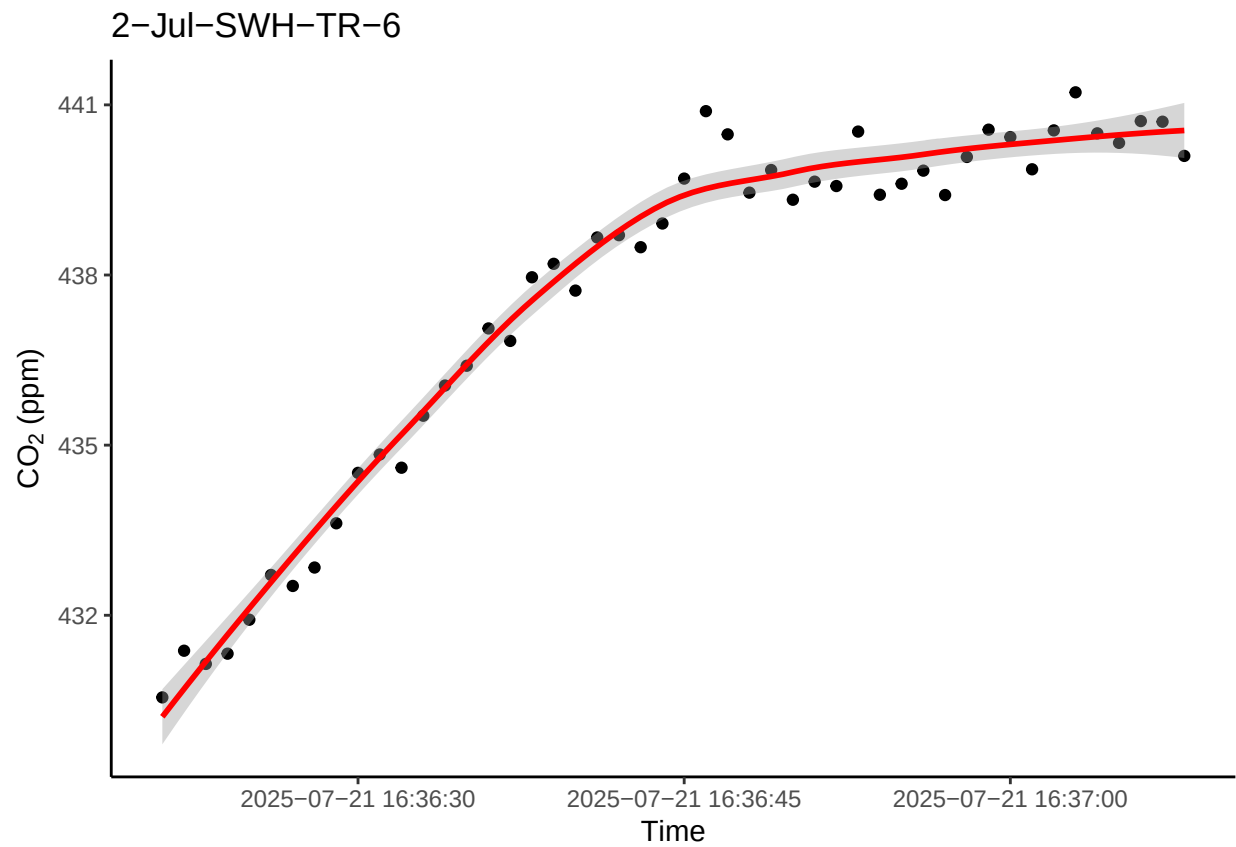
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



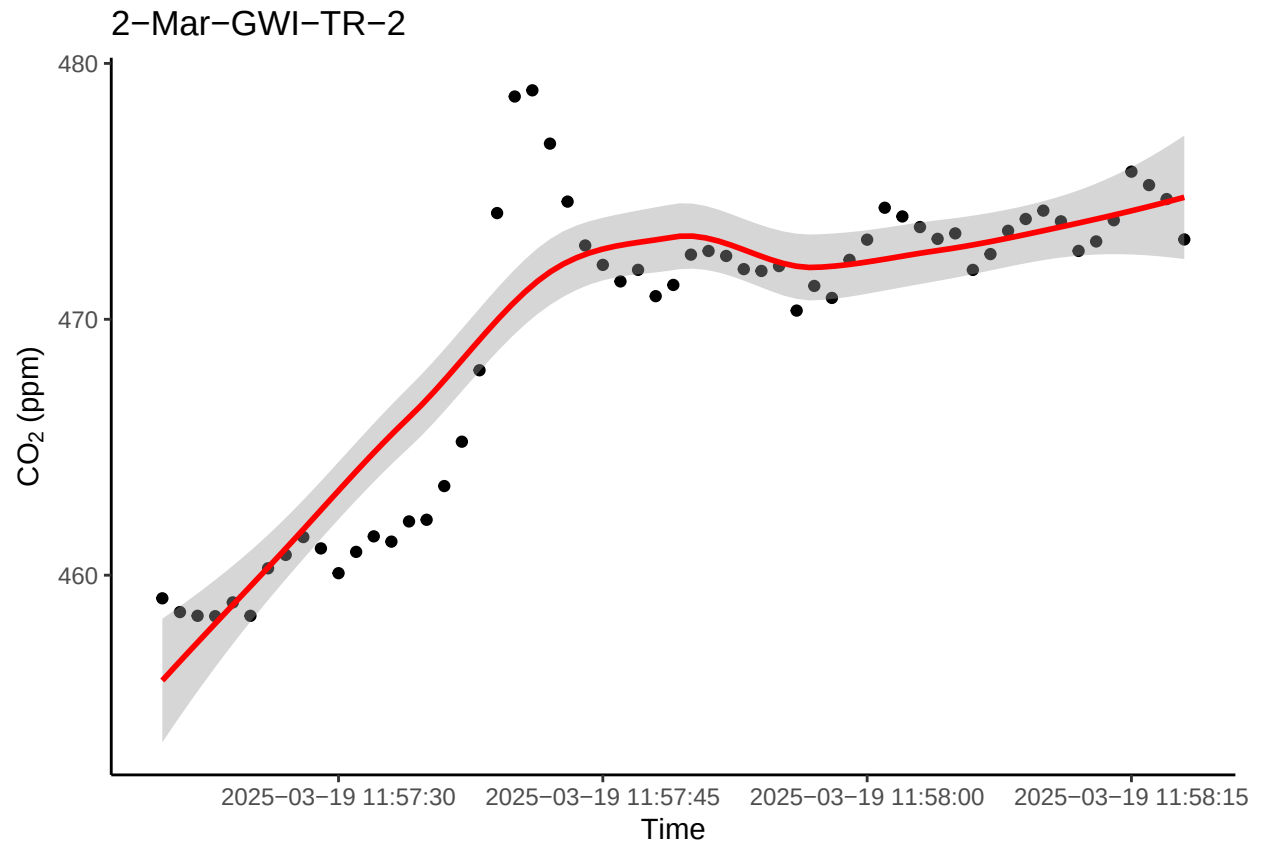
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



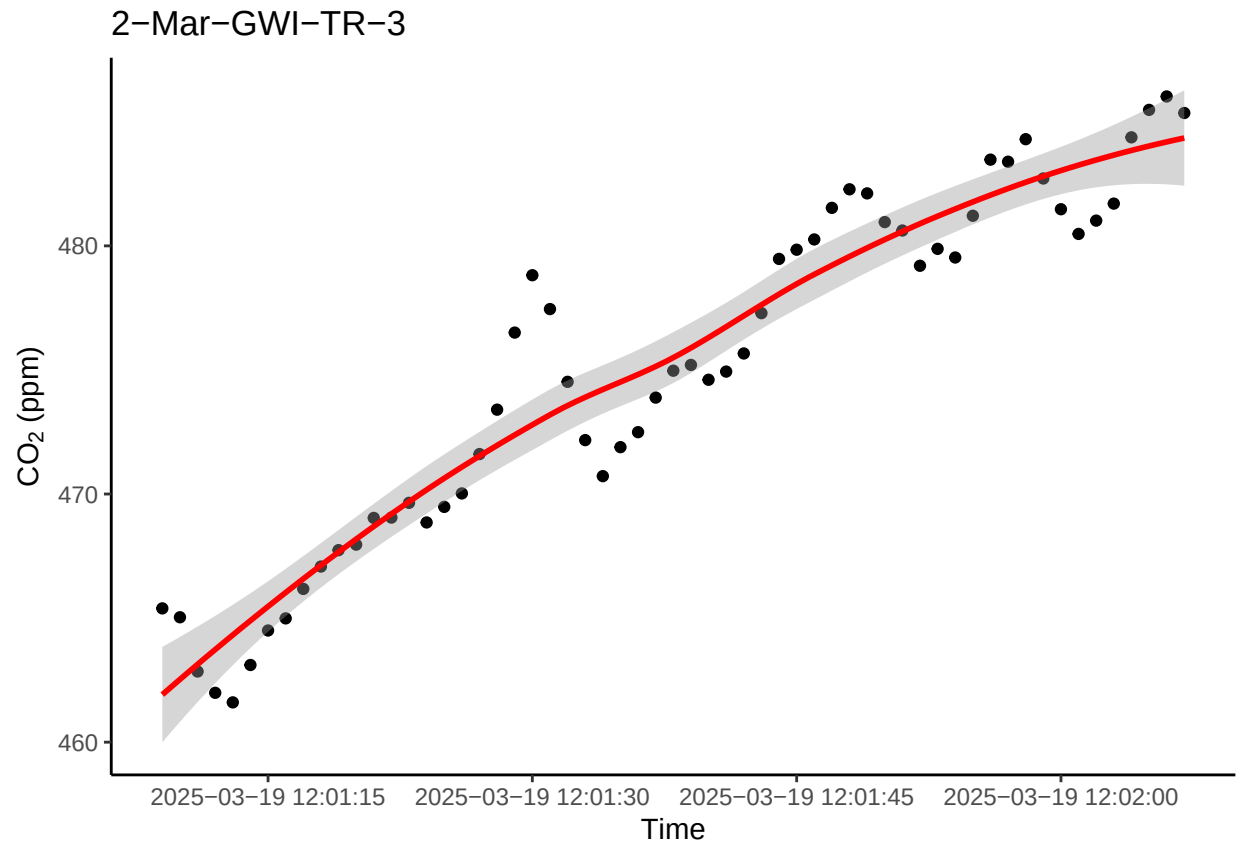
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



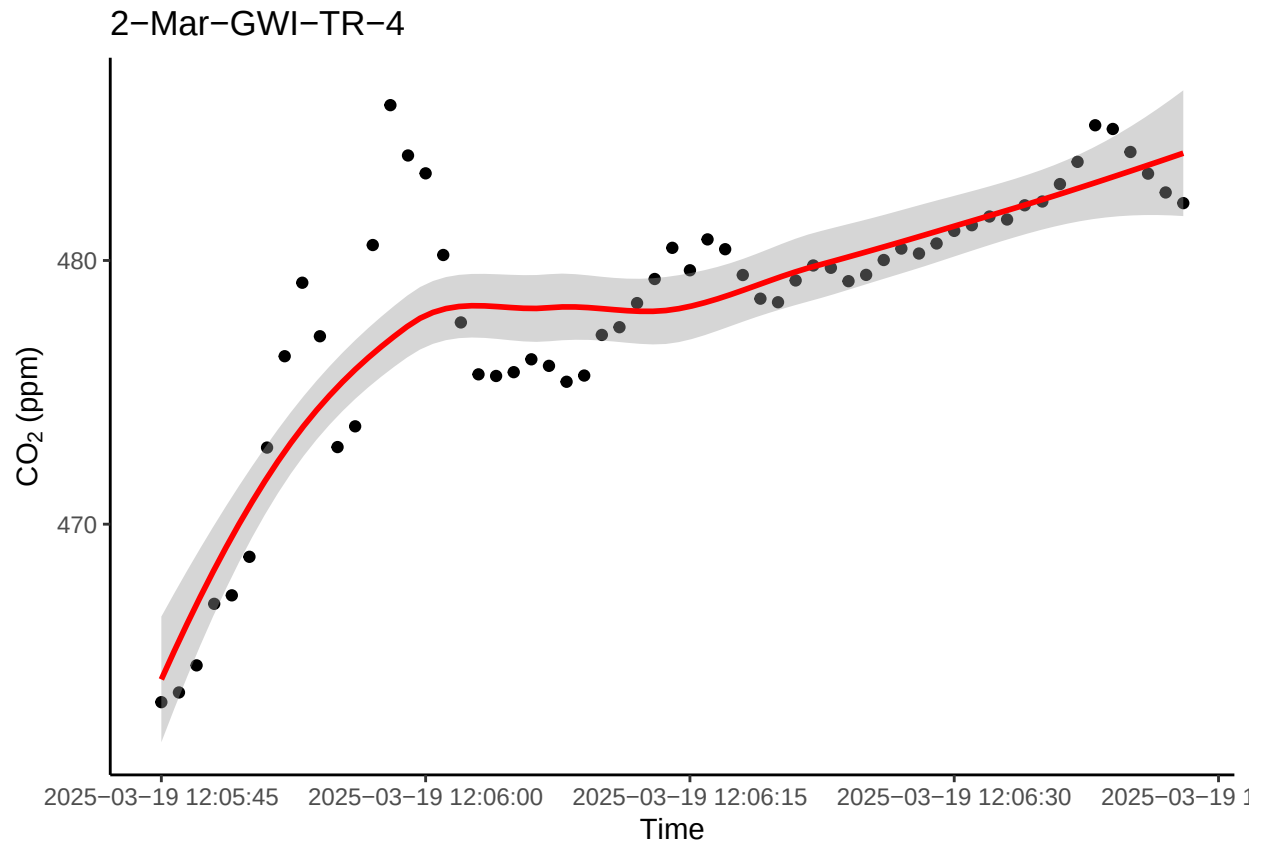
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



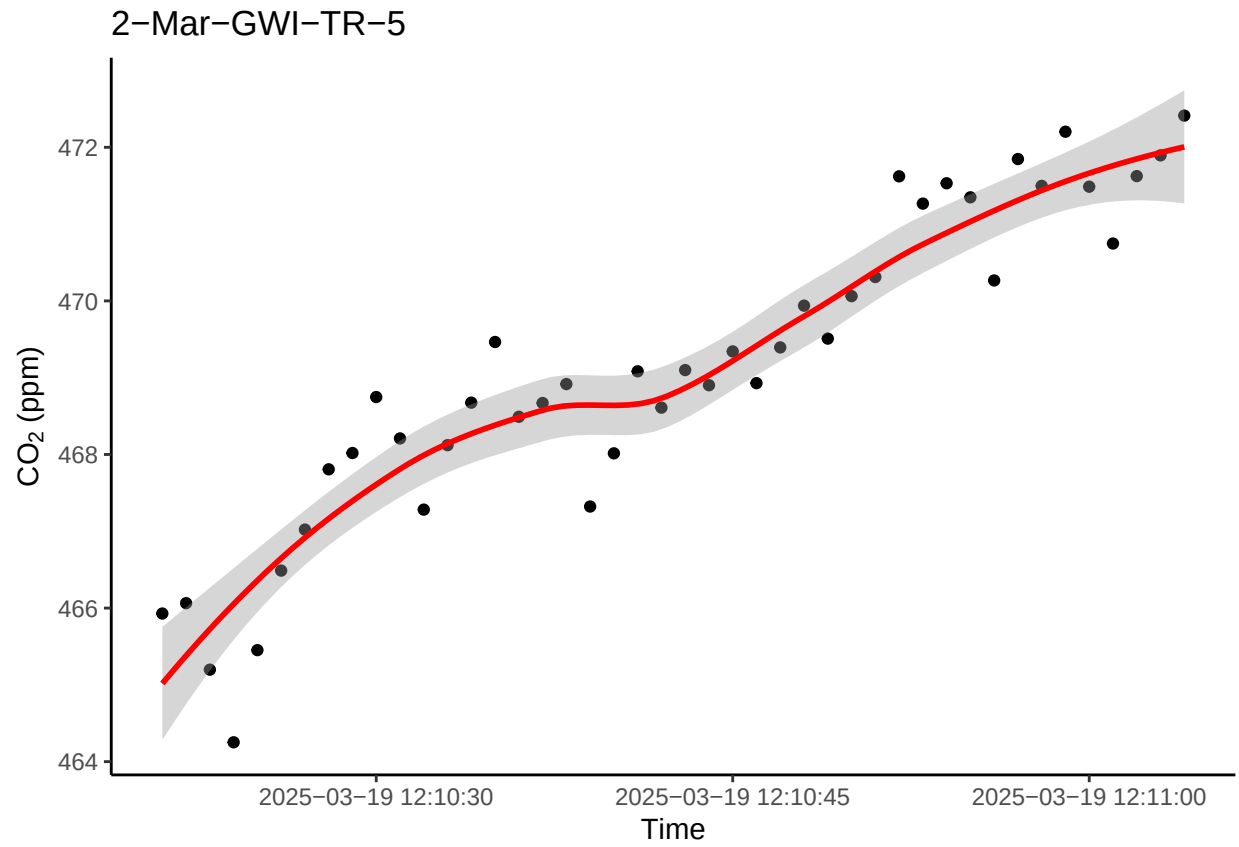
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



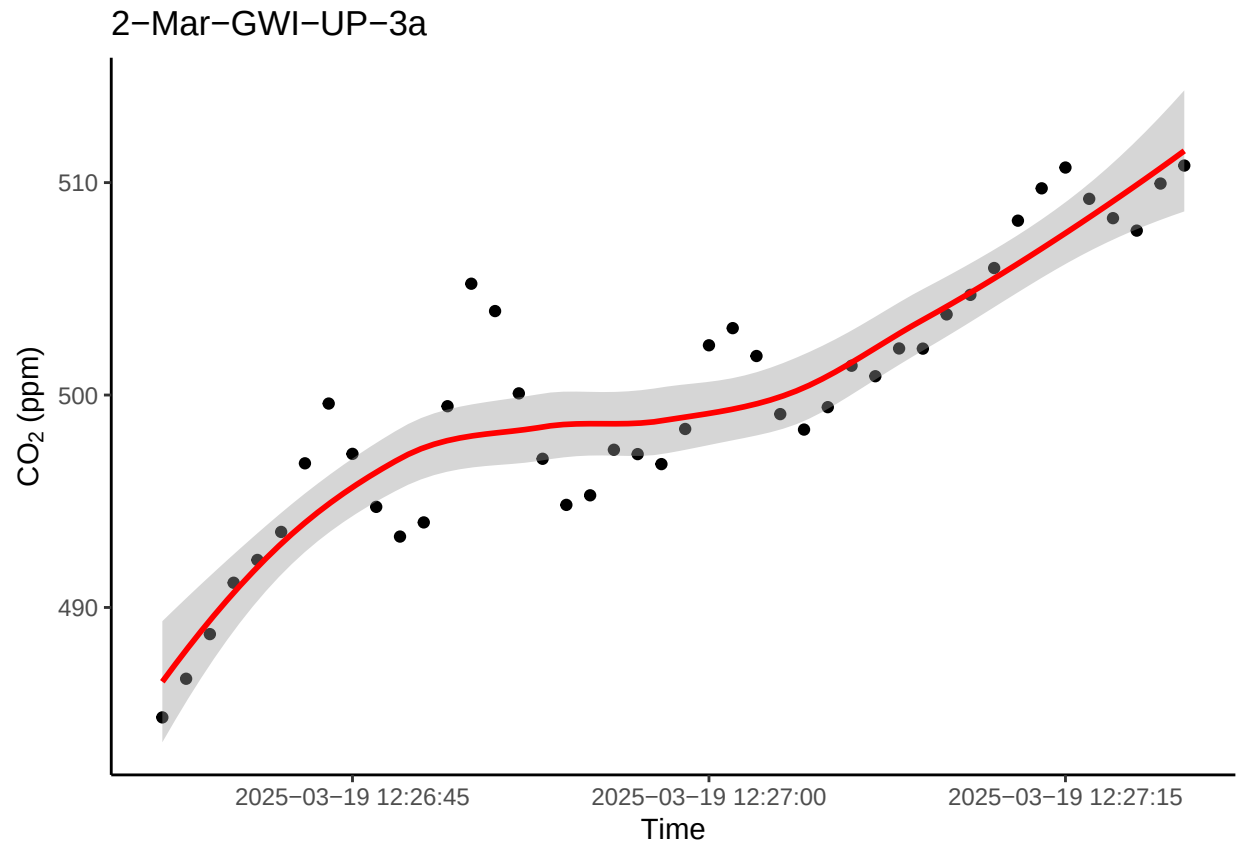
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



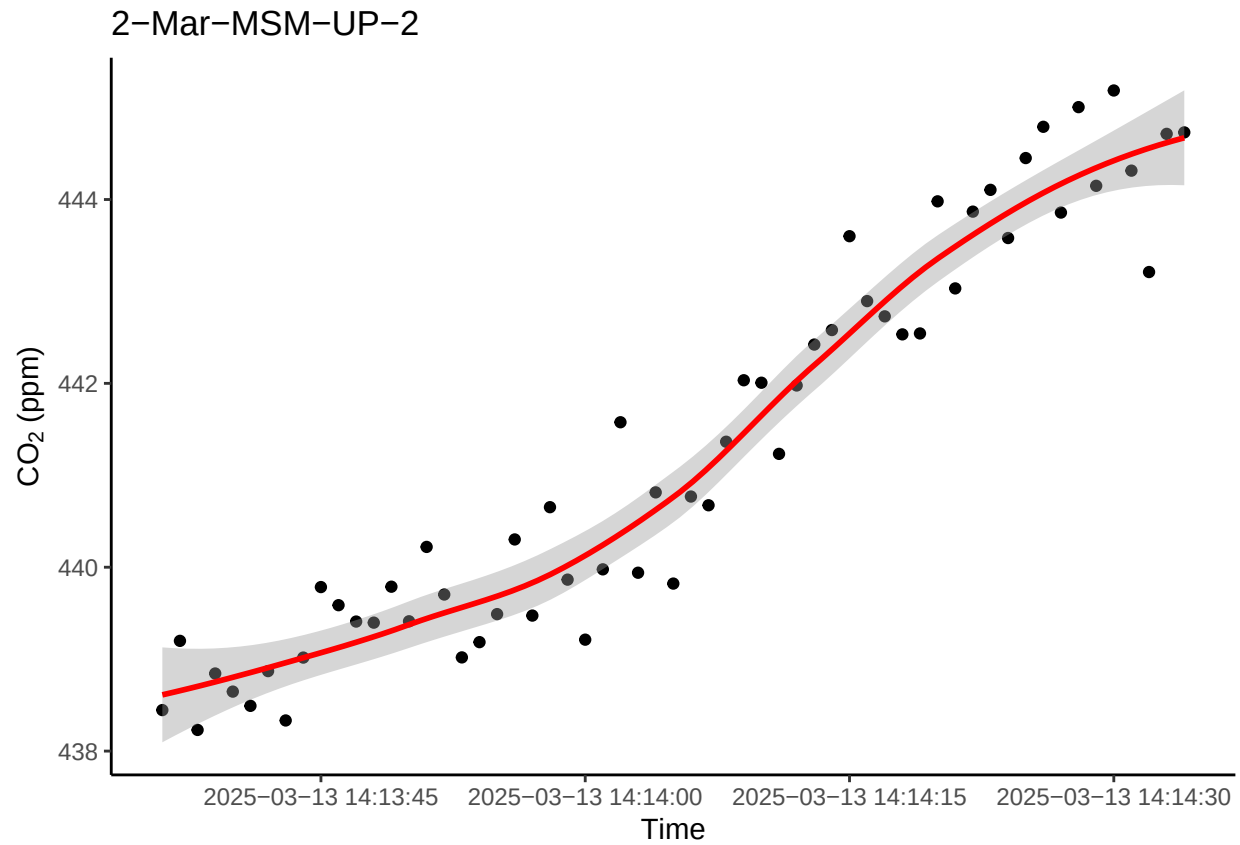
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

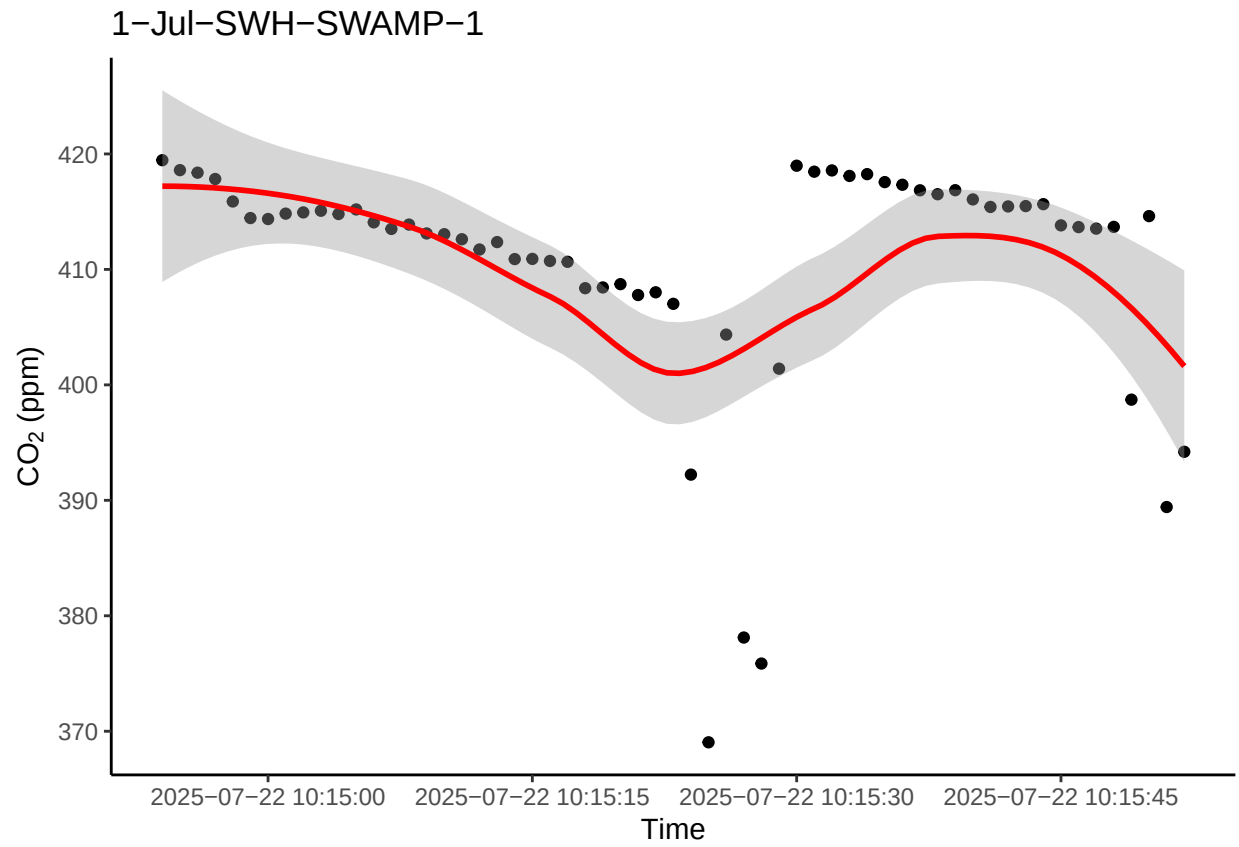


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

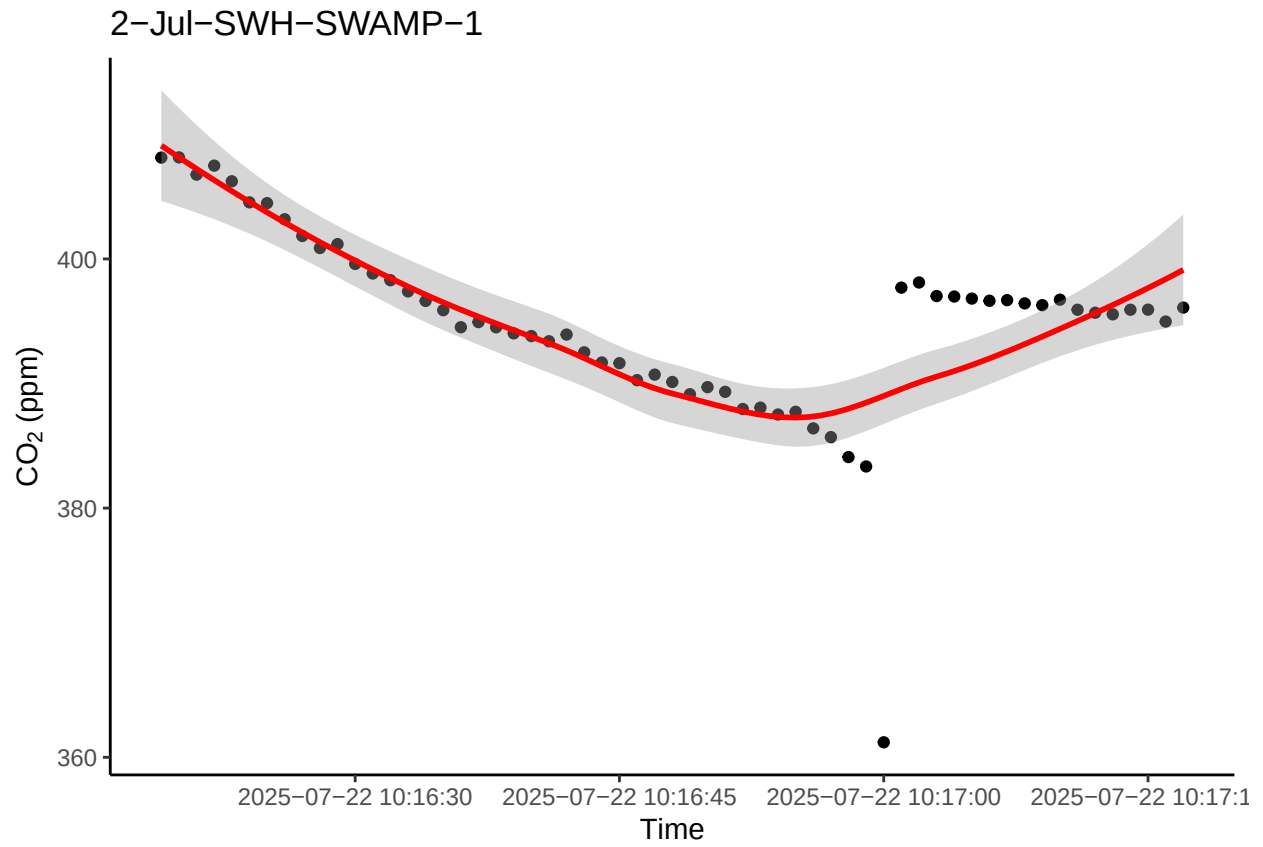


2 Look at any CO₂ fluxes that are negative and remove those plots

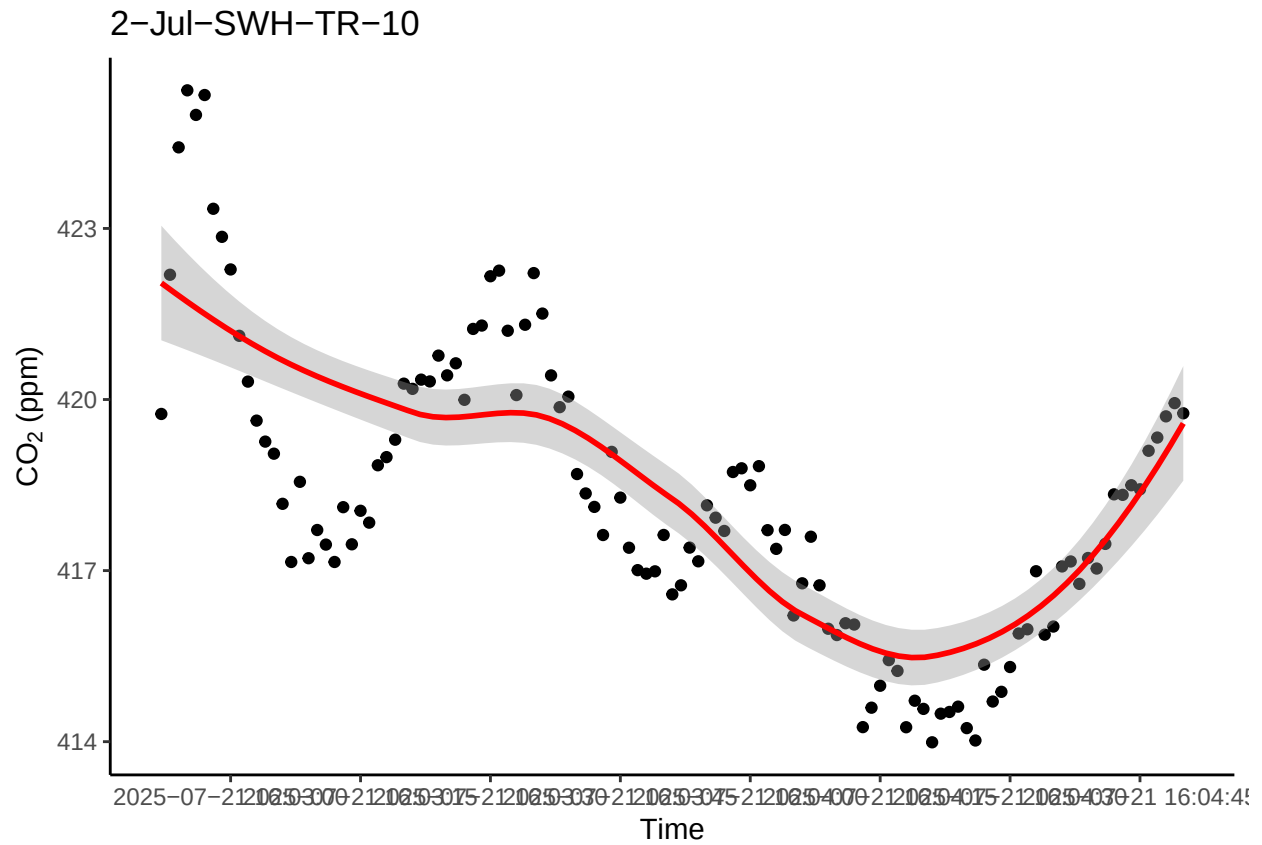
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

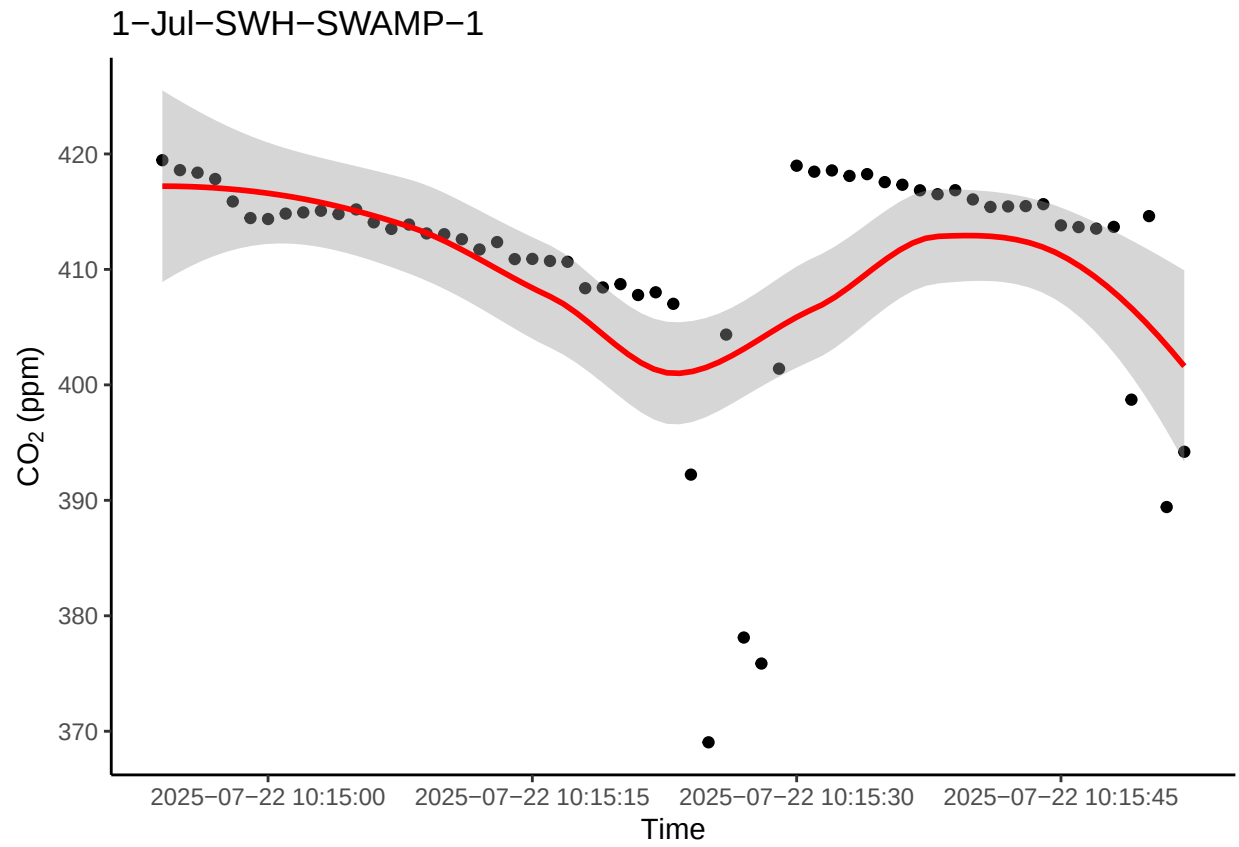


```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```

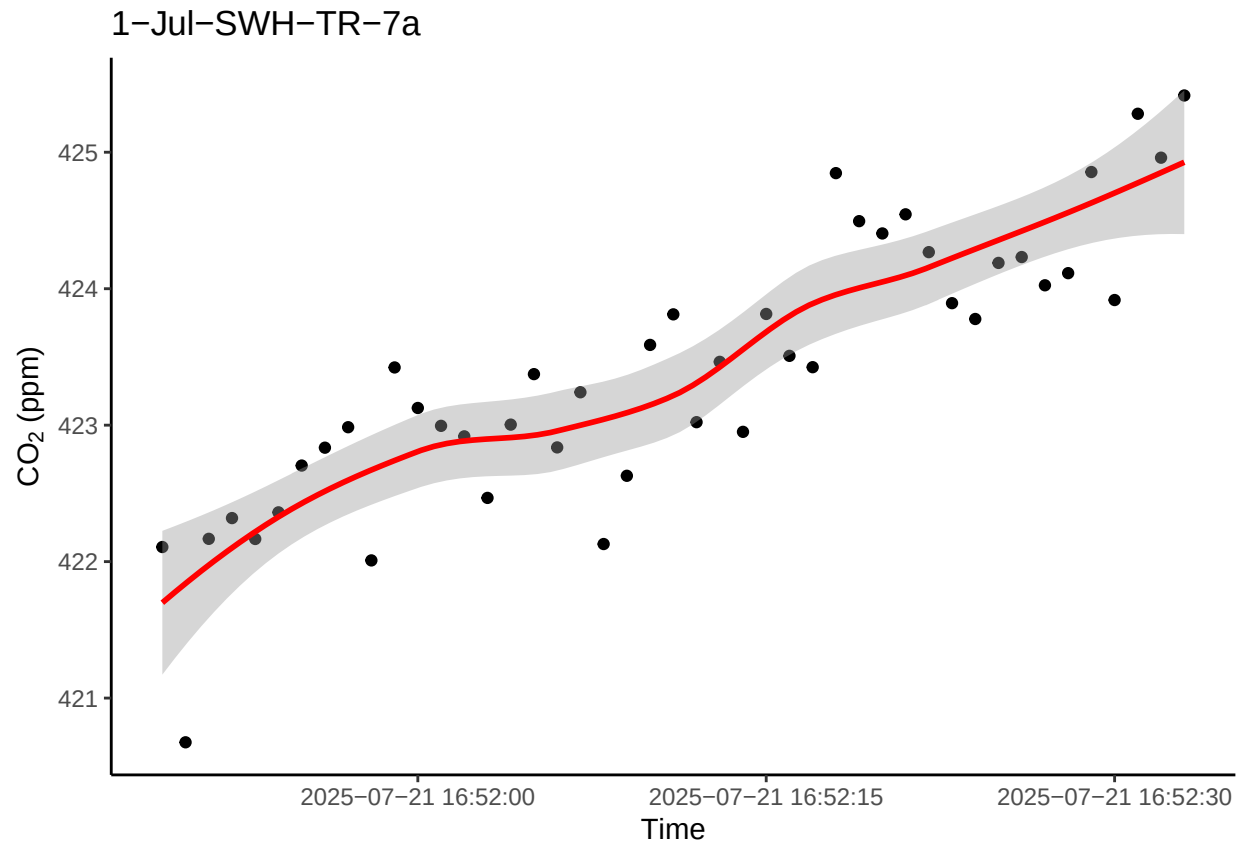


2.1 Look at the really high or really low ones to see if there is a timing issue or something

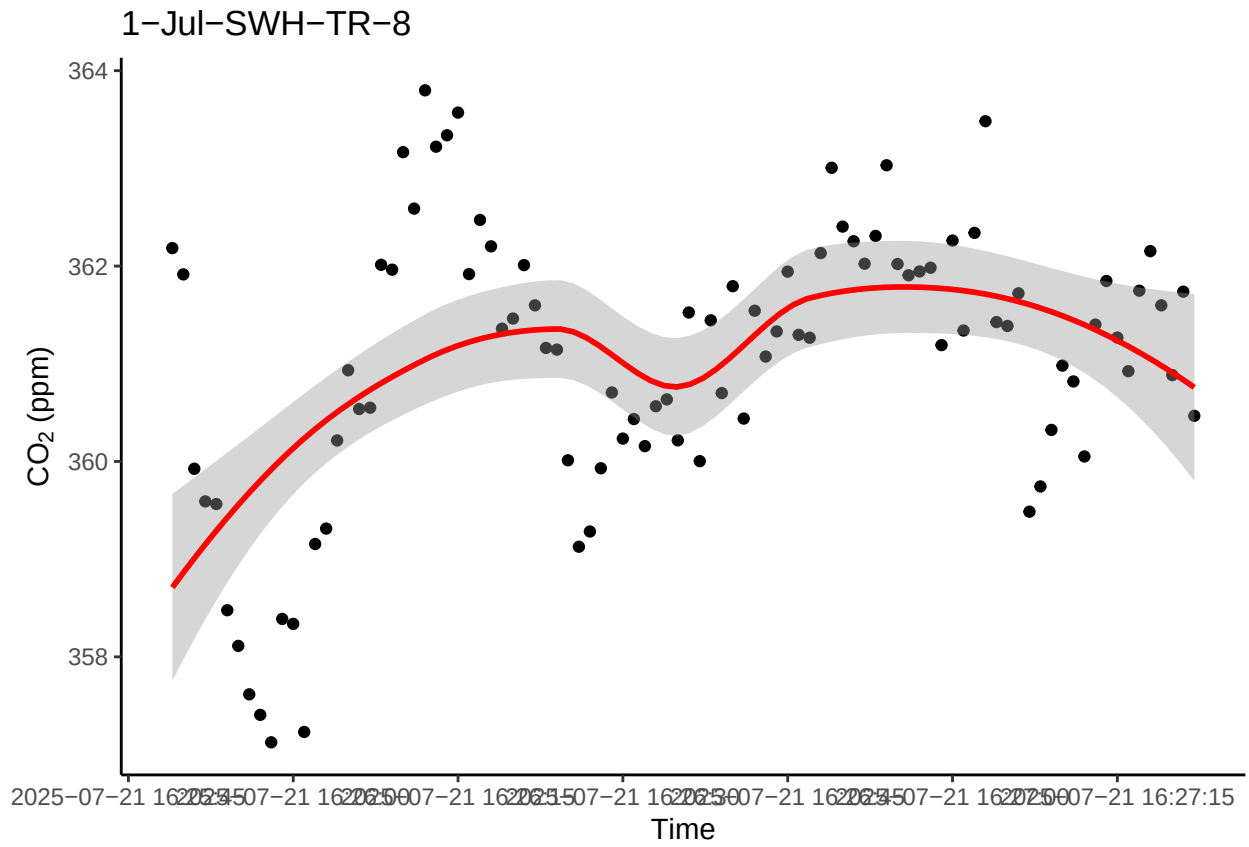
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



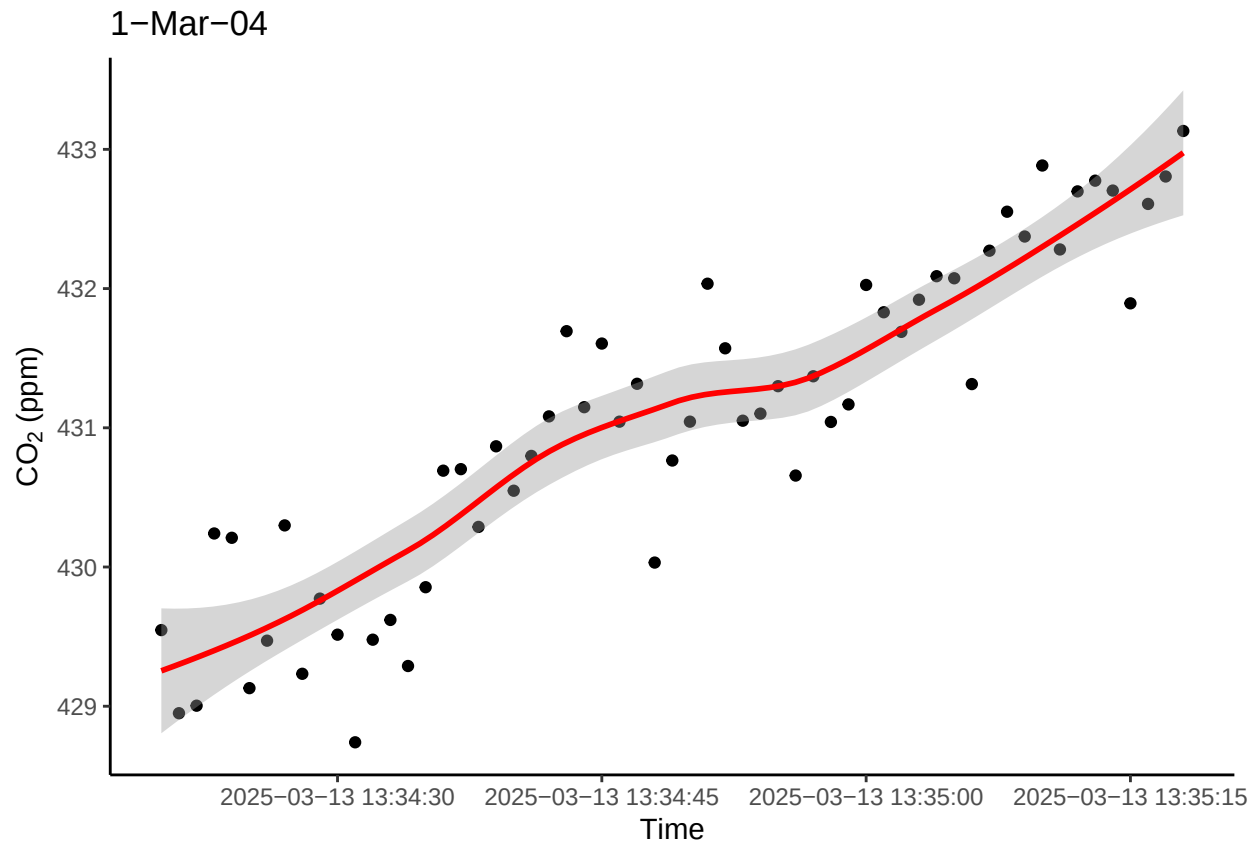
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



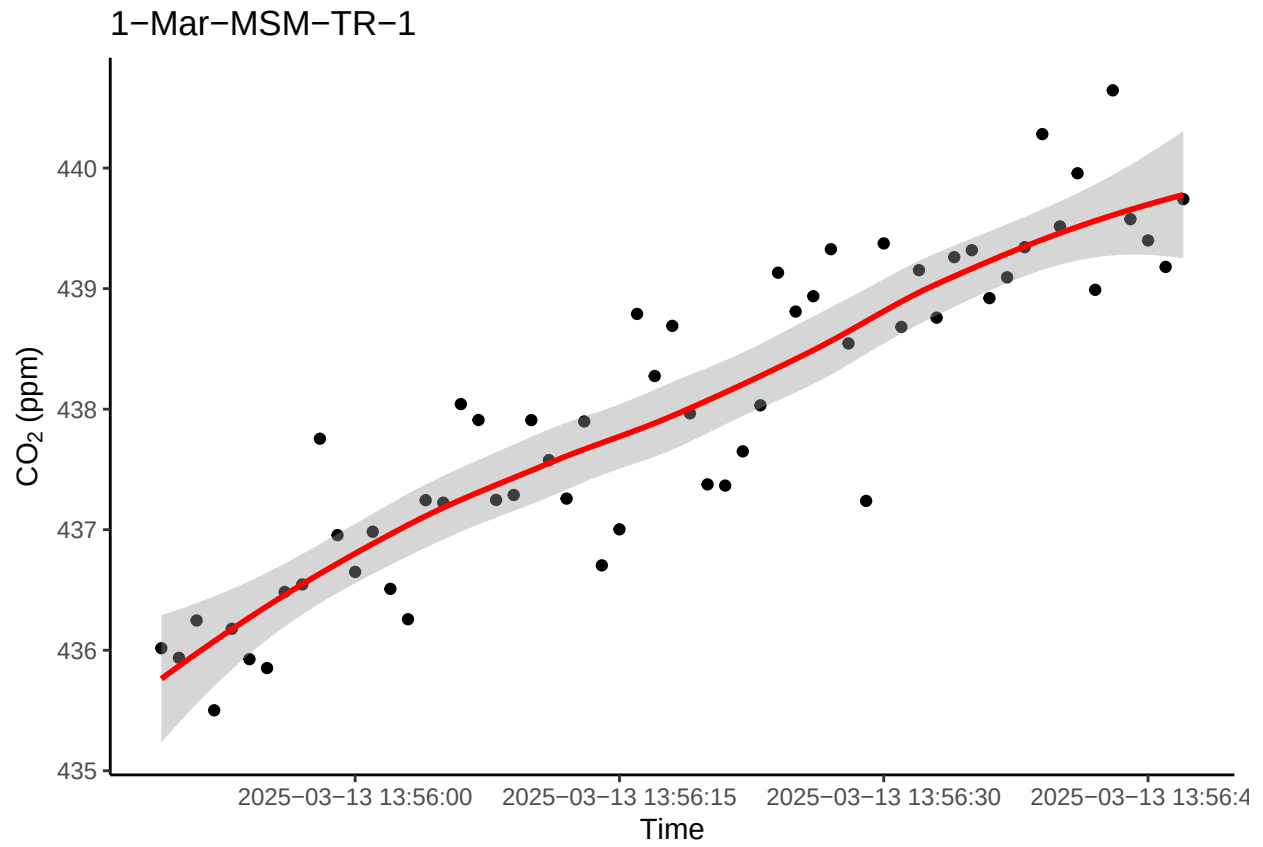
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



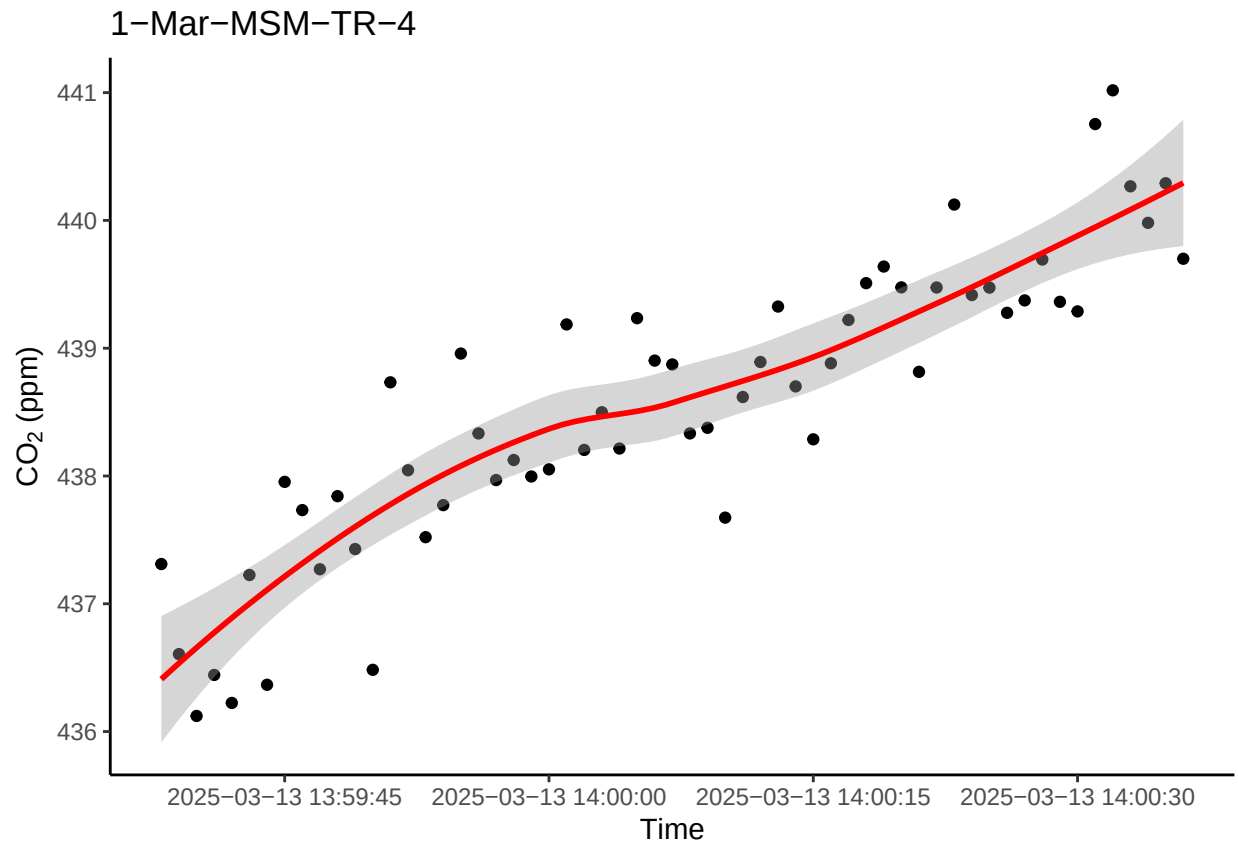
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



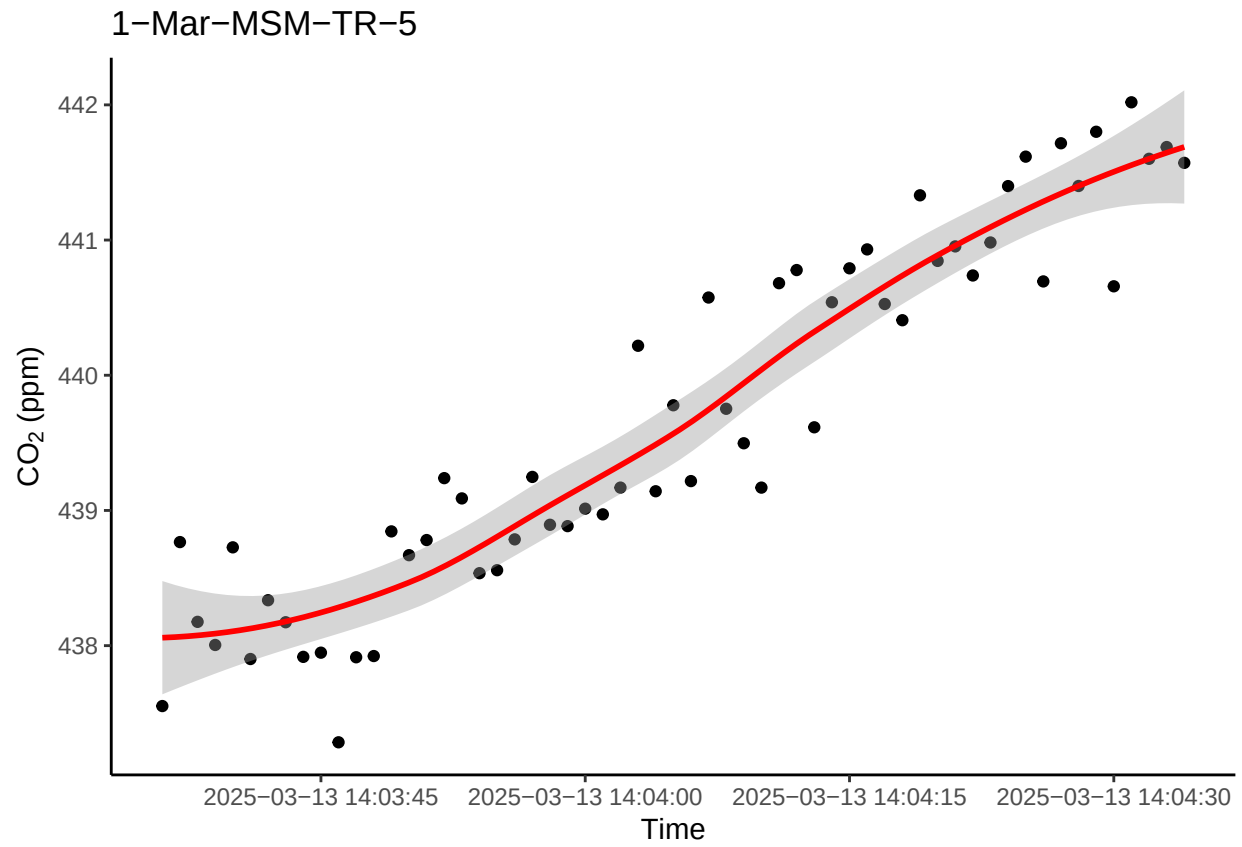
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



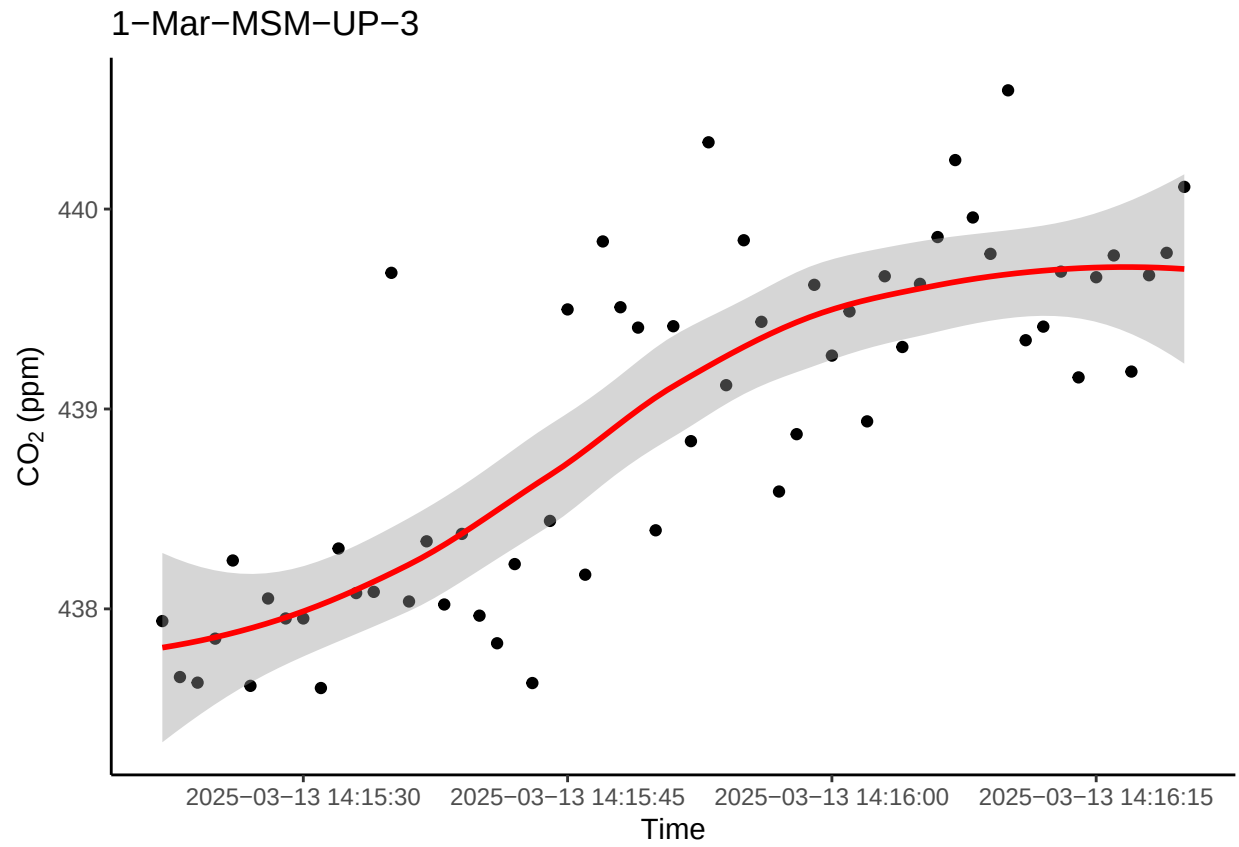
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



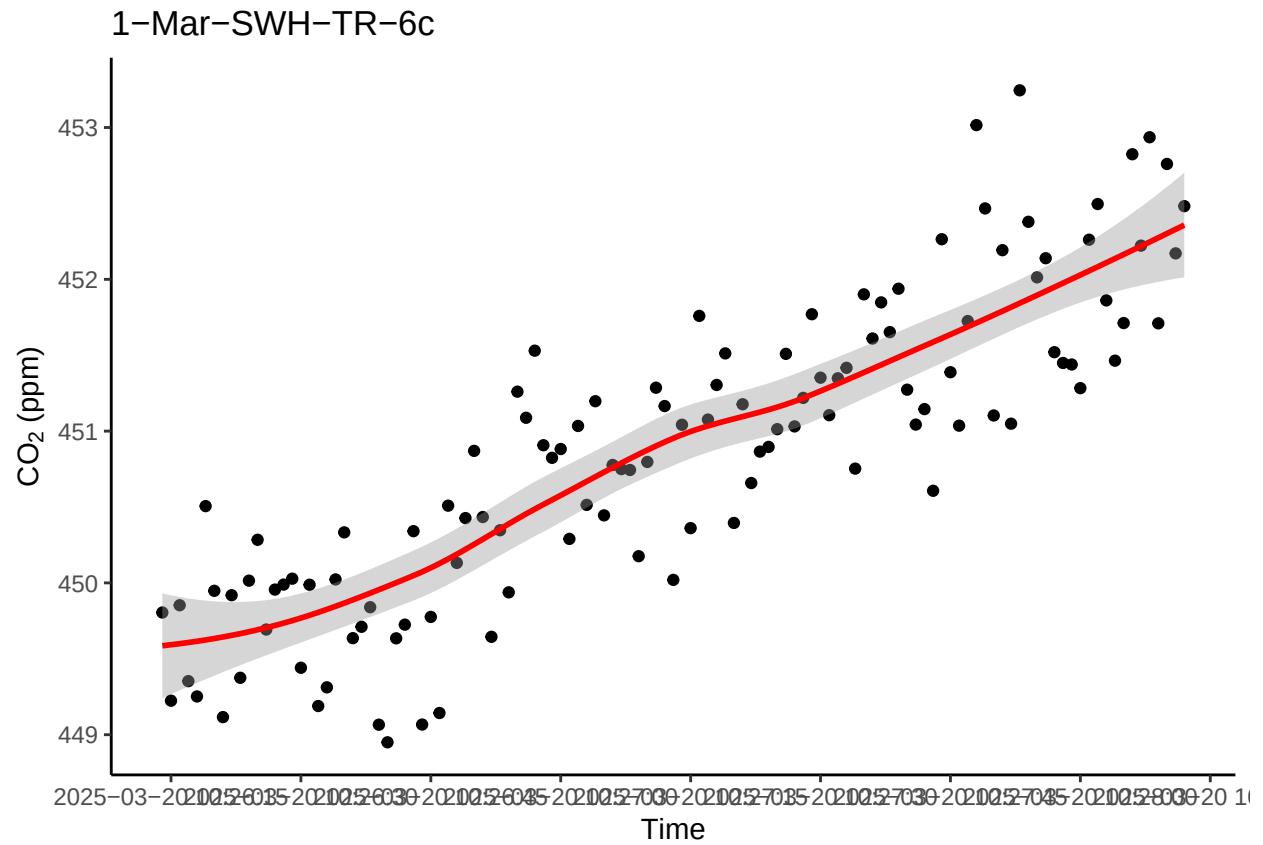
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



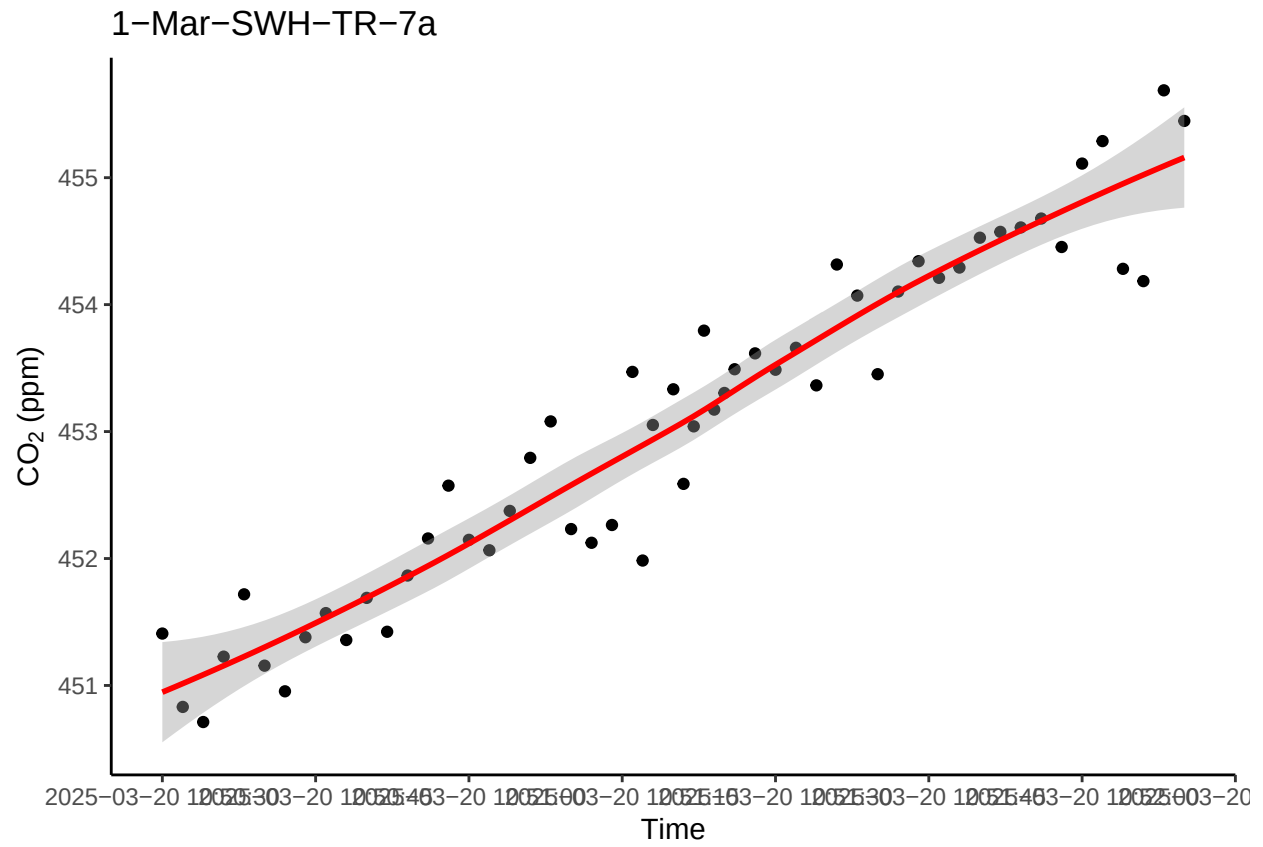
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



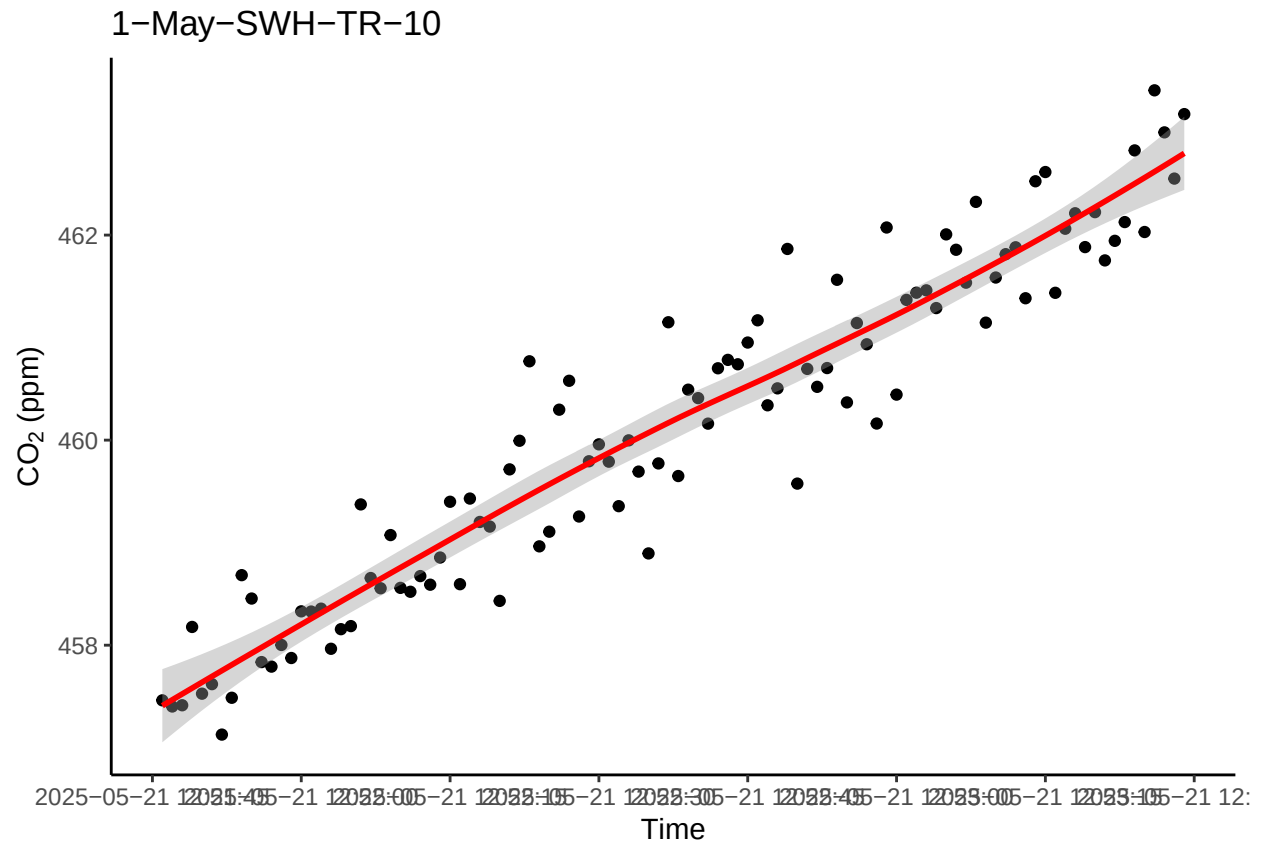
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



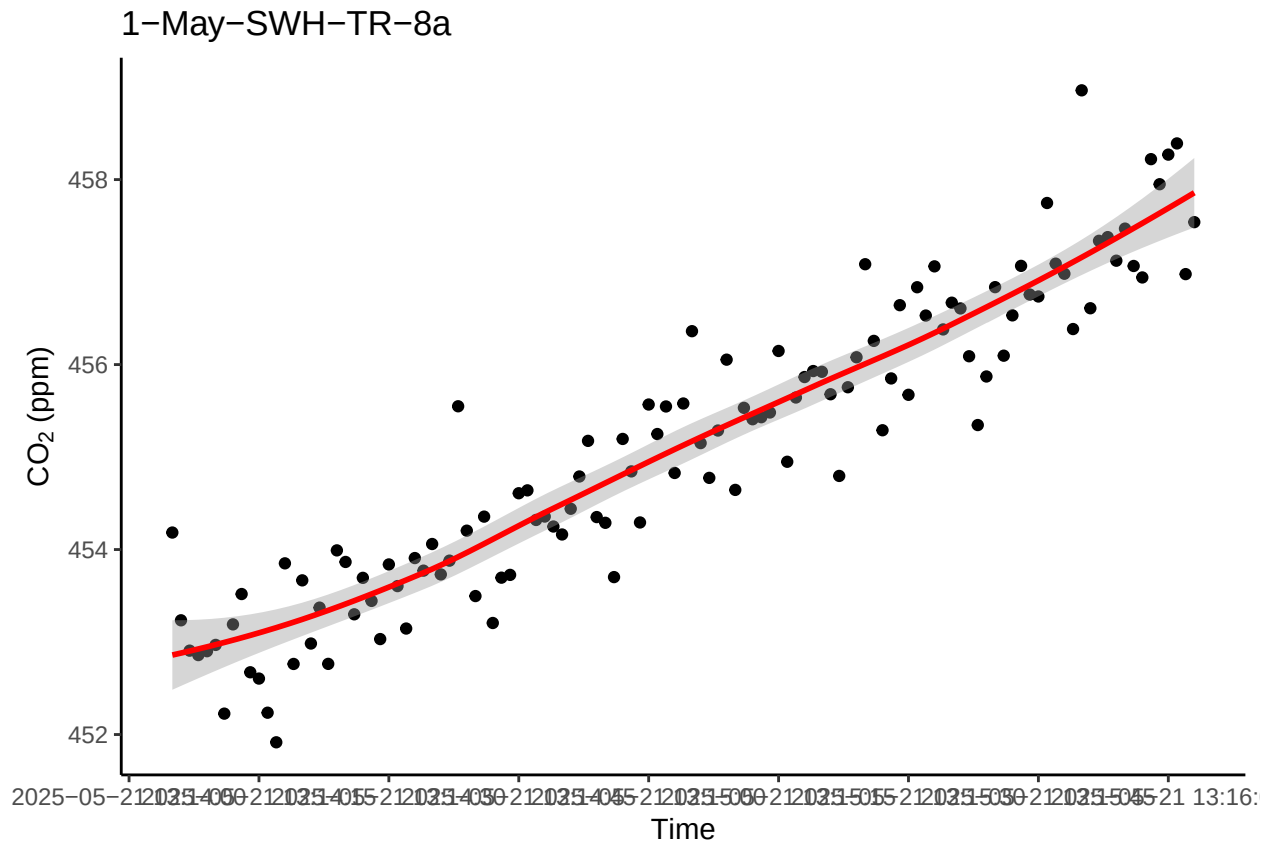
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



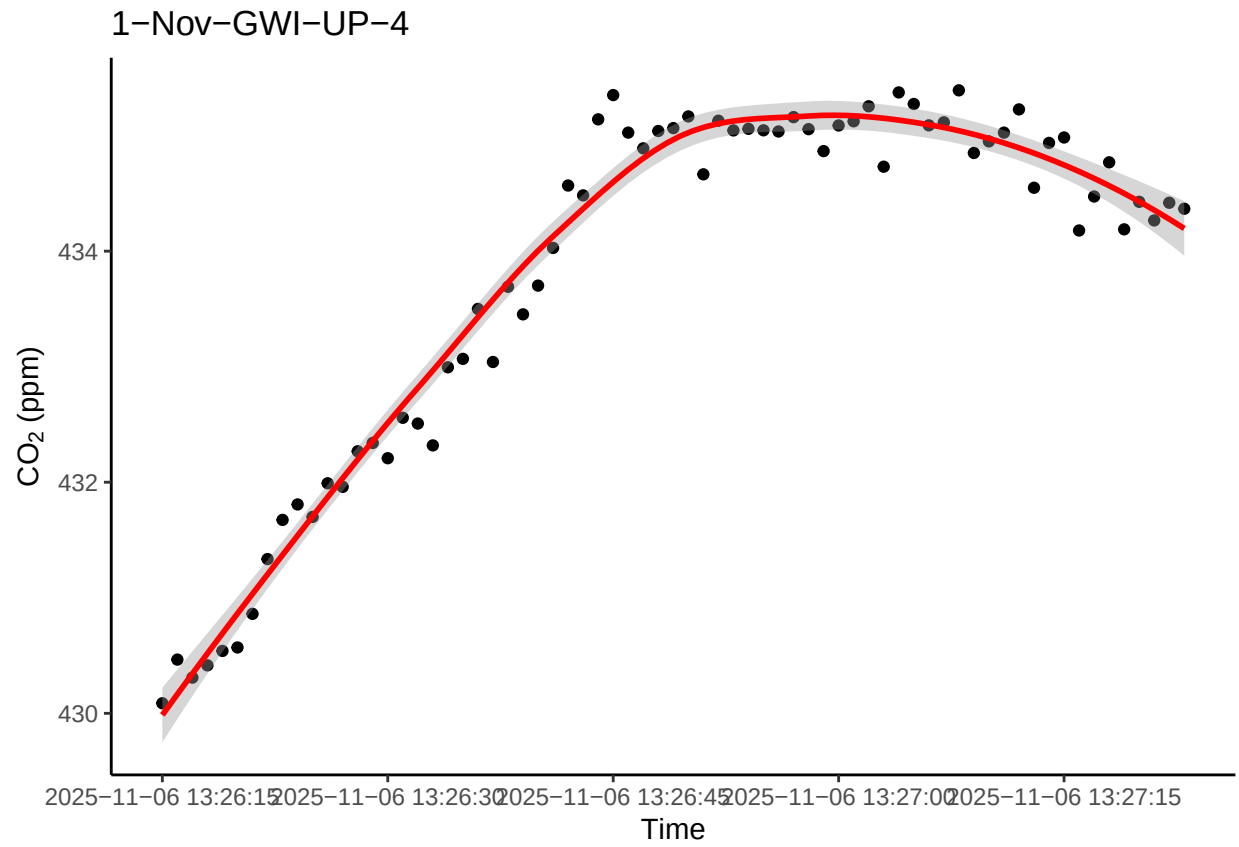
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



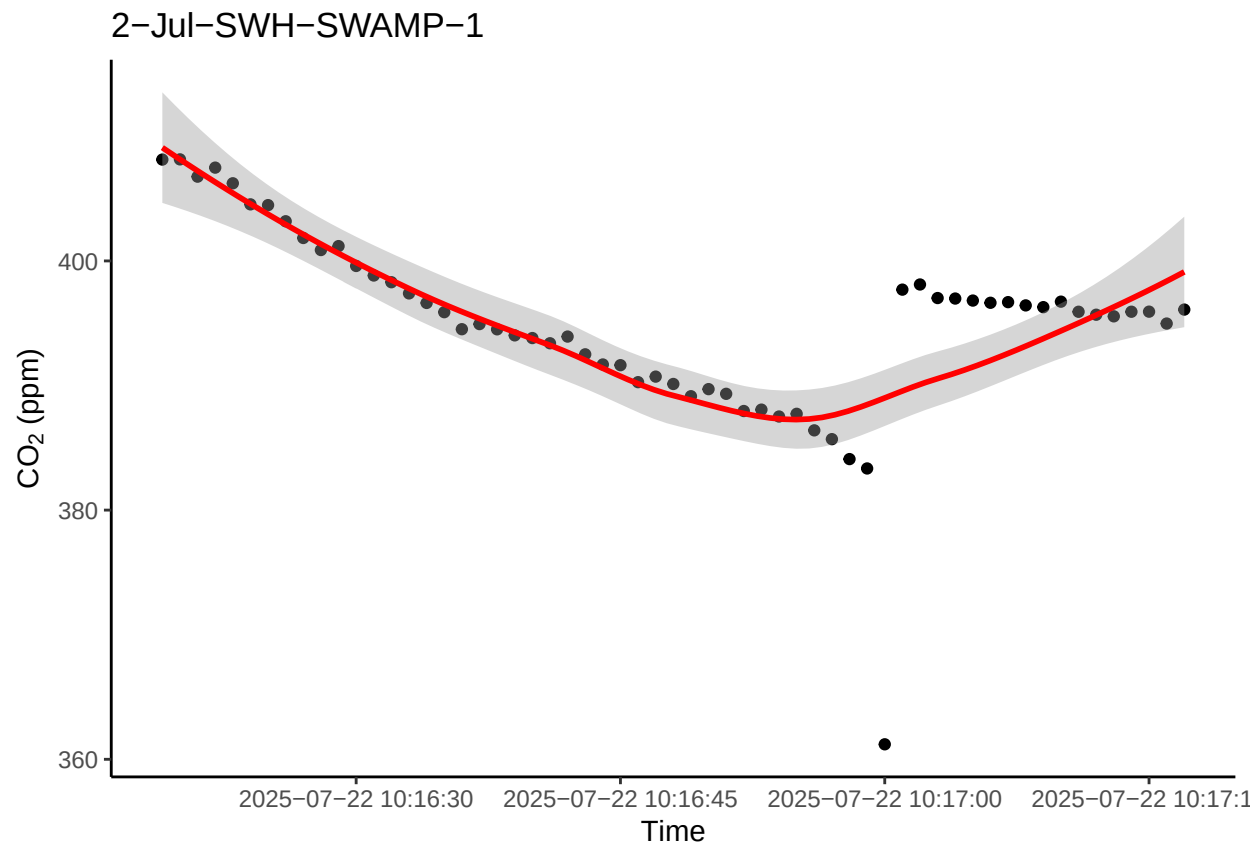
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



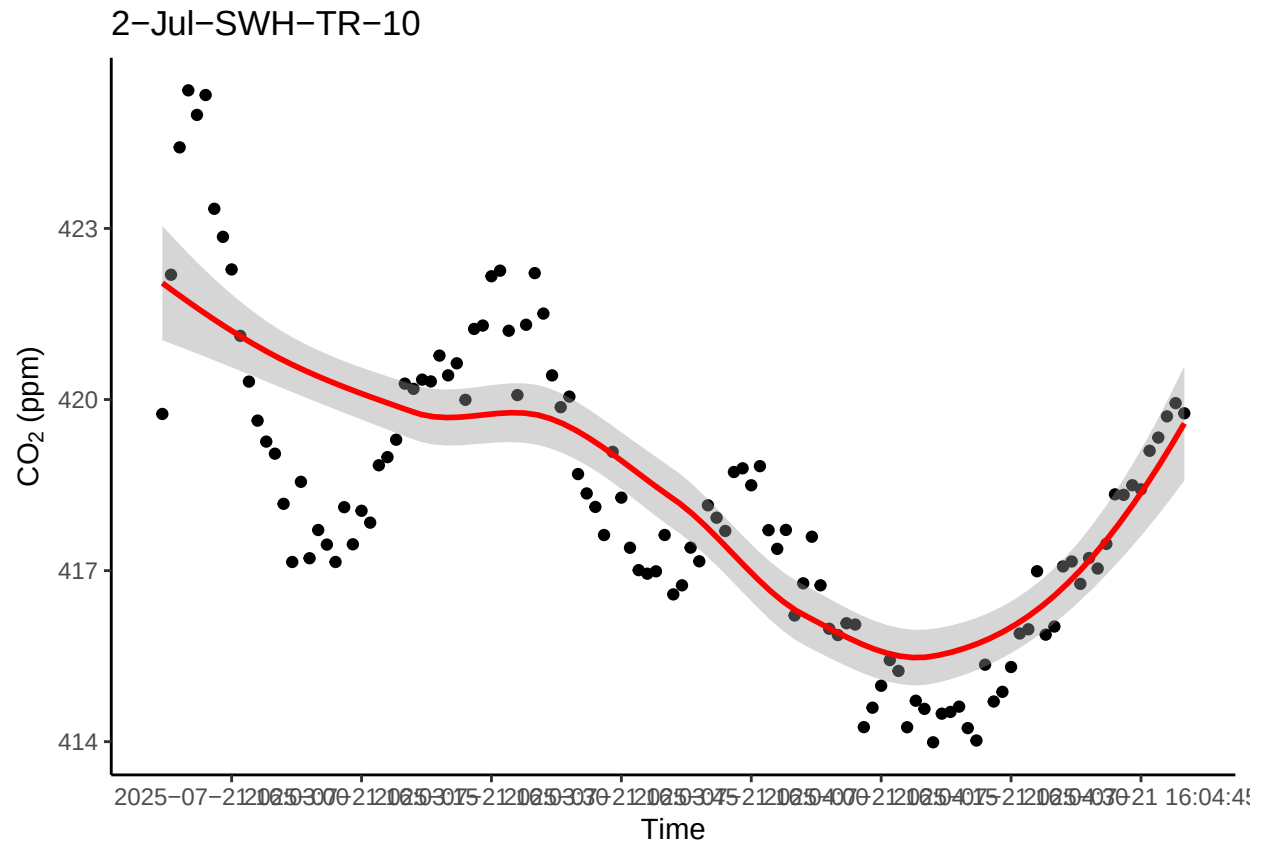
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



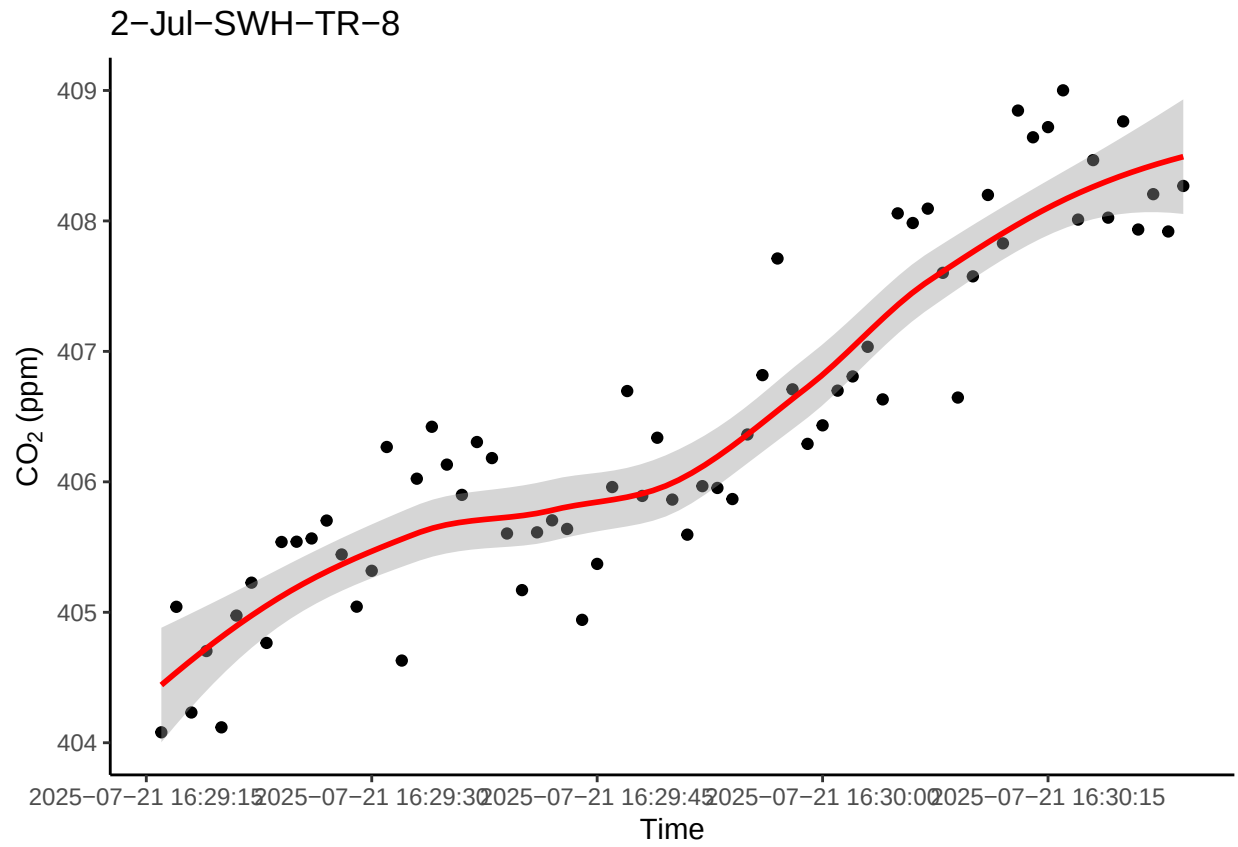
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



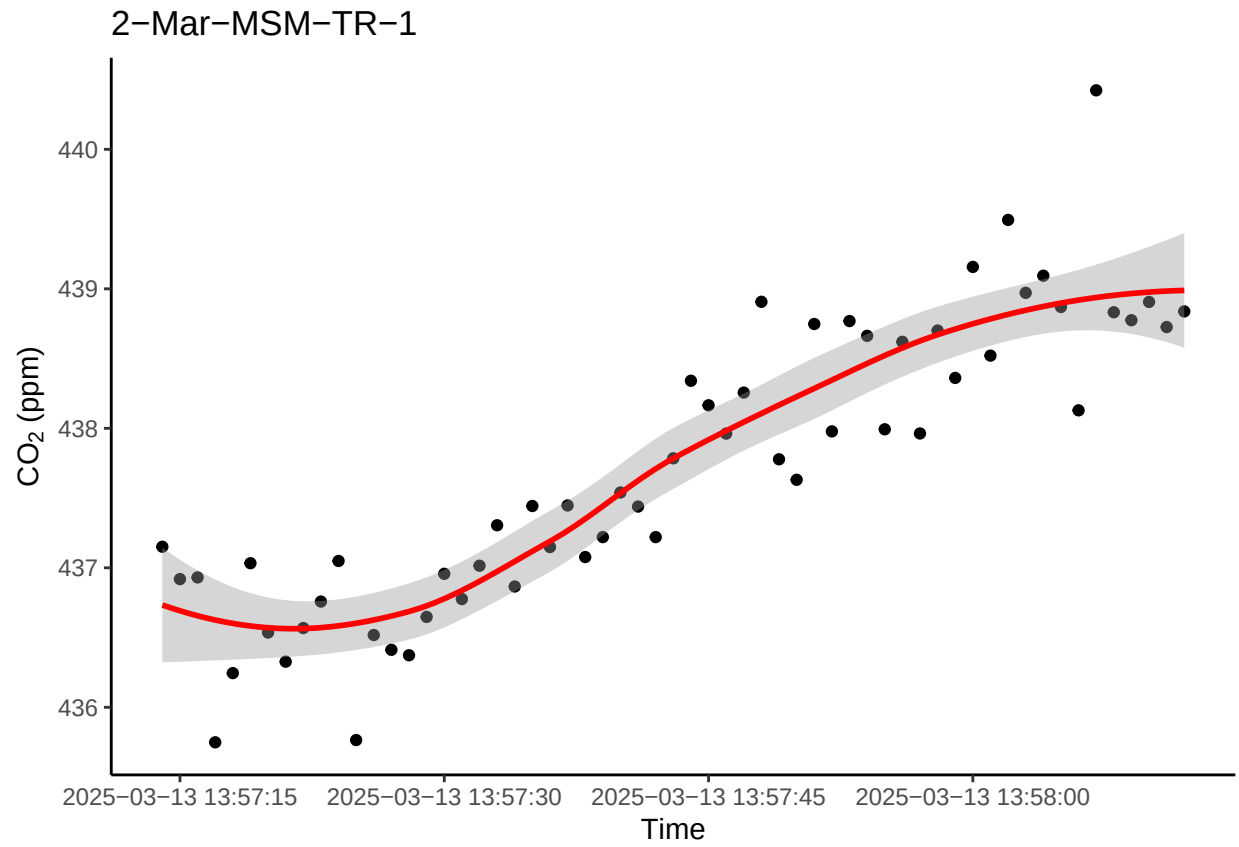
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



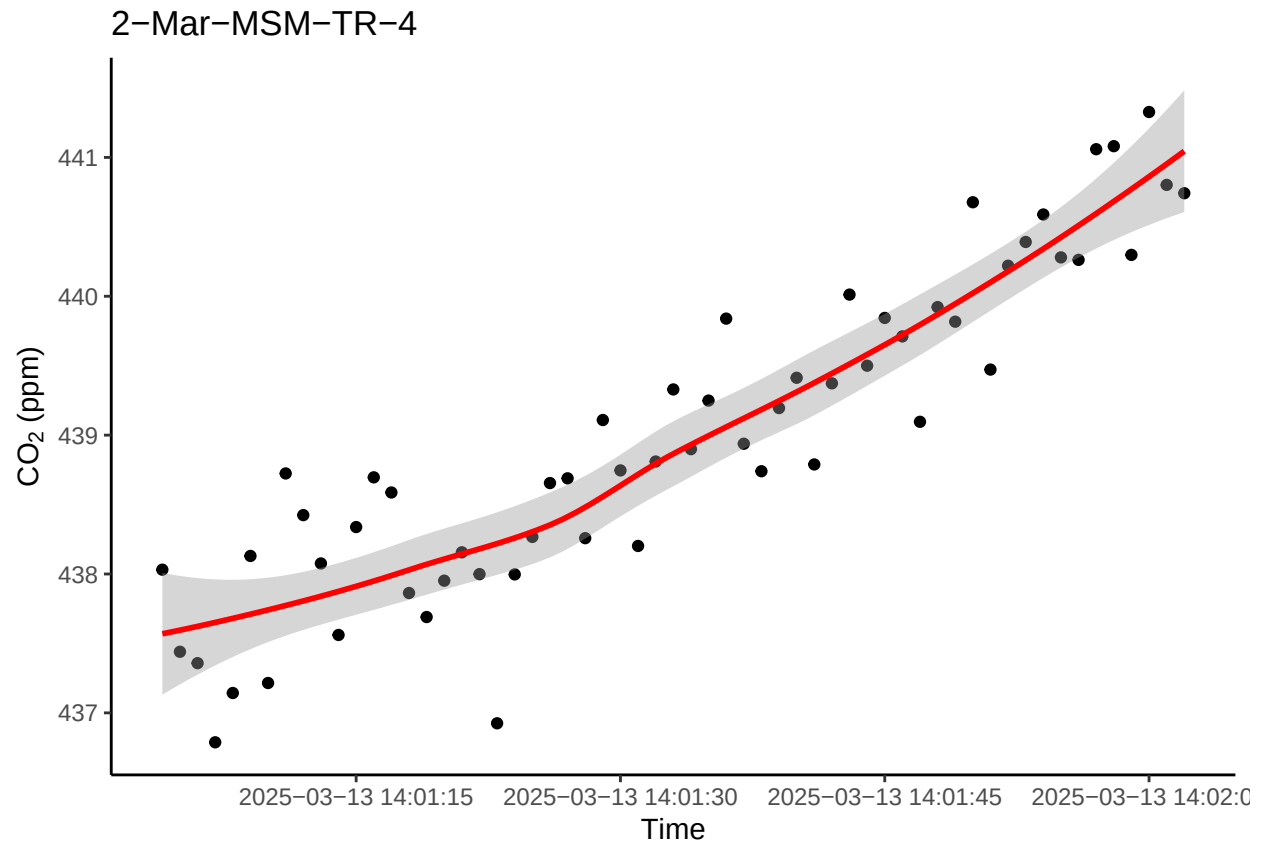
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



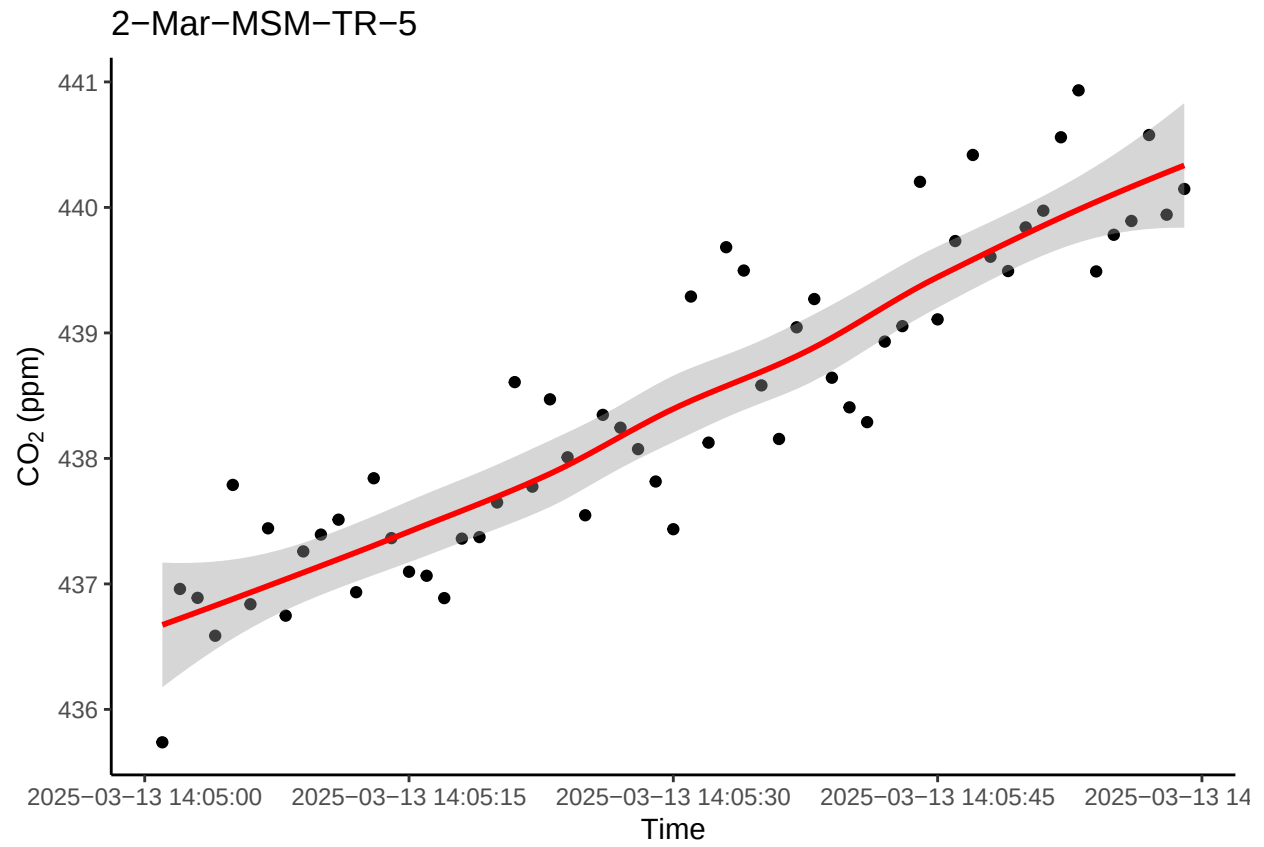
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



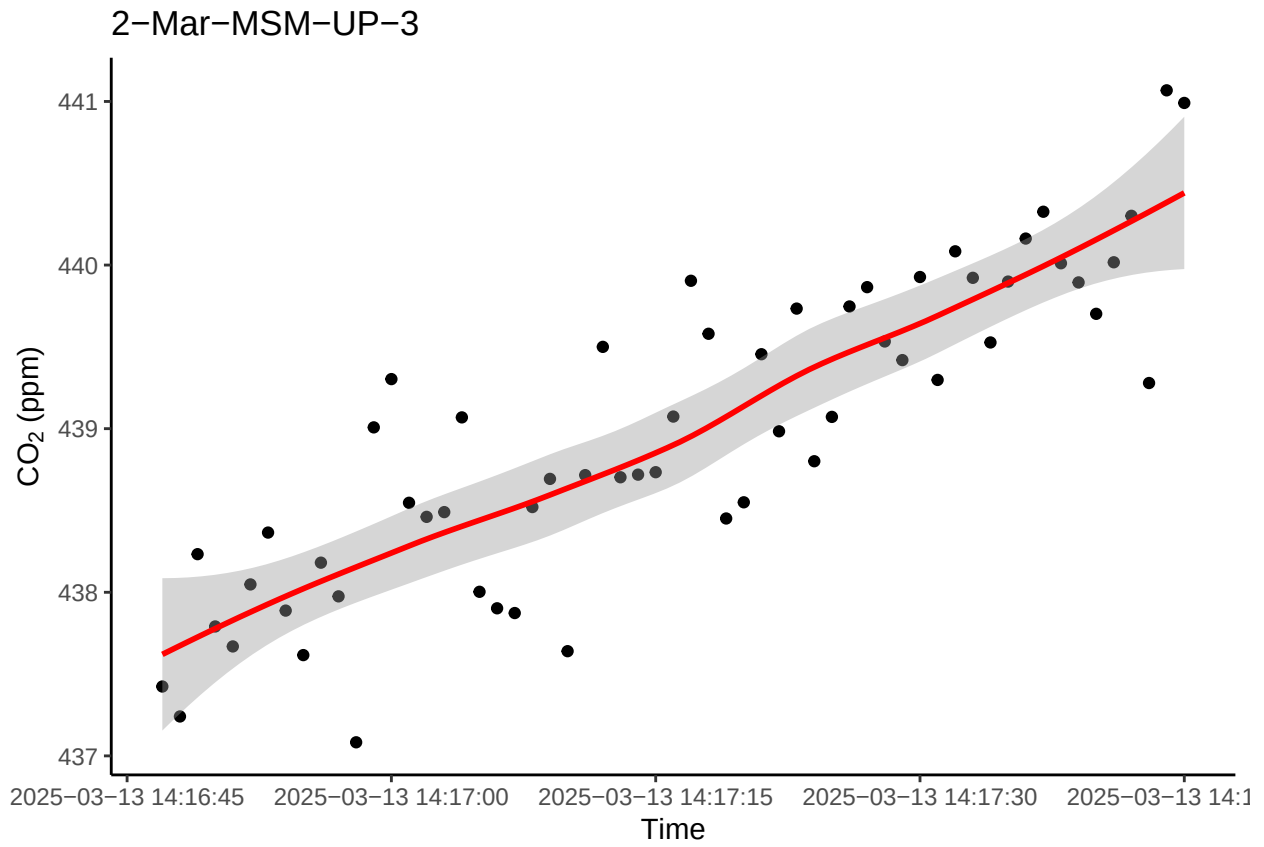
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



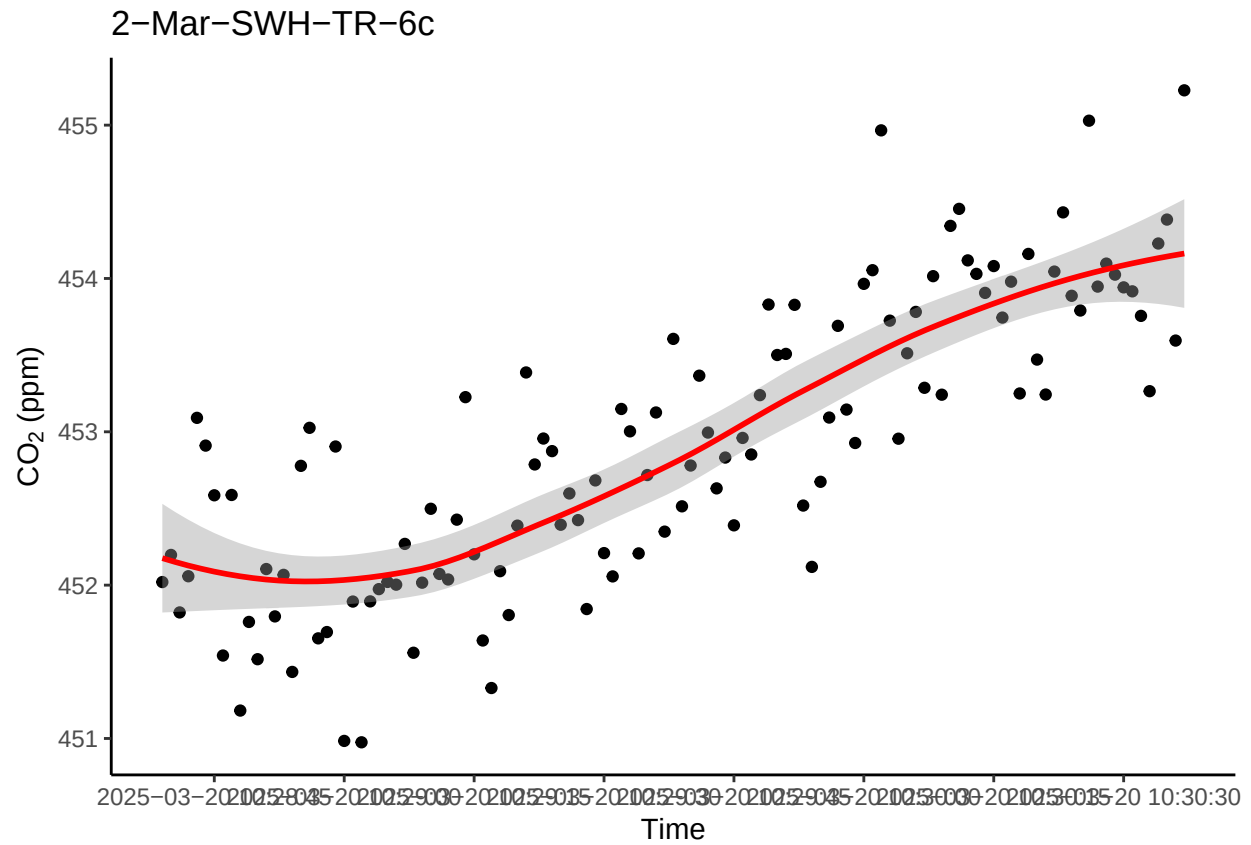
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



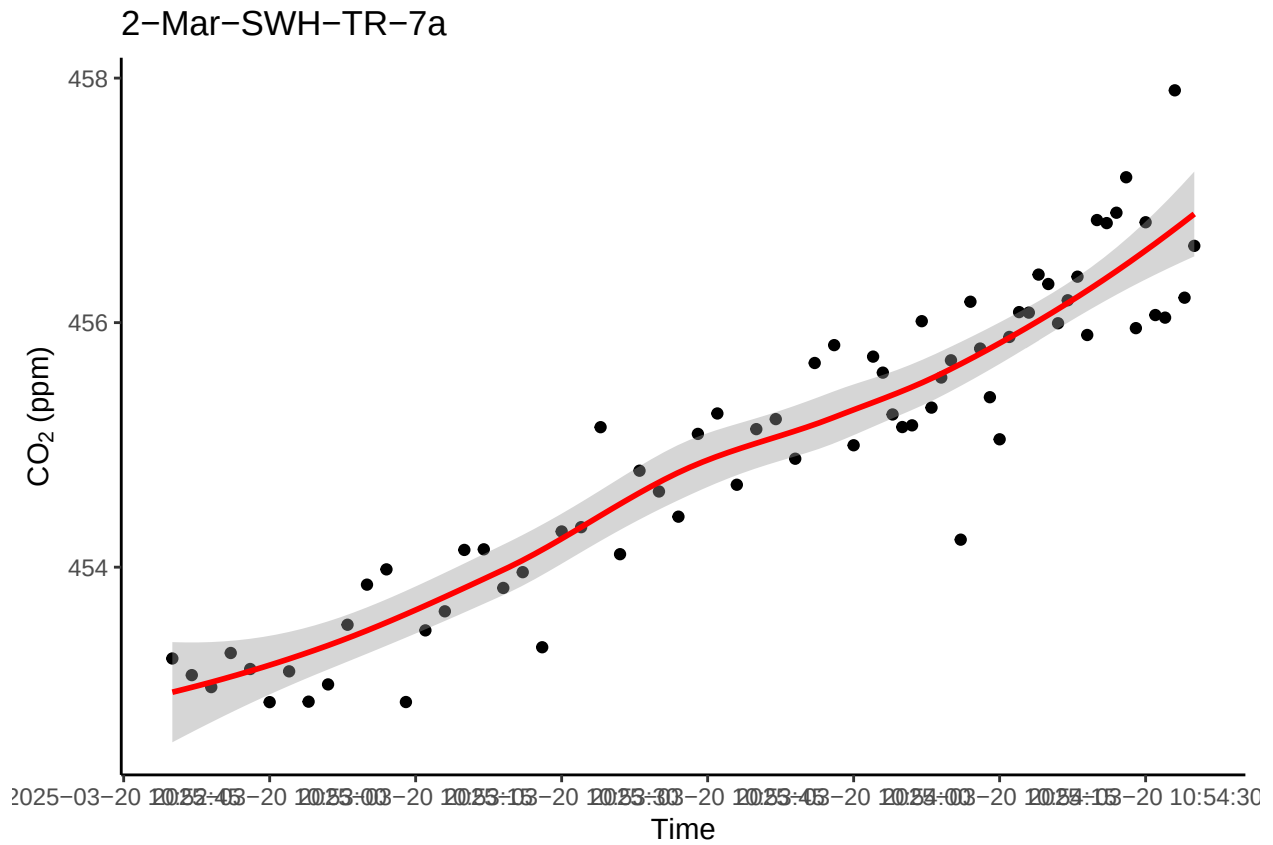
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



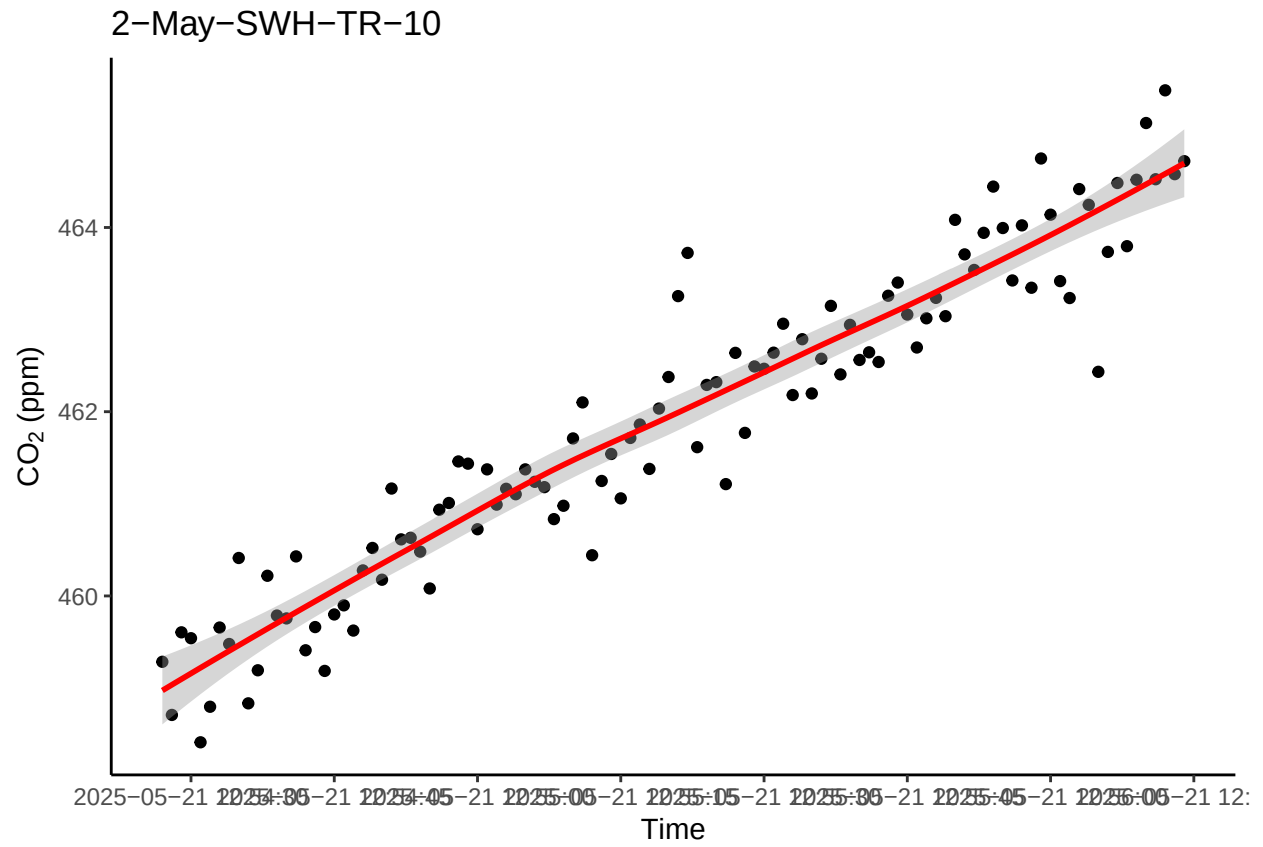
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



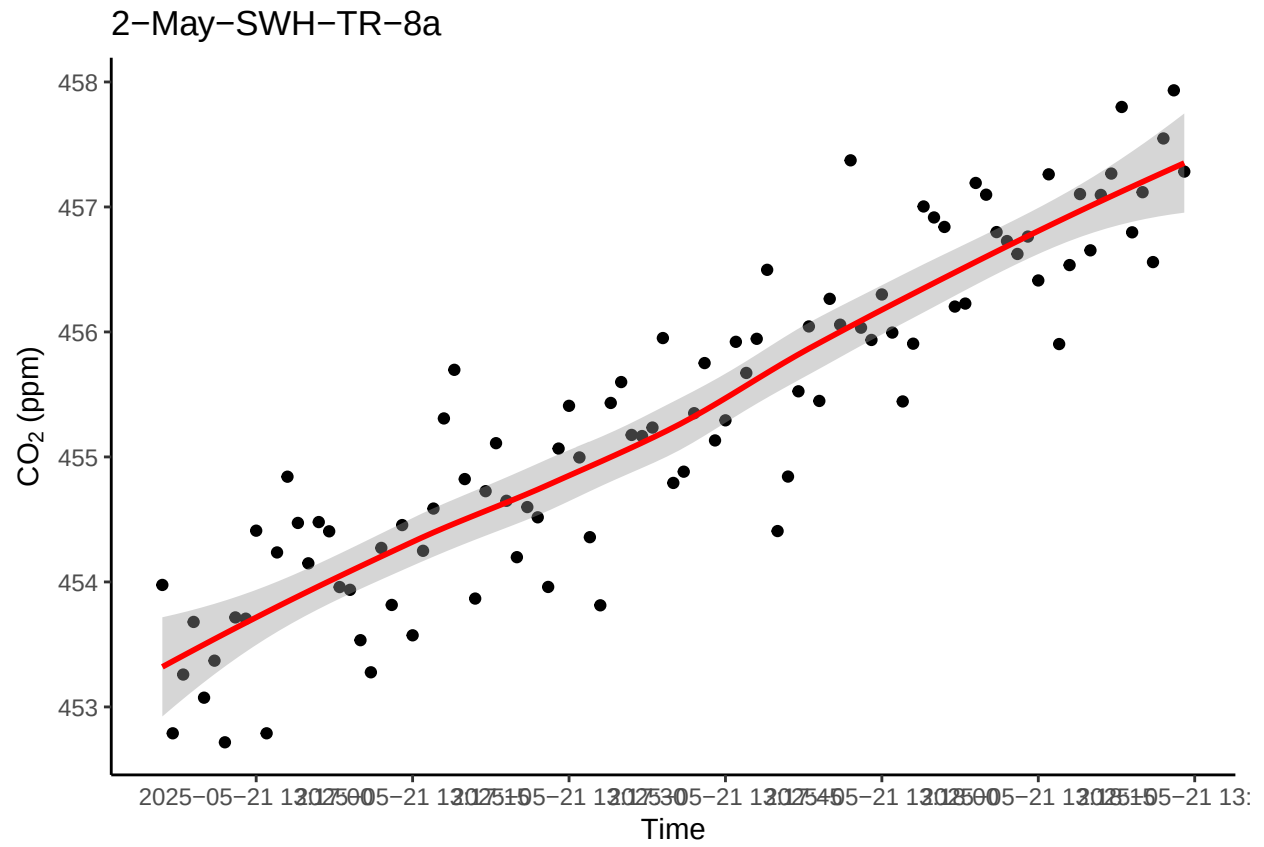
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



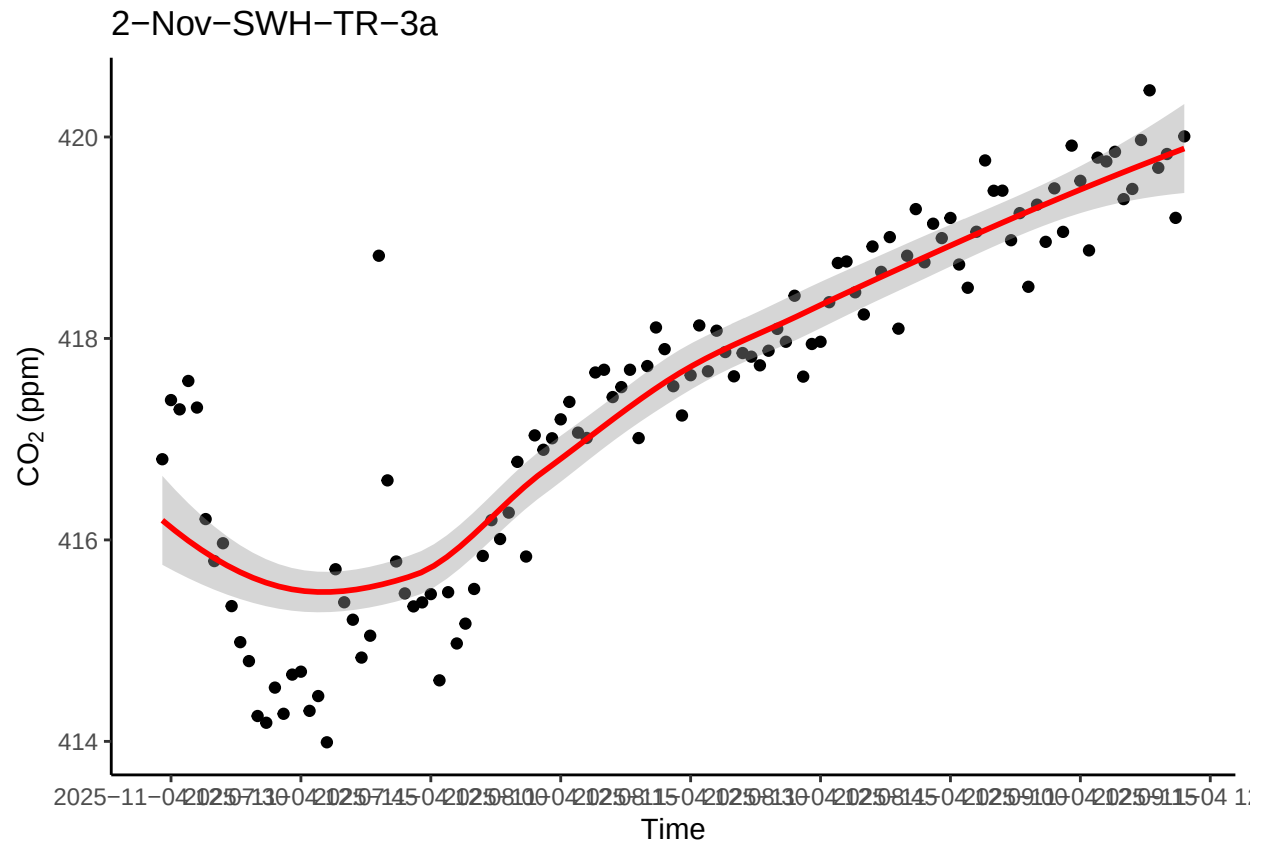
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



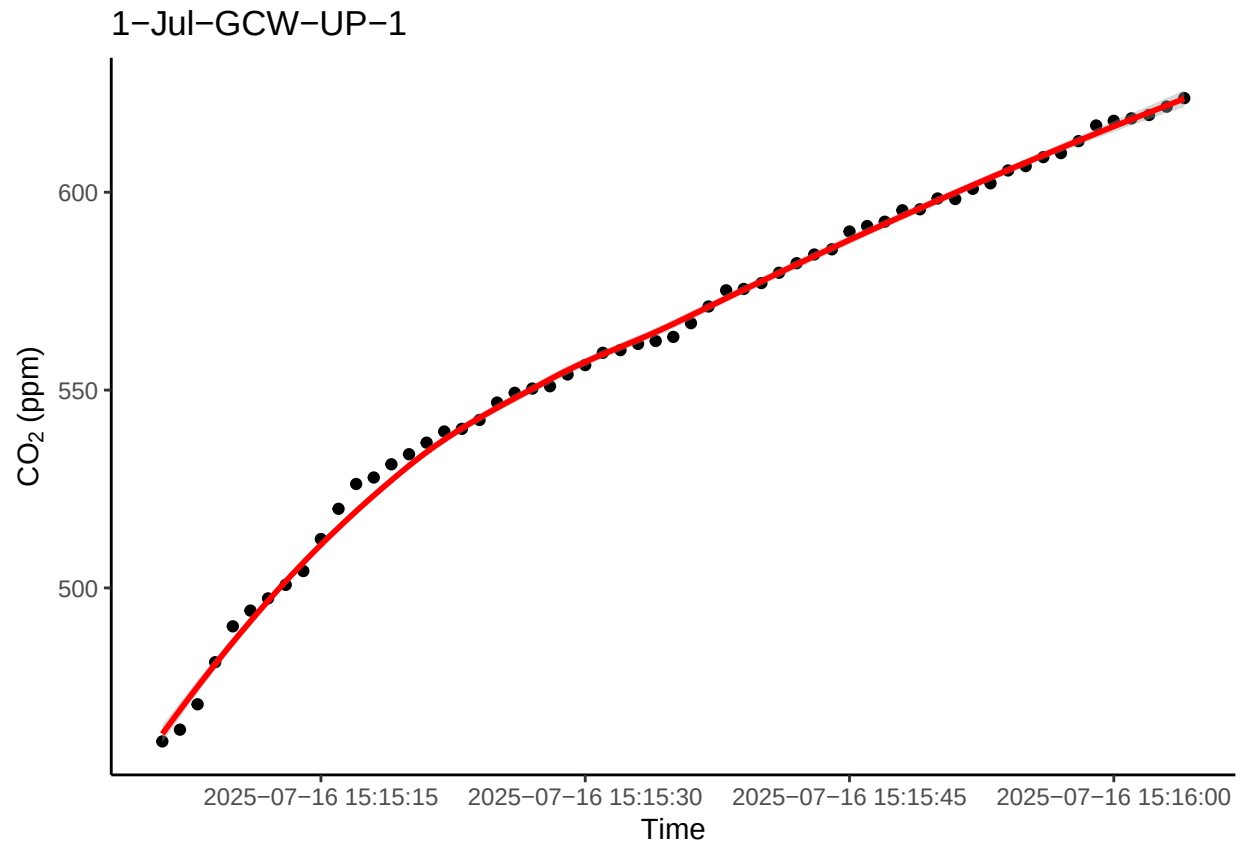
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



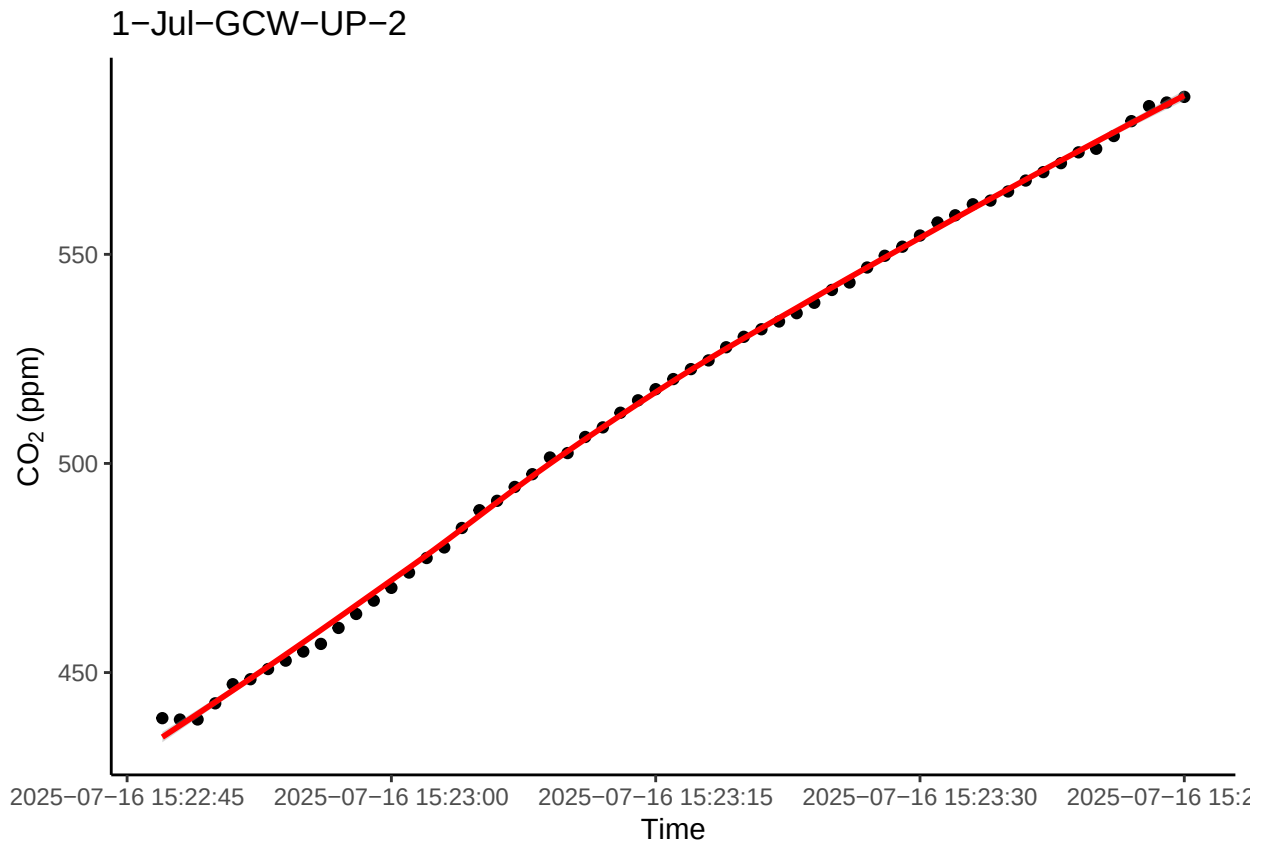
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



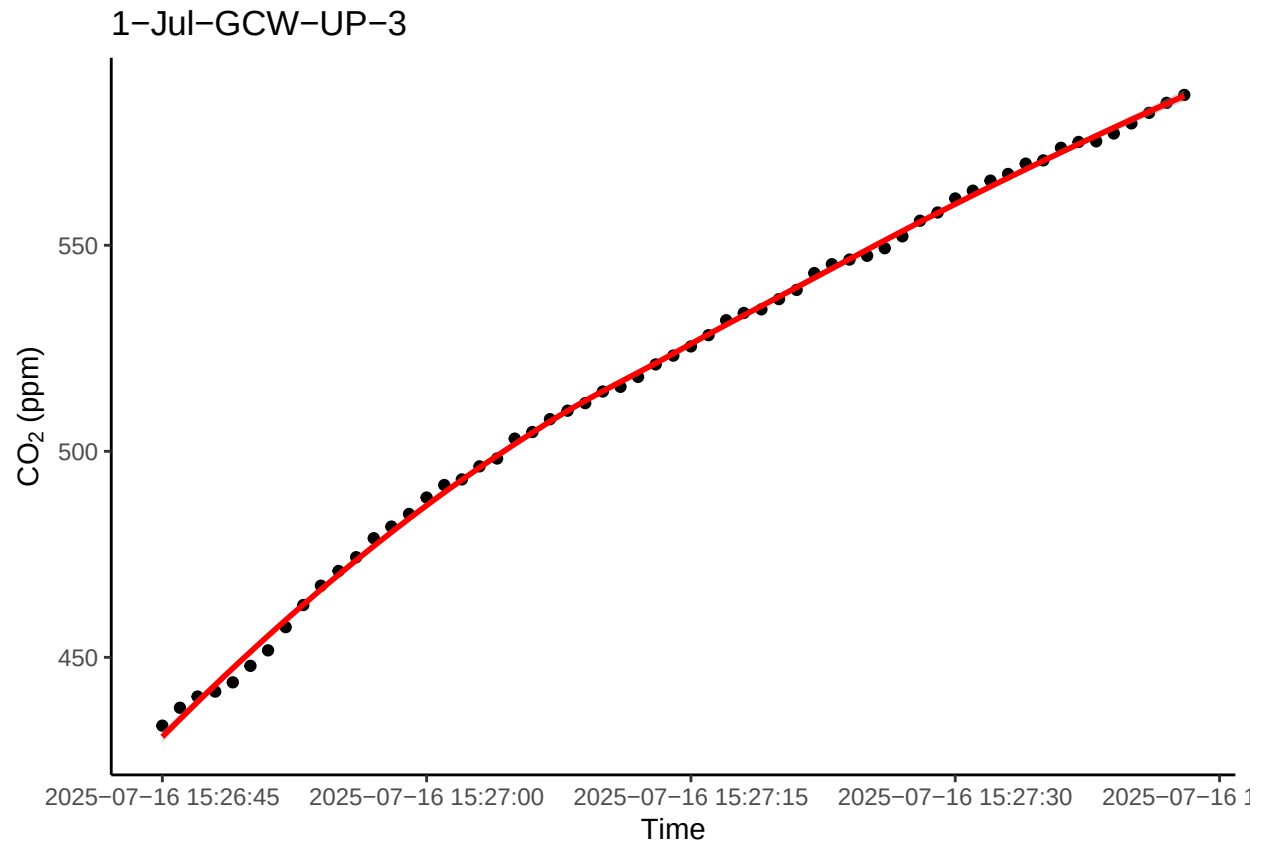
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



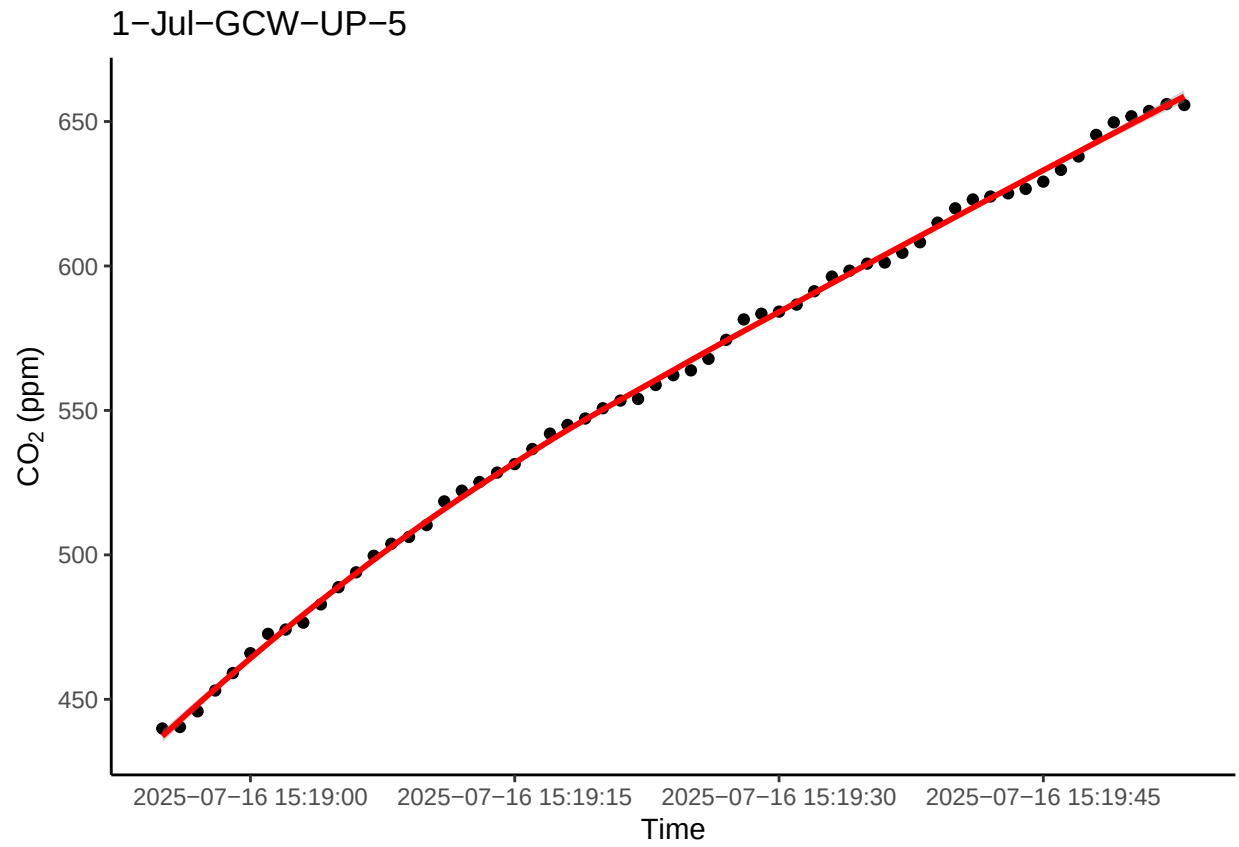
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



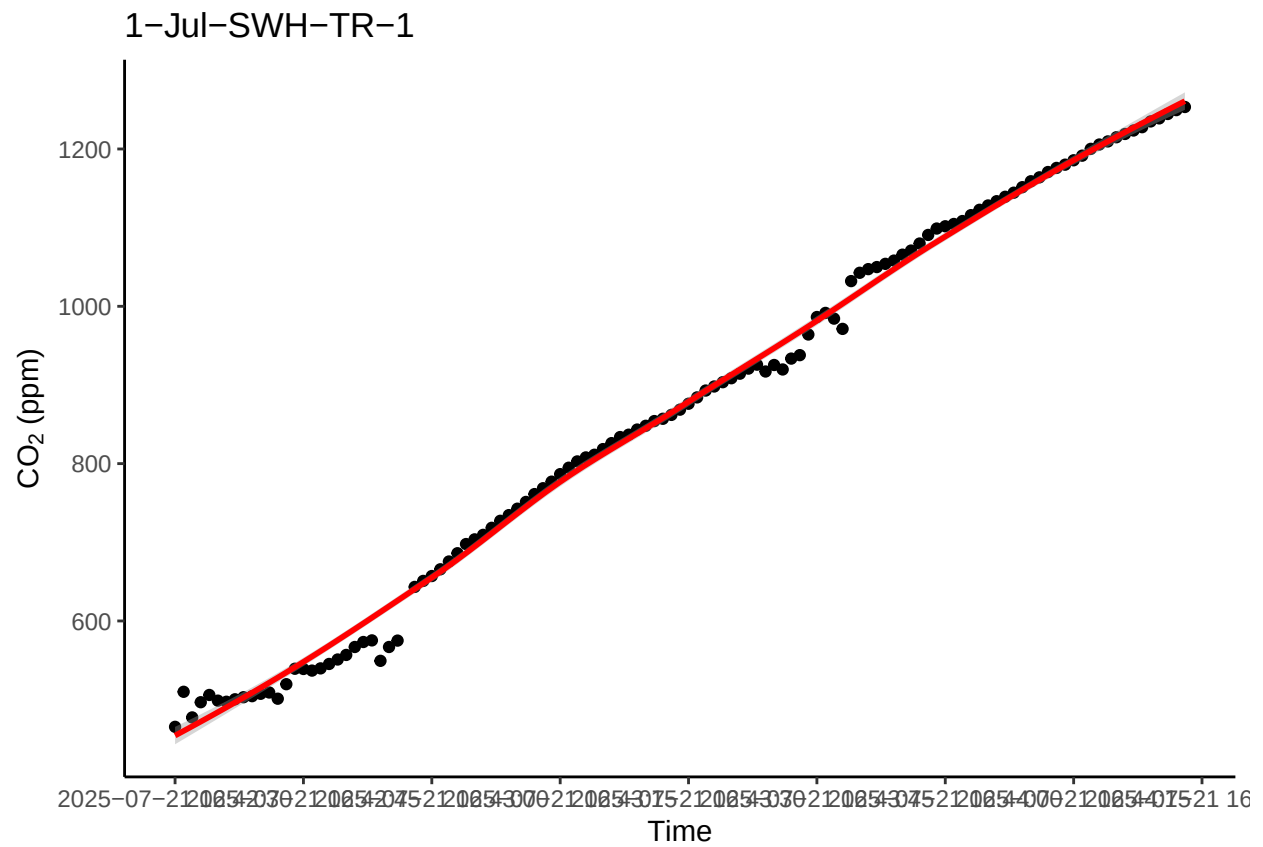
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



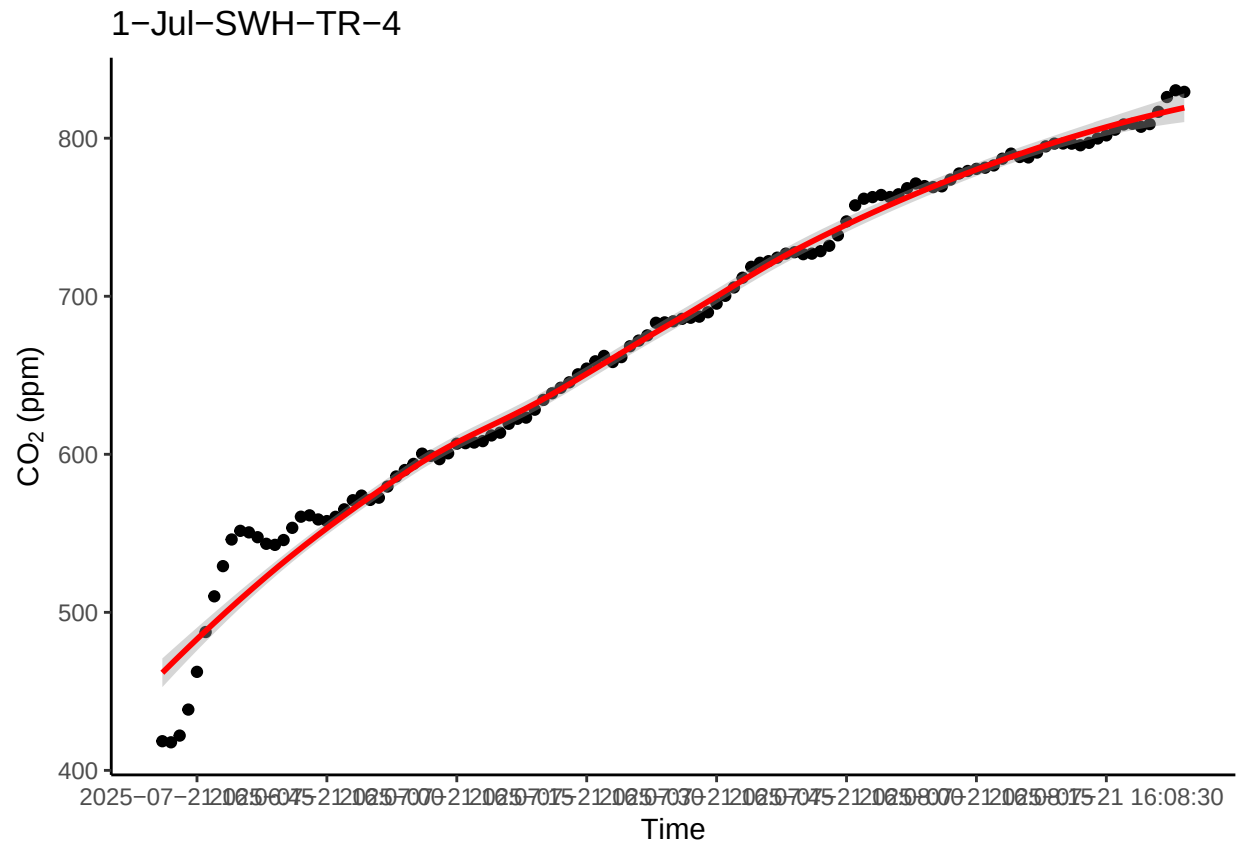
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



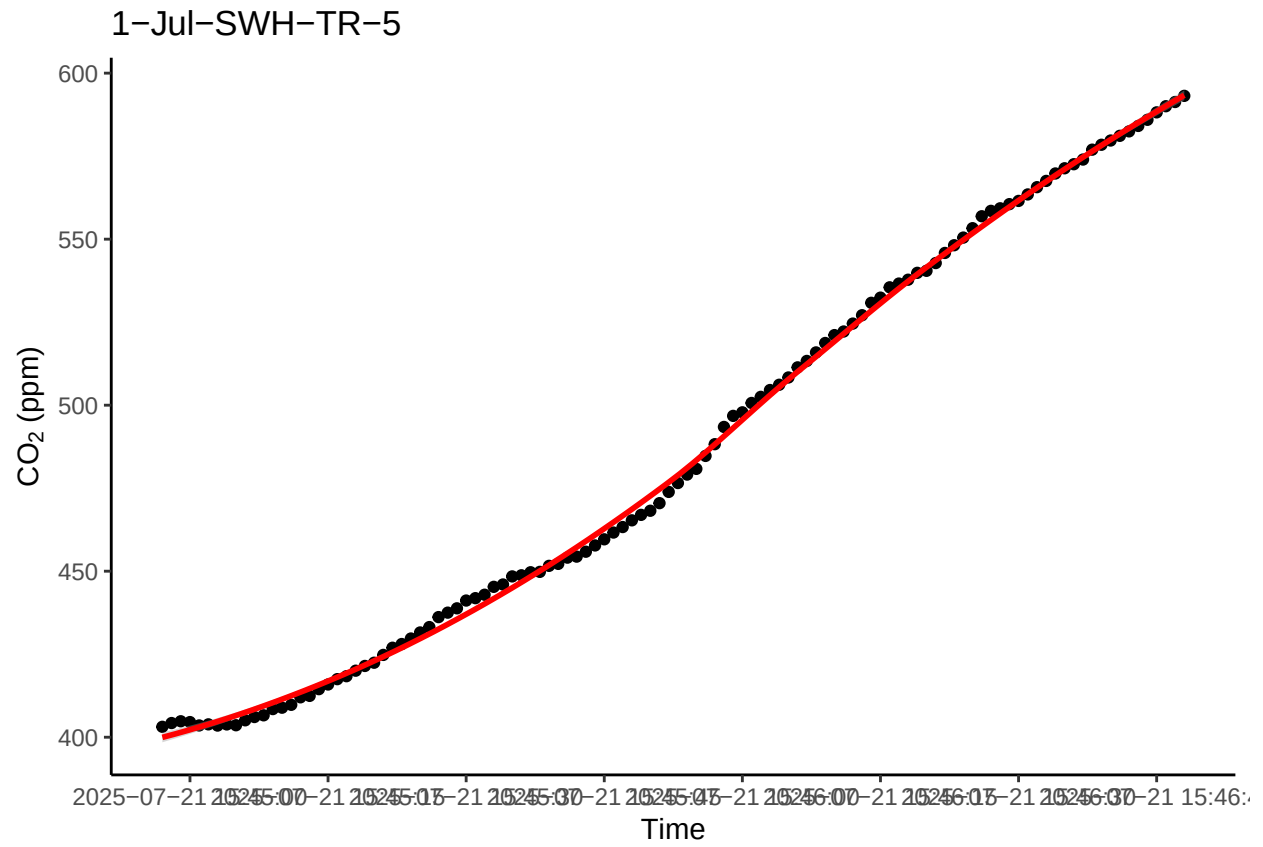
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



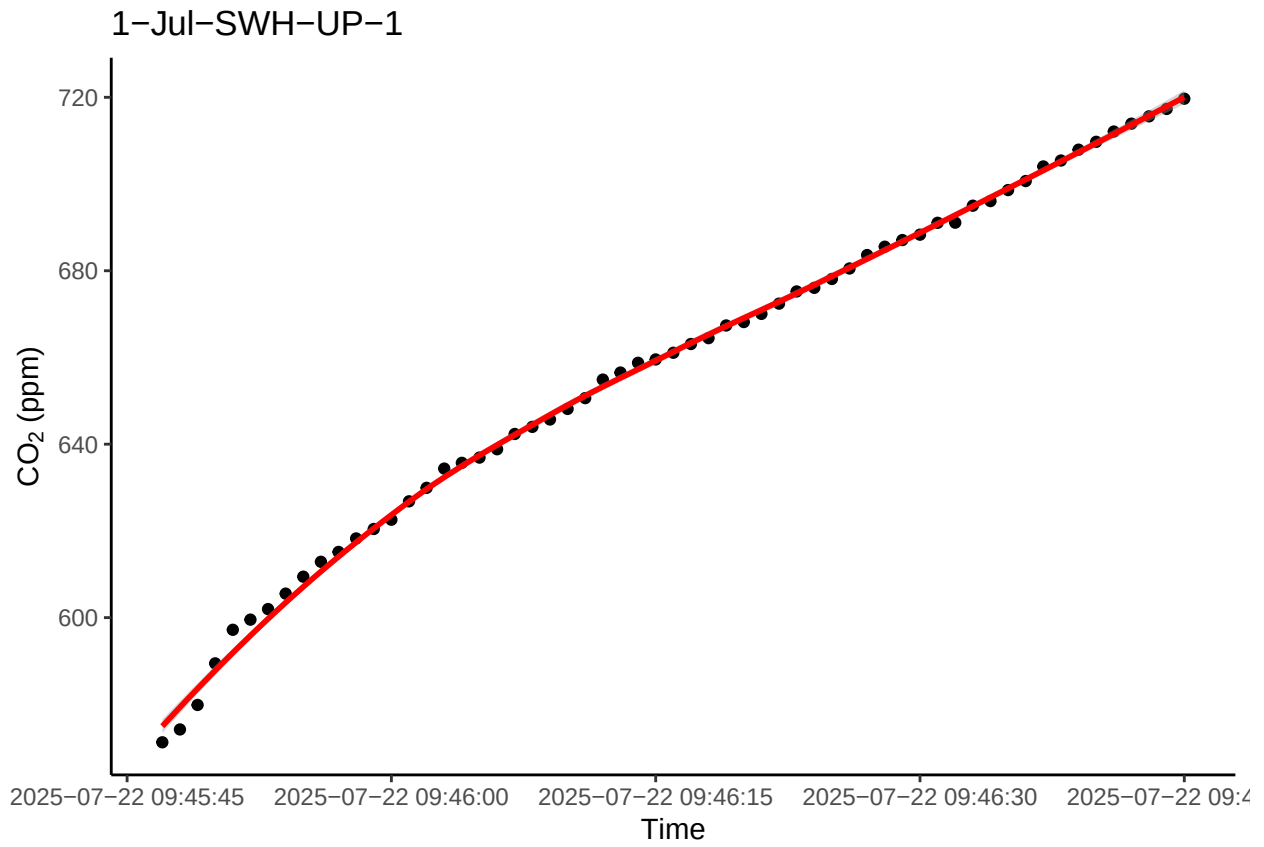
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



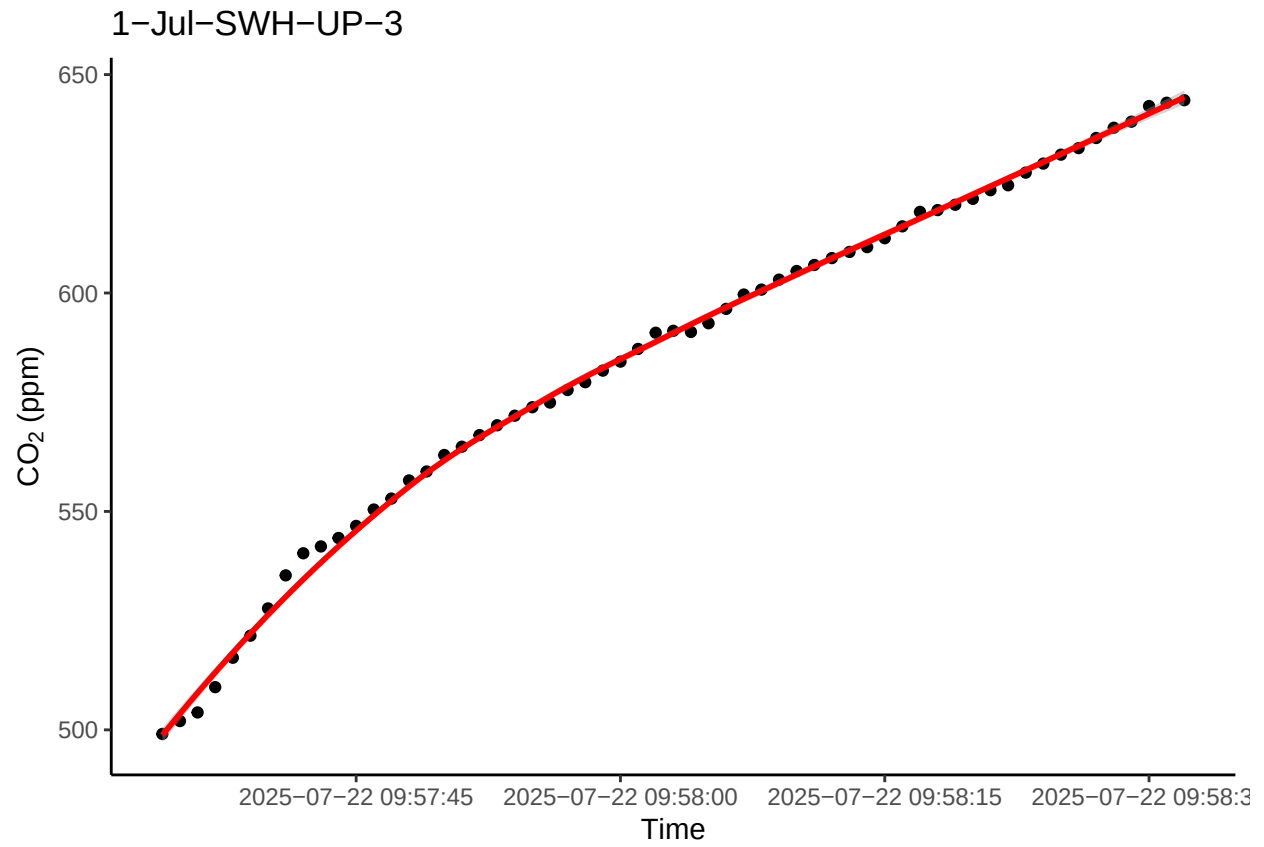
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



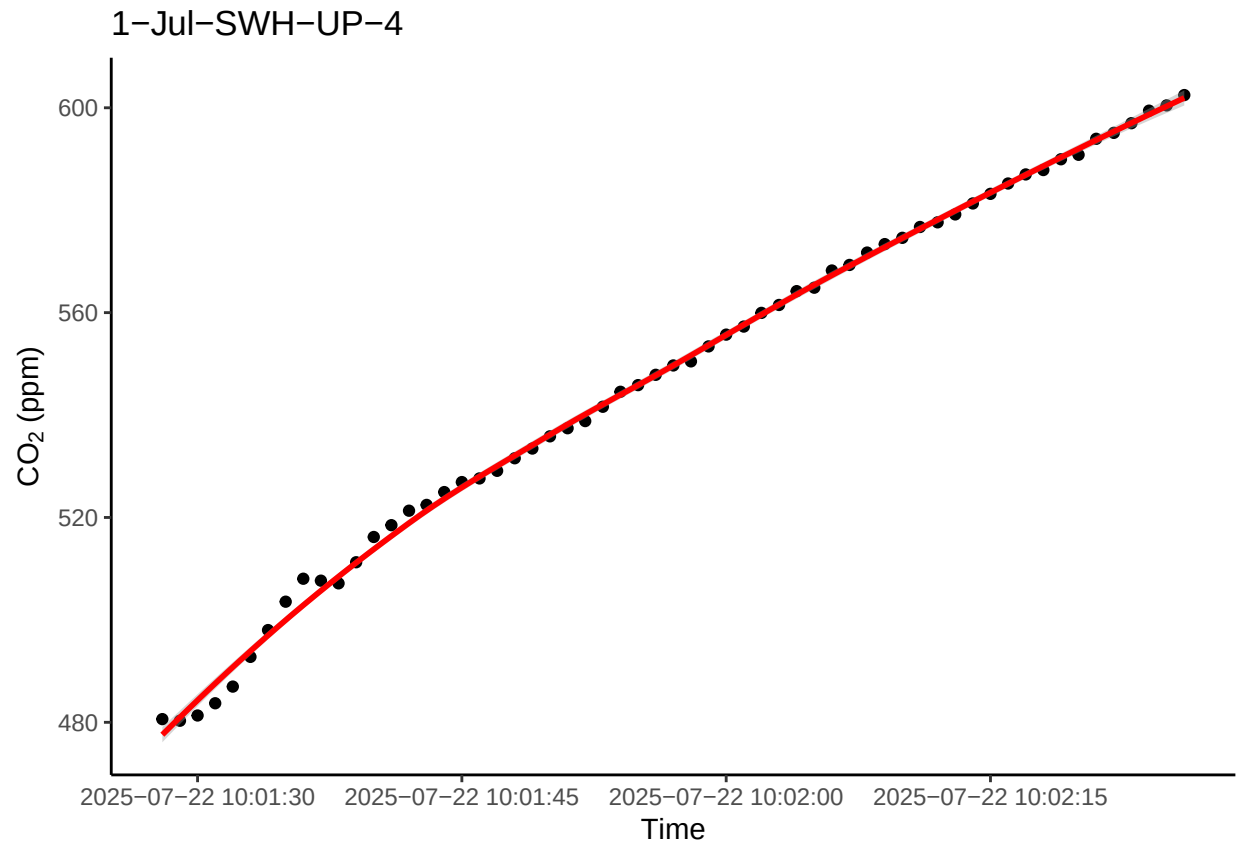
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



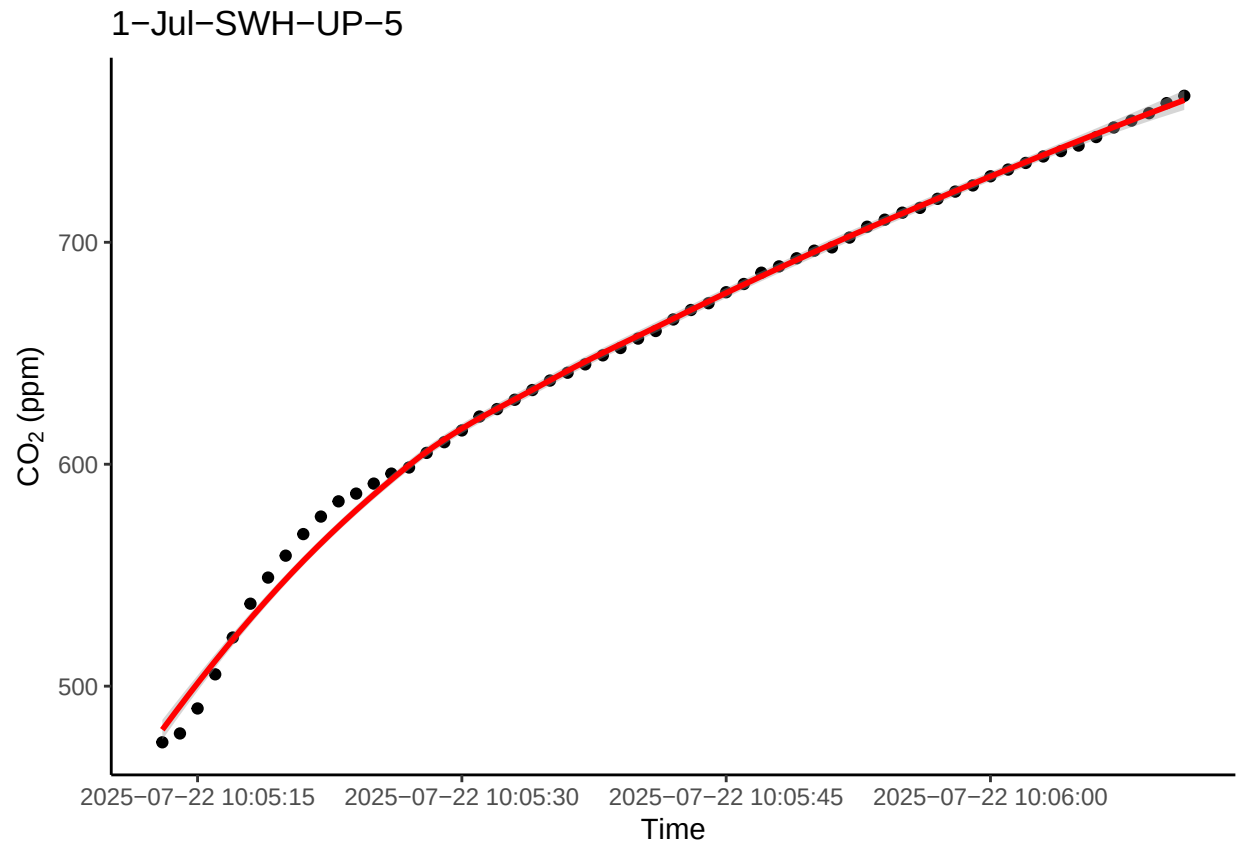
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



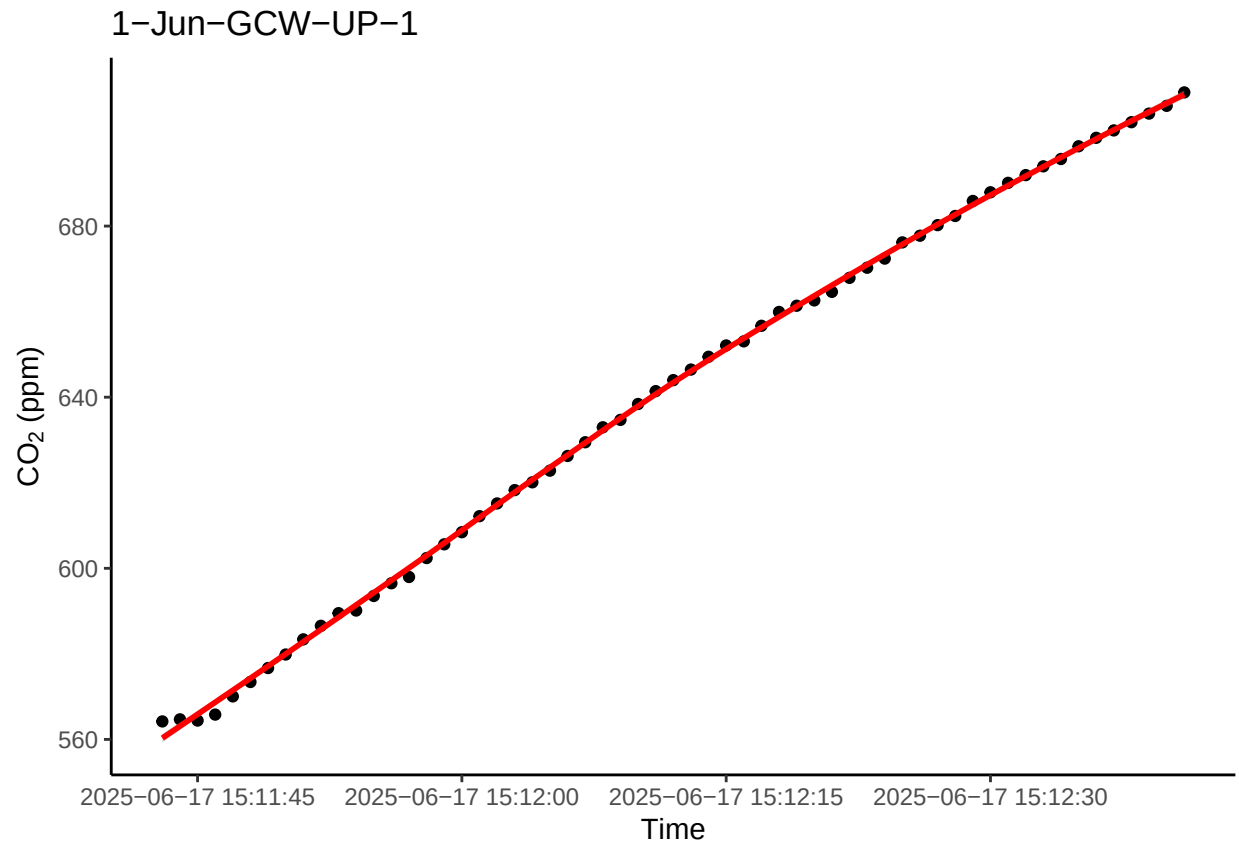
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



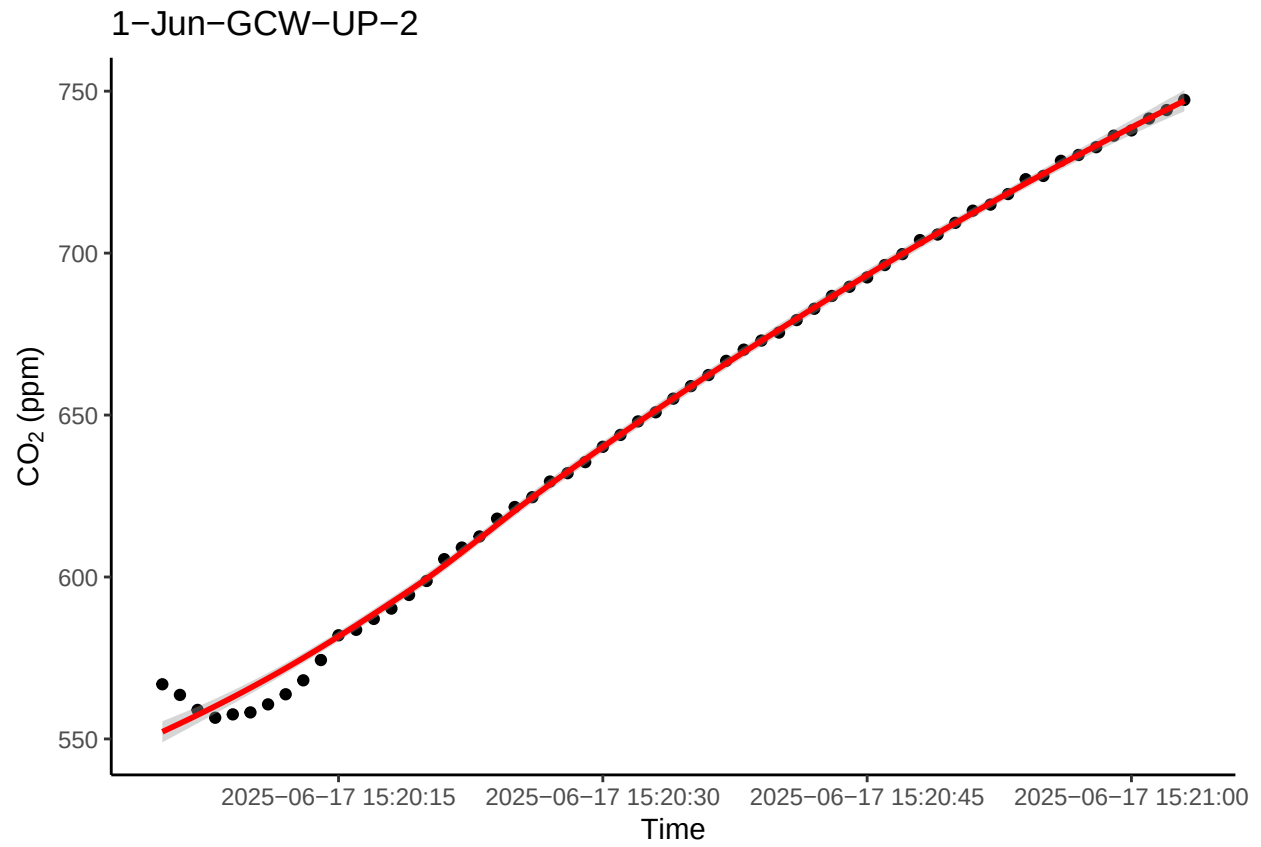
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



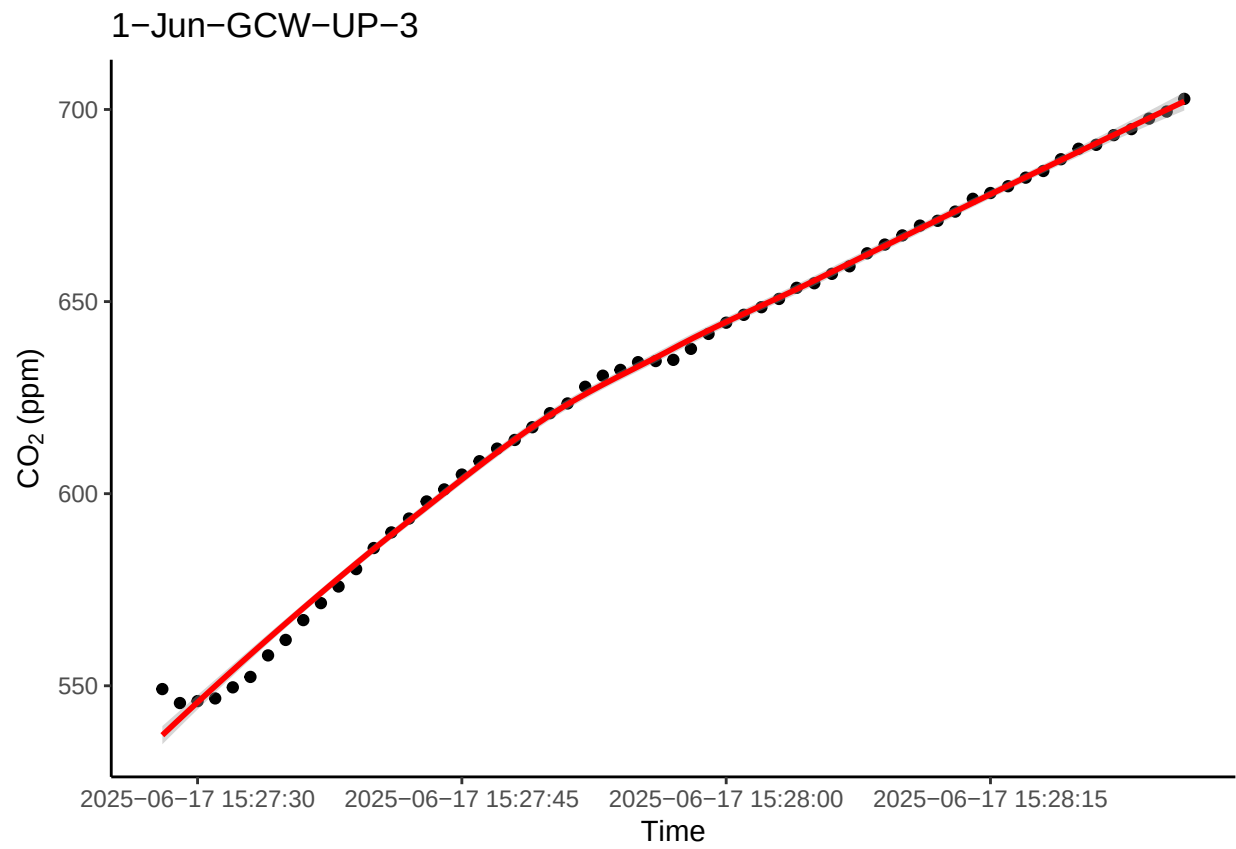
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



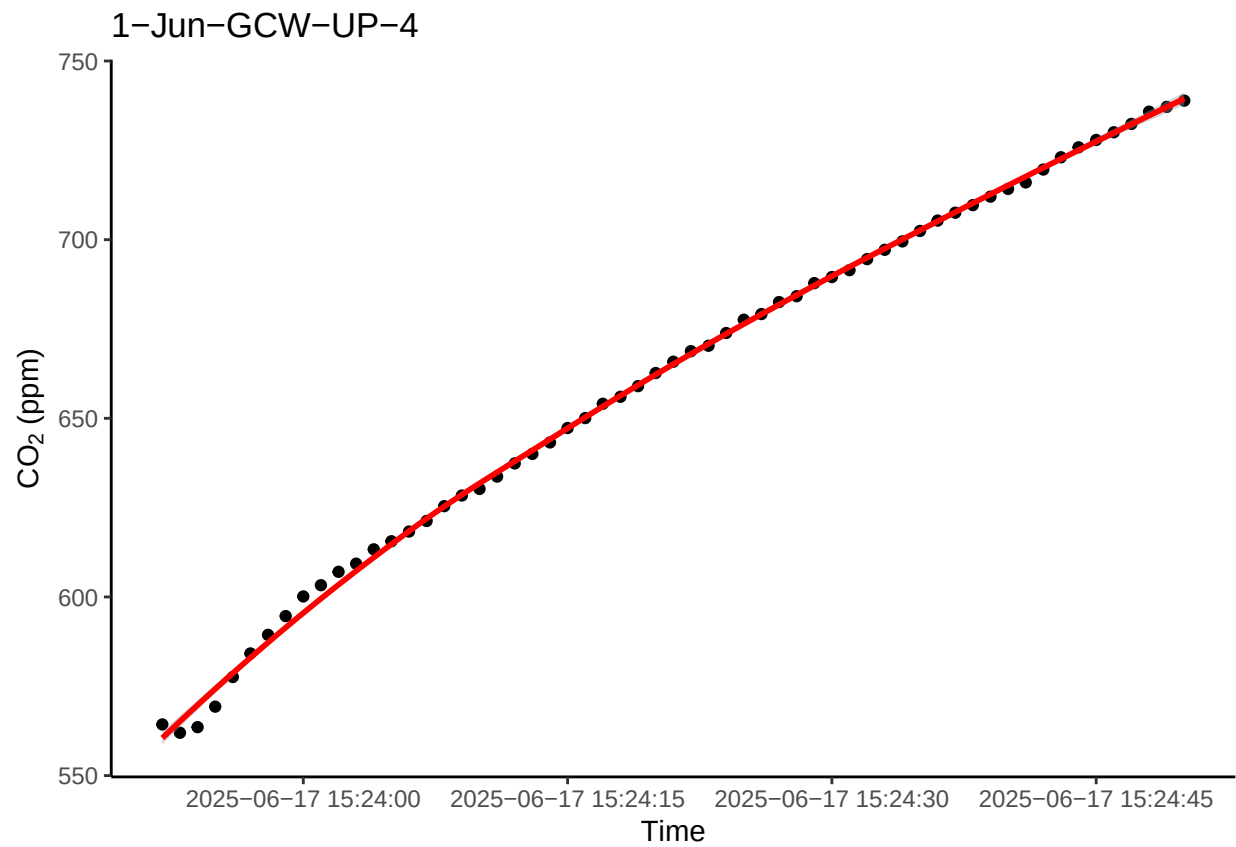
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



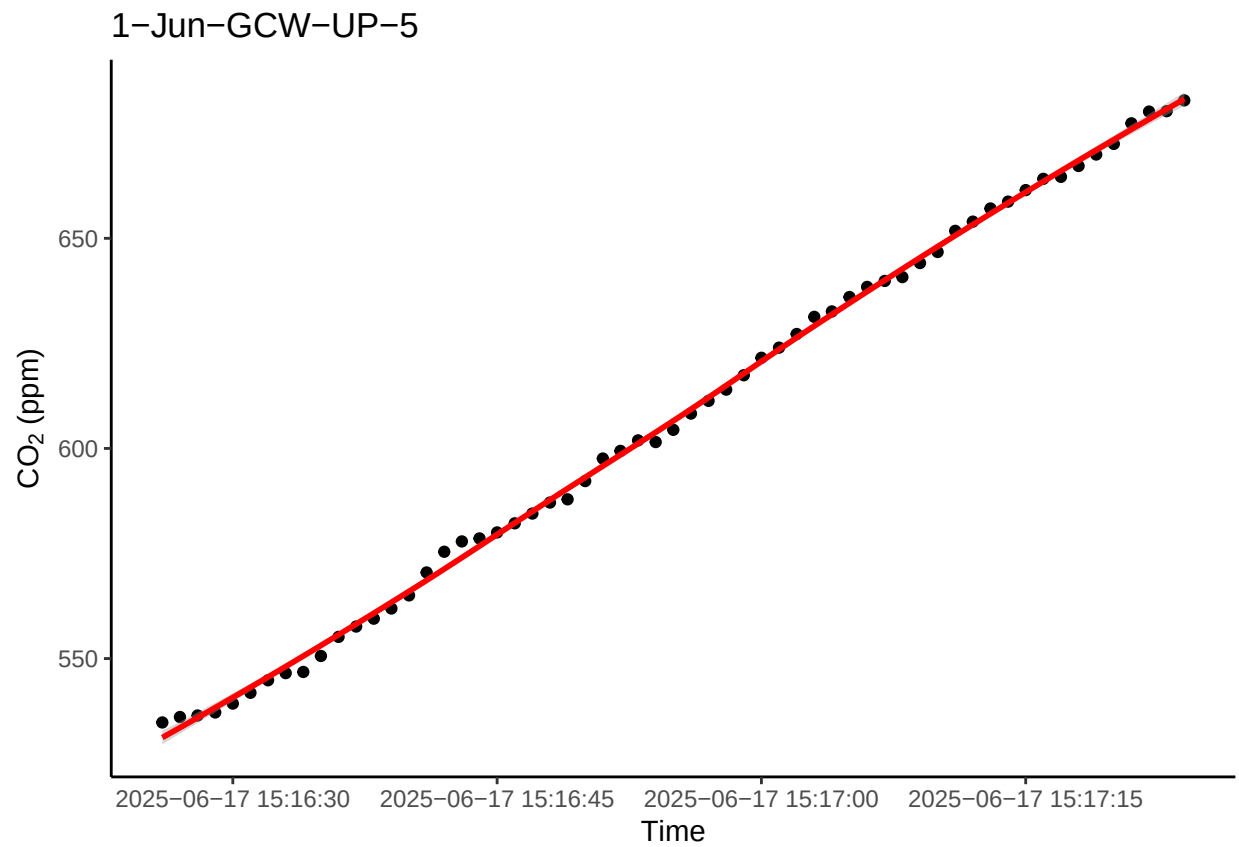
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



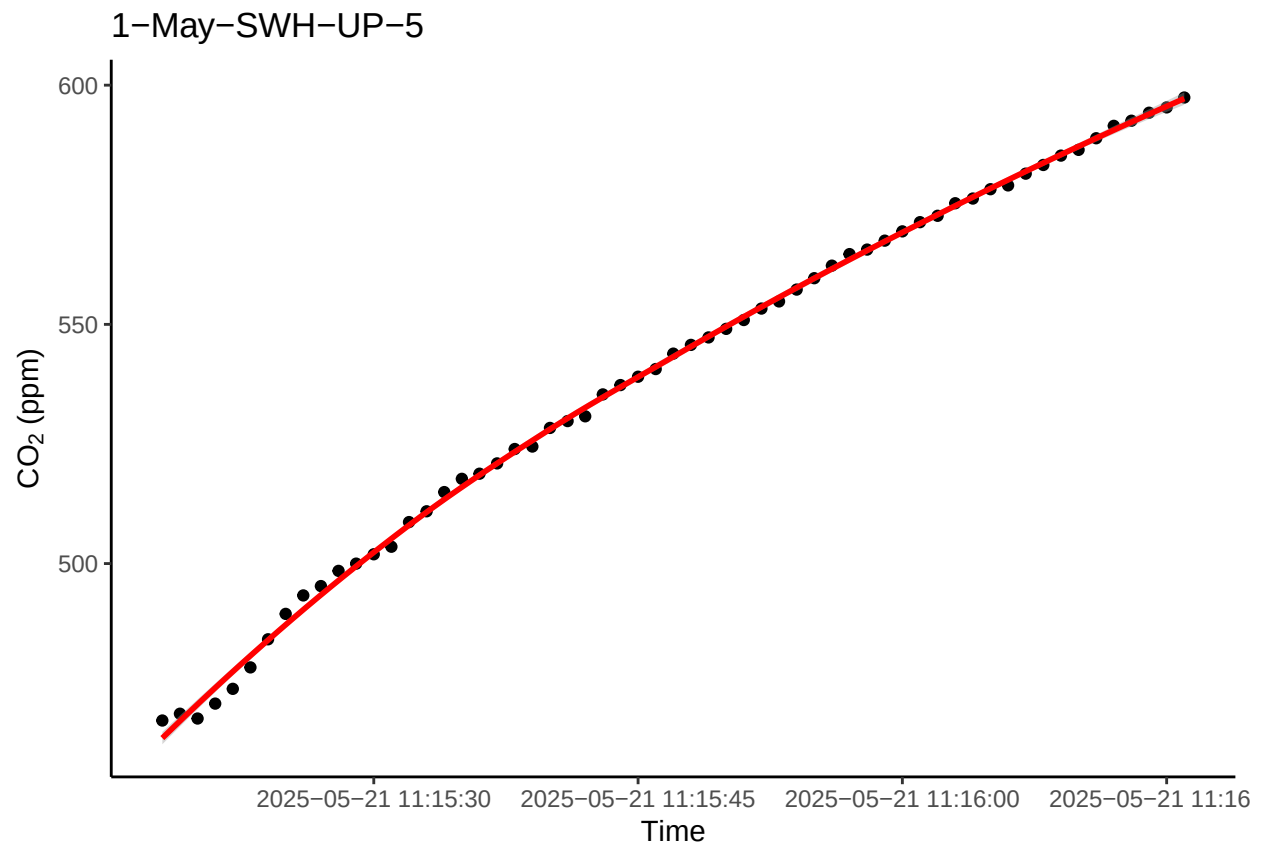
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



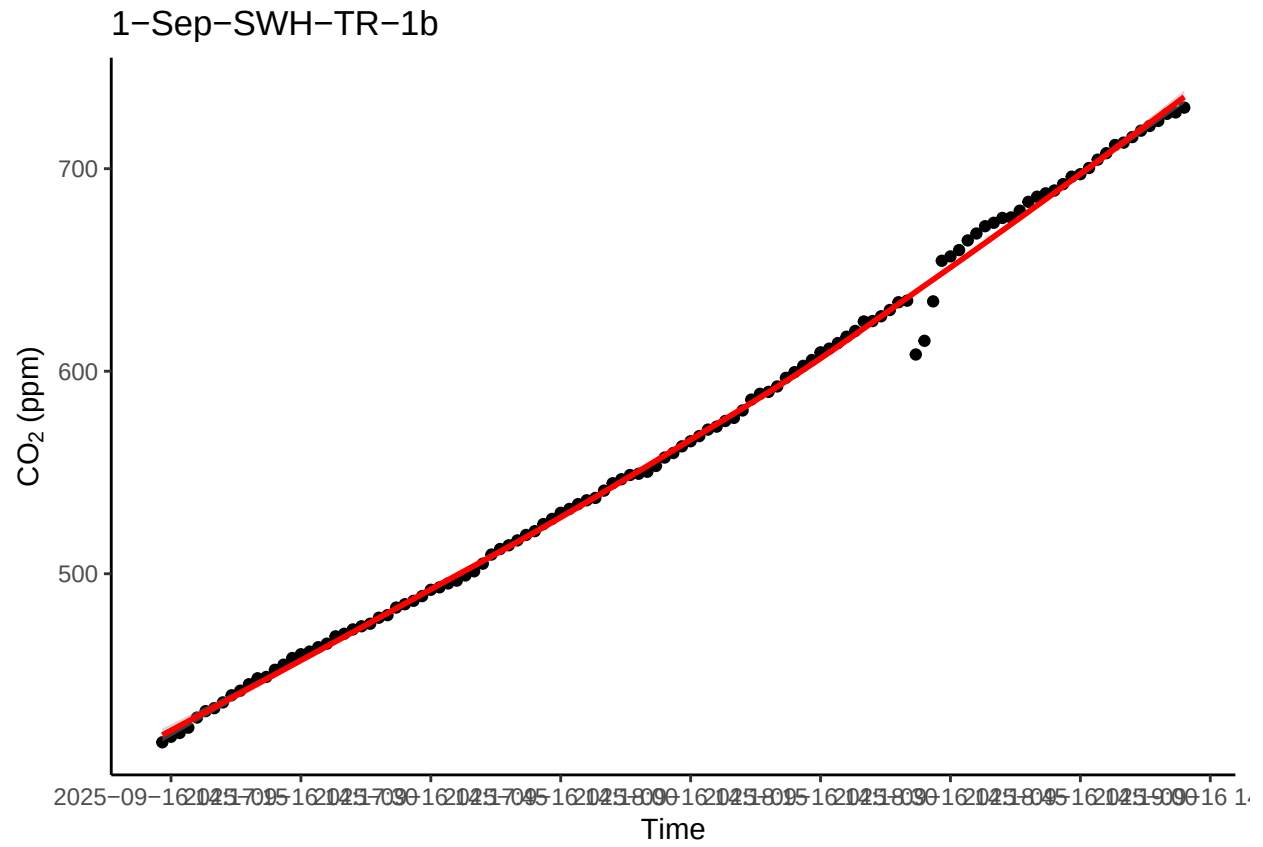
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



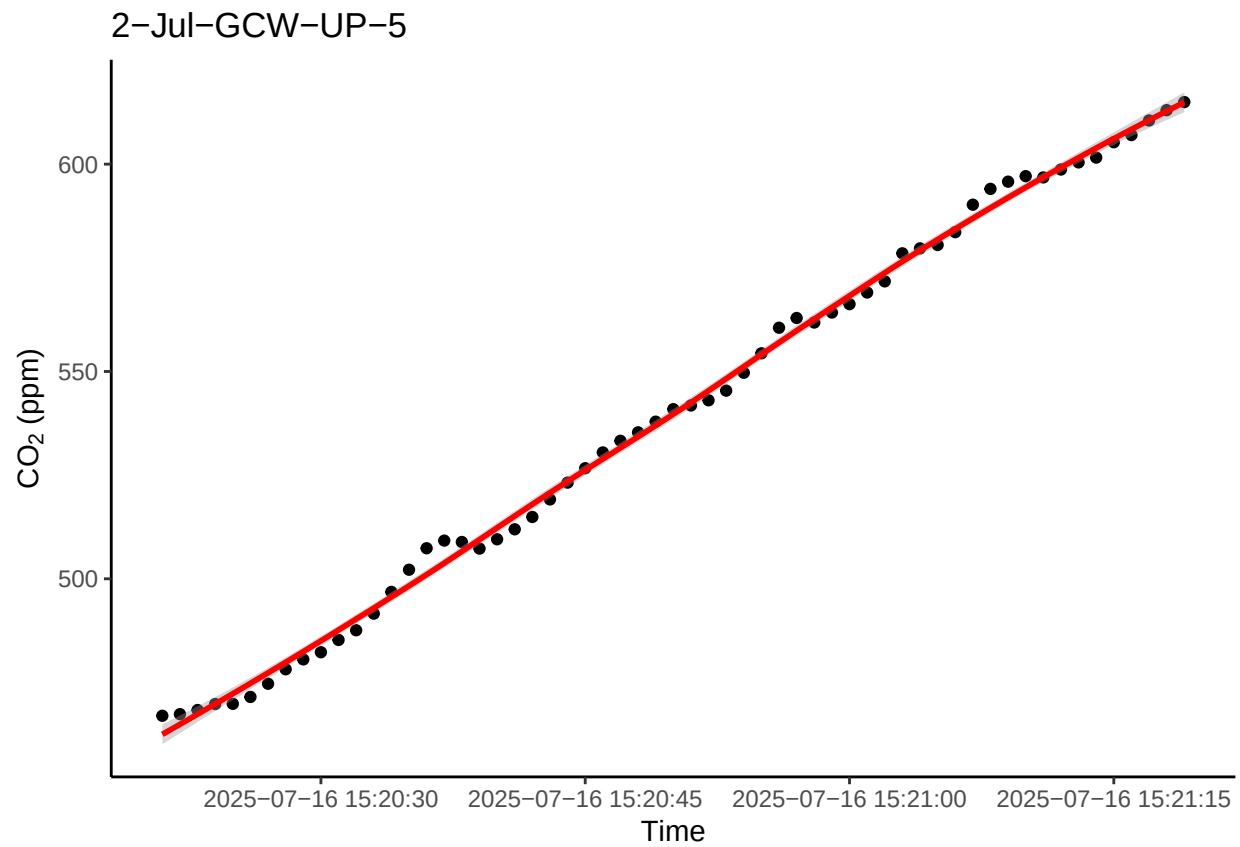
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



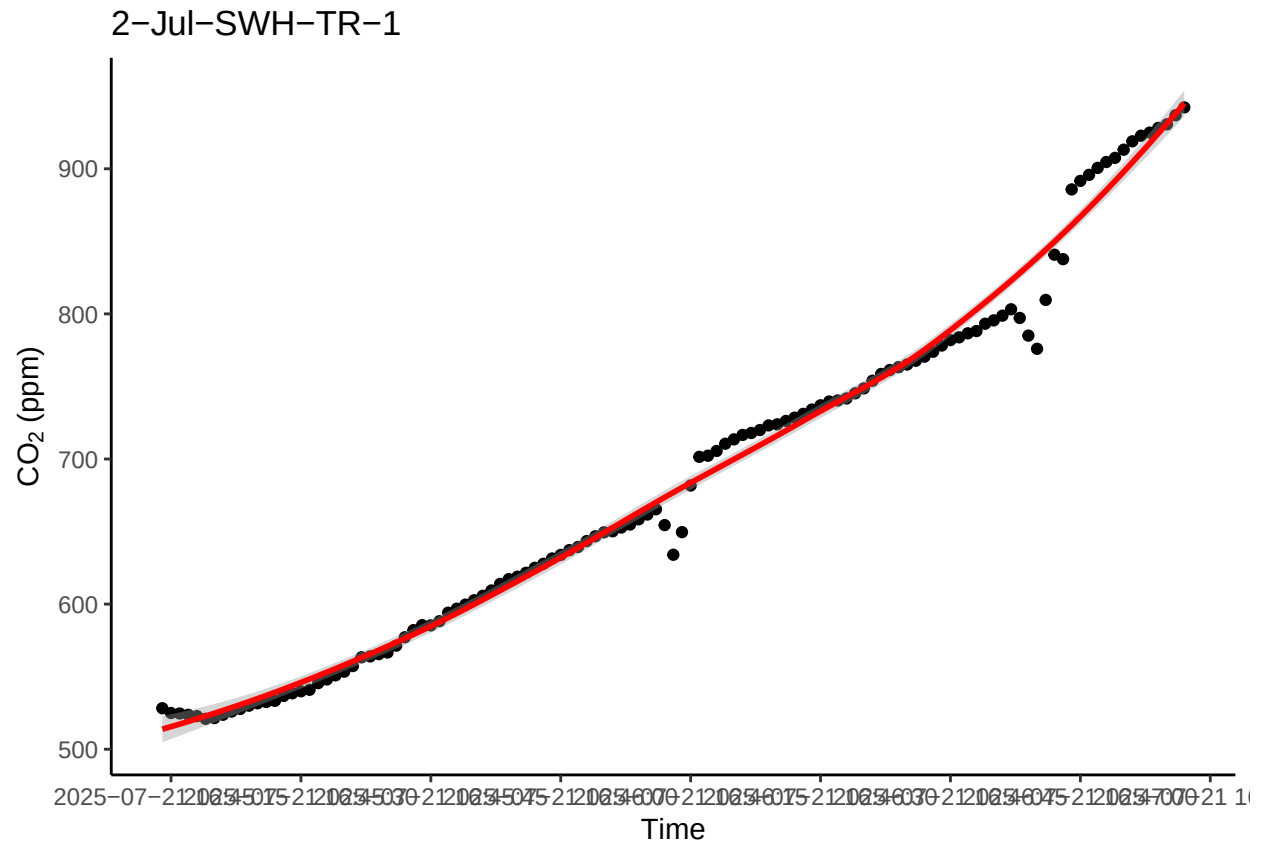
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



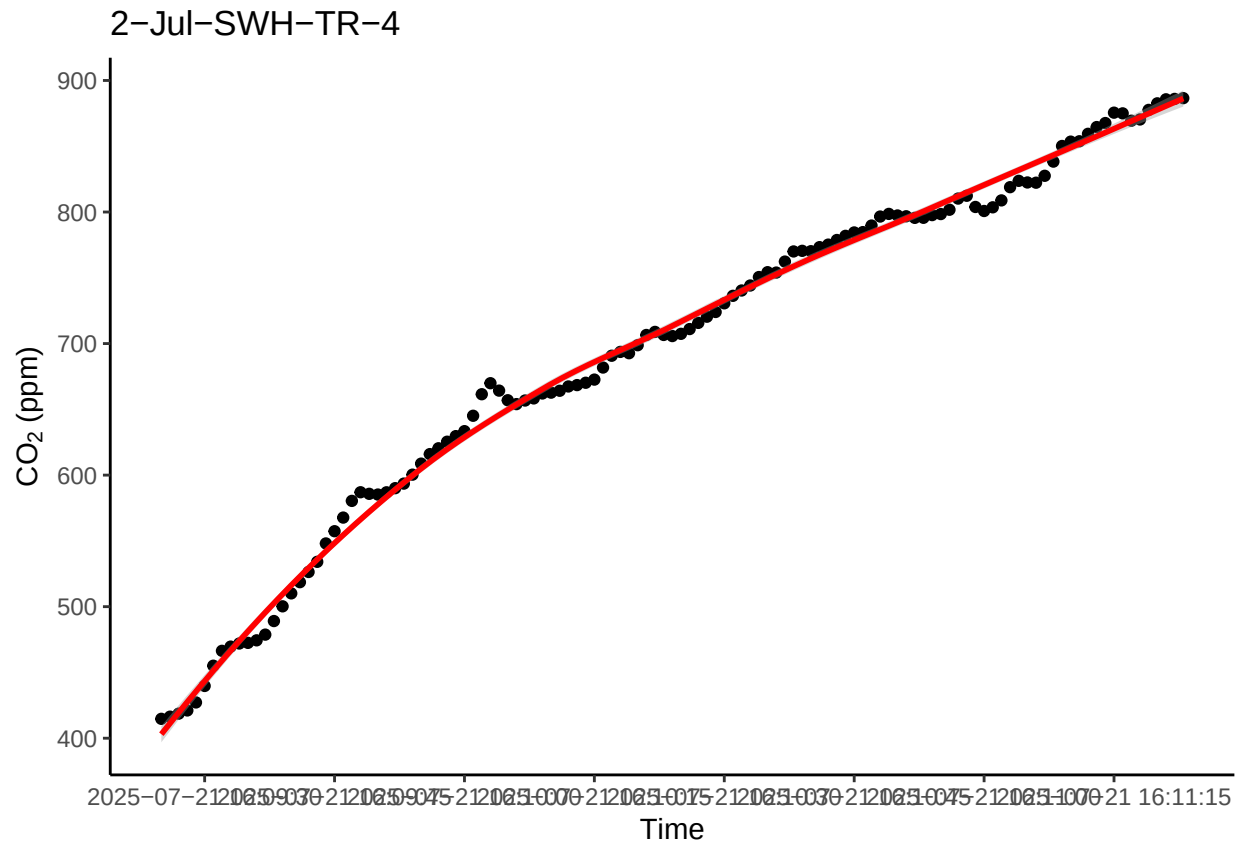
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



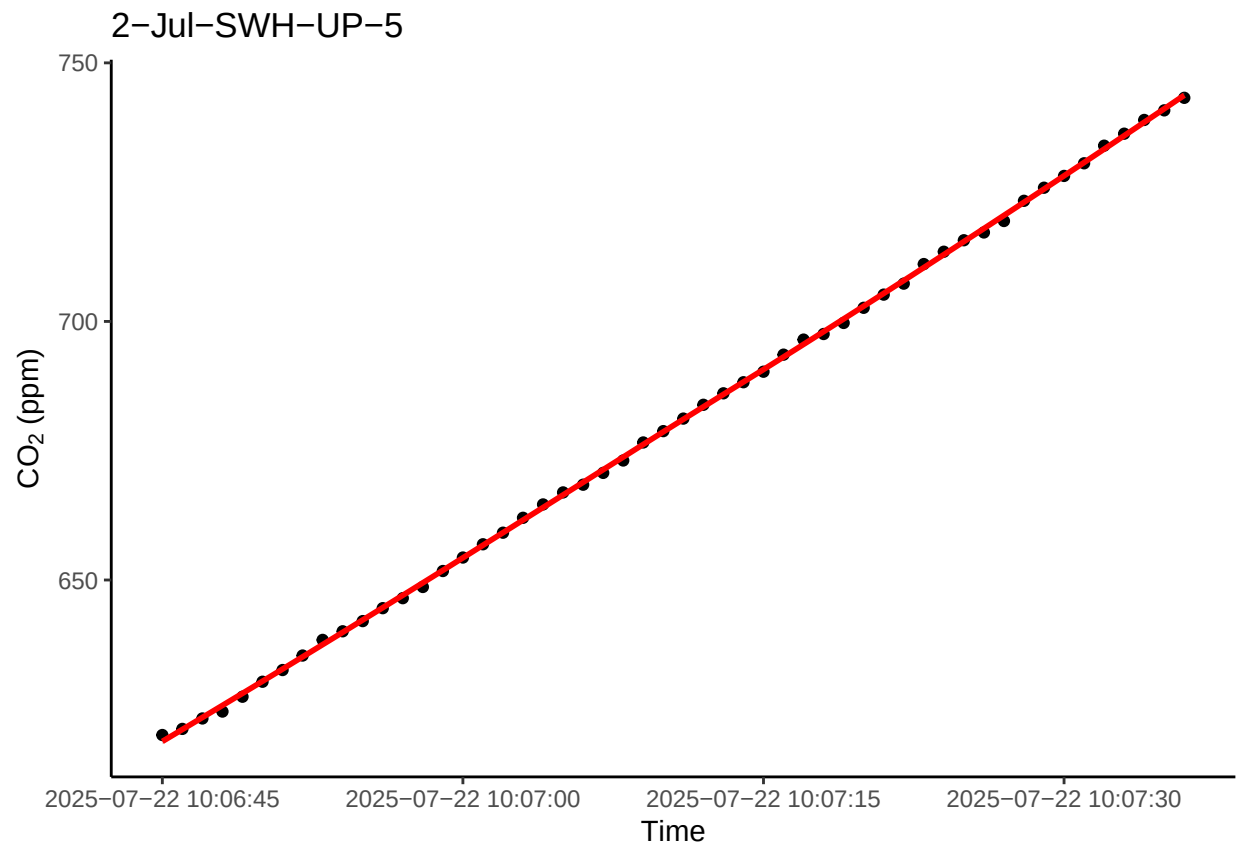
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



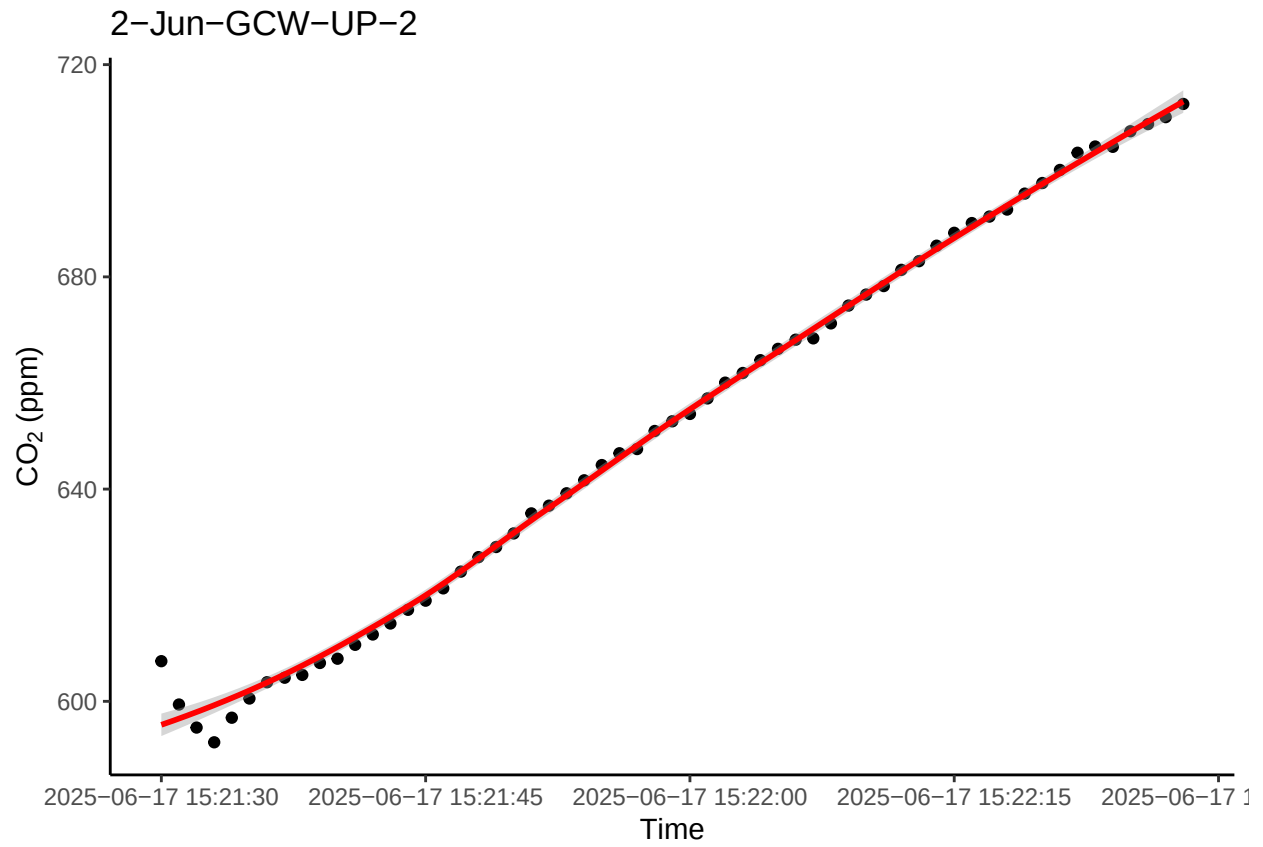
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



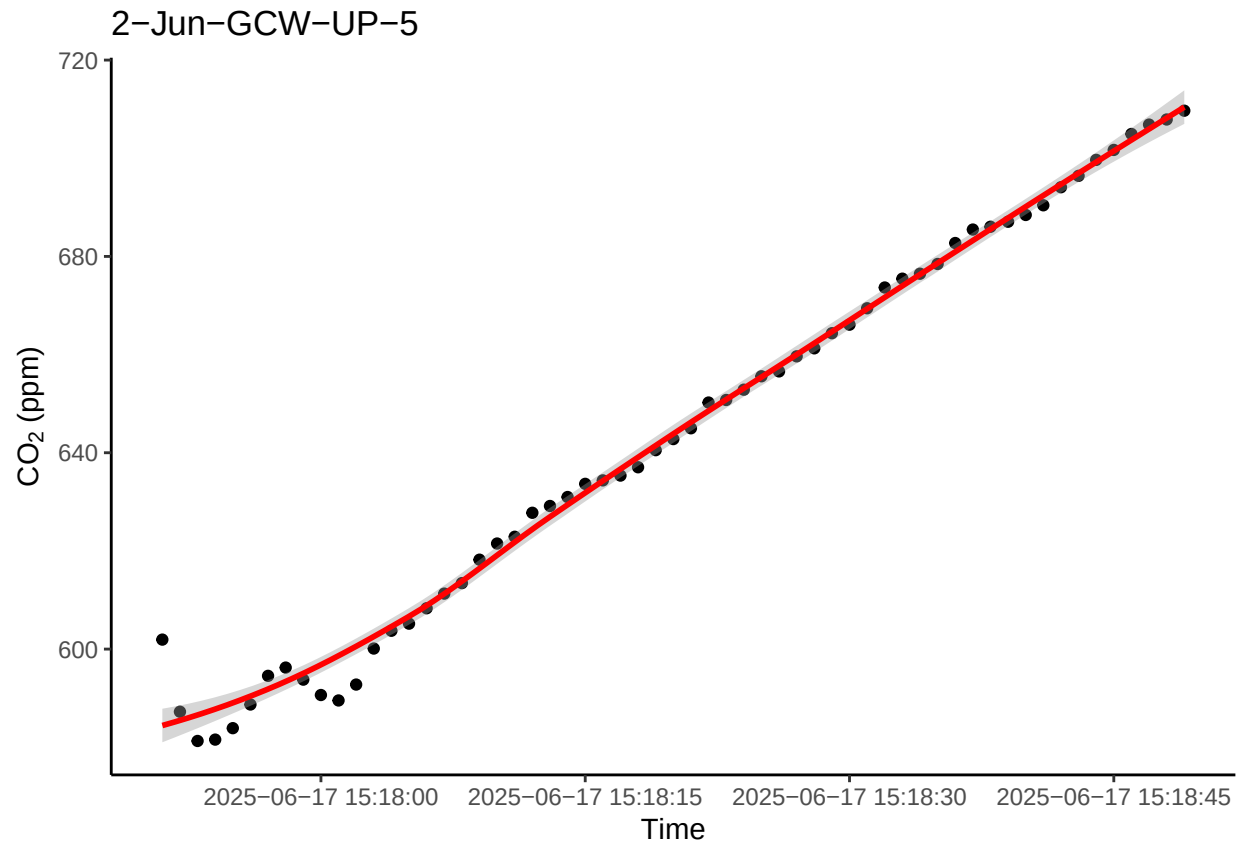
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



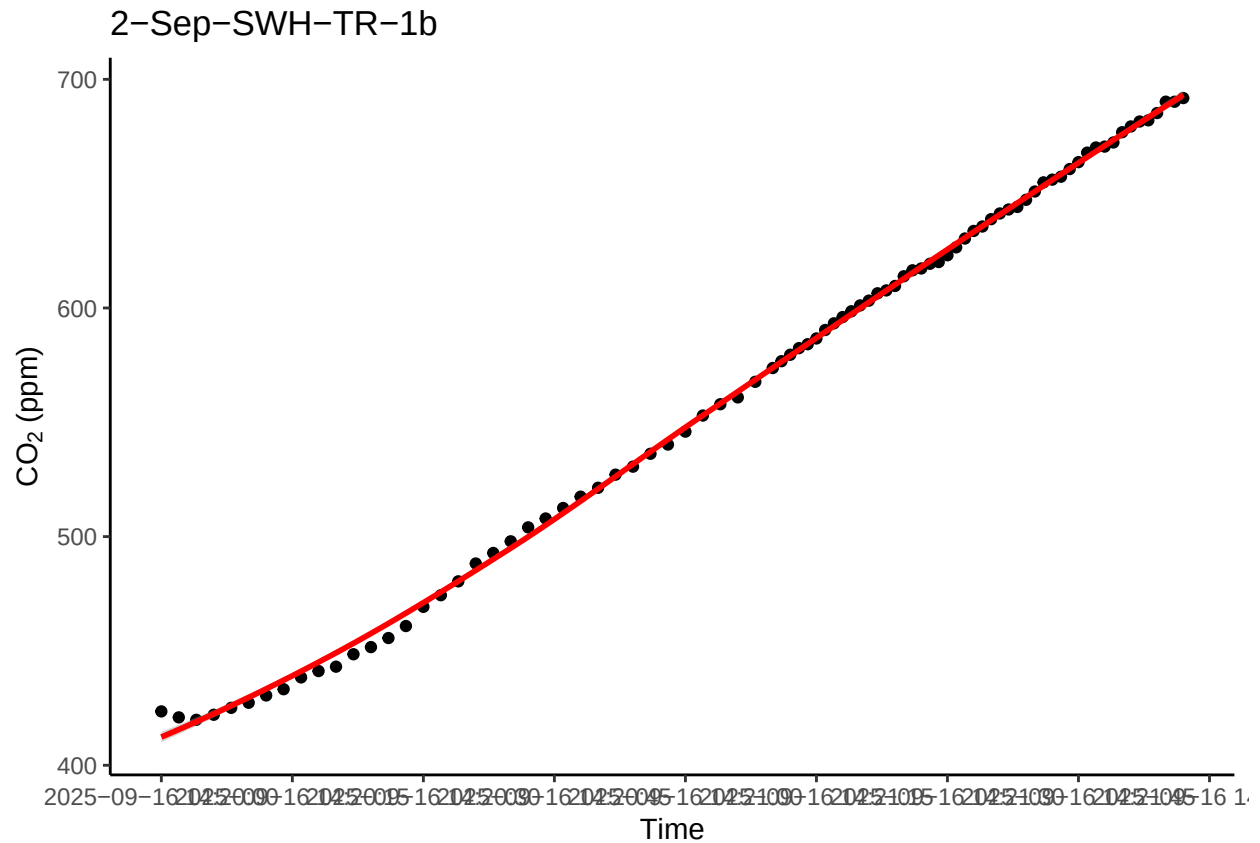
```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



```
## 'geom_smooth()' using method = 'loess' and formula = 'y ~ x'
```



2.2 Check all fluxes

```

““{#r, echo=FALSE}

##Put all fluxes together fluxes_all <- as.data.frame(subset(allflux, select = Plot:TIMESTAMP_max))
all_plots <- unique(fluxes_all$Plot)

#Loop to run five random plots subset above for(P in all_plots){ r1 <- subset(alldat, Plot == P) p1
<- ggplot(r1, aes(TIMESTAMP, co2)) + geom_point() + theme_classic() + scale_x_datetime(breaks
= "15 sec") + ylab(expression(paste( CO [2], " (ppm)"))) + xlab("Time") + labs(title = P) +
geom_smooth(color="red") print(p1) }

## Flag any fluxes that do not fit criteria

```


		Plot	TIMESTAMP	HM81_AIC	HM81_flux.estimate	HM81_p.value
2.3						
2.4	1	1-Apr-GCW-TR-1	2025-04-16 10:58:18	NA	NA	NA
2.5	2	1-Apr-GCW-TR-2	2025-04-16 11:07:21	415.1026	2.767439	1.620905e-31
2.6	3	1-Apr-GCW-TR-3	2025-04-16 11:11:25	417.9395	2.715836	2.236664e-31
2.7	4	1-Apr-GCW-TR-4	2025-04-16 11:15:46	350.5254	1.579223	7.544911e-33
2.8	5	1-Apr-GCW-TR-5	2025-04-16 11:20:03	407.1443	1.859655	7.860953e-29
2.9	6	1-Apr-GCW-UP-1	2025-04-16 11:25:26	NA	NA	NA
2.10				HM81_r.squared	HM81_RMSE	lin_AIC lin_flux.estimate lin_flux.std.error
2.11	1	NA	NA	471.5188	6.246571	0.16032365
2.12	2	0.9288755	10.890805	299.8162	2.560309	0.03269938
2.13	3	0.9279904	11.180673	314.5950	2.609335	0.03749459
2.14	4	0.9367803	5.989341	308.6082	1.477810	0.03547271
2.15	5	0.9098118	10.117133	302.4367	2.107285	0.03350251
2.16	6	NA	NA	292.1896	1.791610	0.03046990
2.17				lin_int.estimate	lin_int.std.error	lin_p.value lin_r.squared lin_RMSE
2.18	1	3745.930	5.634565	3.725386e-40	0.9668802	18.362133
2.19	2	3820.197	1.149218	1.227954e-55	0.9915894	3.745114
2.20	3	3865.564	1.317745	5.325900e-53	0.9893771	4.294318
2.21	4	3821.678	1.246686	1.273818e-41	0.9709106	4.062748
2.22	5	3867.925	1.177444	9.630206e-51	0.9870270	3.837098
2.23	6	3823.874	1.070863	3.055808e-49	0.9851825	3.489769
2.24				poly_AIC	poly_r.squared	poly_RMSE rob_AIC rob_converged
				rob_flux.estimate		
2.25	1	277.0248	0.9991611	2.980210	472.0052	TRUE 6.340314
2.26	2	285.7144	0.9939849	3.229907	299.8216	TRUE 2.558427
2.27	3	275.5805	0.9952104	2.940620	314.5963	TRUE 2.609710
2.28	4	292.6486	0.9798994	3.444087	308.7654	TRUE 1.466874
2.29	5	296.0562	0.9892958	3.554485	302.4814	TRUE 2.106470
2.30	6	295.6443	0.9853314	3.540957	292.2965	TRUE 1.784351

```
## Do some visualization of QAQC plots to assess fluxes
<!-- --> ).**

```
<!-- -->
```

```
Read out final dataframe and export for use elsewhere
```

2.47 **Warning in rbind(c("1", "Apr", "GCW", "TR", "1"), c("1", "Apr", "GCW", "TR", :**

2.48 **number of columns of result is not a multiple of vector length (arg 61)**

2.49 **RepNumber Month Site Zone Replicate**

2.50 **1 1 Apr GCW TR 1**

2.51 **2 1 Apr GCW TR 2**

2.52 **3 1 Apr GCW TR 3**

2.53 **4 1 Apr GCW TR 4**

2.54 **5 1 Apr GCW TR 5**

2.55 **6 1 Apr GCW UP 1**

““